

Why Rotating Fluids Make Polygons

Paper III: Gravitational Theory

Gordon Speagle

Abstract

We derive the vacuum Einstein equations $R_{\mu\nu} = \Lambda g_{\mu\nu}$ from three conditions on regular point-vortex polygons, all of which are mathematical theorems: (H1) the angular eigenvalues have the Havelock form $\lambda_m = C_1 - m(N-m)/2$ ($\csc^2$ uniqueness); (H2) the polygon entropy functional has a unique maximum at constant curvature (strict concavity + Osgood–Phillips–Sarnak spectral determinant theorem); (H3) the Hadamard parametrix holds. Together these force $R = \text{const}$ (the vacuum Einstein equation in 2+1D). A uniqueness theorem establishes the correspondence: the point vortex system is the unique 2D Hamiltonian with polynomial tangential eigenvalues, and $\text{SL}(2, \mathbb{R})$ Chern-Simons is the unique 2+1D topological theory producing the same polynomial structure. The match is forced, not chosen. The graviton emerges at $N = 7$ as a spin-2 boundary Virasoro mode (the unique polygon with $j = 2$ by the Pell equation). The cosmological constant is determined by N through the Wheeler-DeWitt equation on the breathing mode. In 3+1D, the full Riemann tensor (all 10 Weyl components) is determined by N (Theorem ??).

1 Introduction and summary

Papers I and II develop the Havelock decomposition as pure mathematics and fluid dynamics. This paper connects the eigenvalue structure to gravity. The starting point is the observation (Paper II) that N equal point masses on a spatial slice of 2+1D gravity interact via the same Green’s function as point vortices (Proposition 13.1), giving identical Havelock eigenvalues $\lambda_m = C_1(\xi) - m(N-m)/2$. The conformal invariance of the flat-plane backreaction (Proposition 13.2) ensures that $N_{\text{crit}} = 7$ is independent of the deficit angle, a simplification that makes the algebraic-number thresholds of Paper I into gravitational phase transitions.

The argument proceeds as follows. The $N = 7$ marginally stable mode carries spin 2, so the stability threshold is a graviton (§??); the Chern–Simons formulation makes the spin identification exact. The central theorem (§??) then shows that conditions (H1)–(H3) force the scalar curvature to be constant via a Jensen inequality on the polygon entropy: this is the vacuum Einstein equation in 2+1D. The Wheeler–DeWitt equation on the breathing mode (§??) fixes the cosmological constant Λ in terms of N , while the polygon–BTZ correspondence (§??) connects the vortex partition function to black hole thermodynamics. In 3+1D, a Kaluza–Klein decomposition and Fefferman–Graham reconstruction determine the full Riemann tensor (all 10 Weyl components) from N alone (Theorem ??, §??).

Main result. Theorem ??: under conditions (H1)–(H3), the polygon entropy functional has a unique maximum at uniform curvature $R = \text{const}$, implying the vacuum Einstein equations $R_{\mu\nu} = \Lambda g_{\mu\nu}$.

The foundational chain. The gravitational partition function for N equal conical singularities on $\mathbf{H}^2$ is a function of the Havelock eigenvalues. This is a theorem, proved by composing three results:

1. *2+1D gravity = $\text{SL}(2, \mathbb{R})$ CS theory (?)*. The Einstein–Hilbert action with $\Lambda < 0$ in 2+1D is identically equal to the $\text{SL}(2, \mathbb{R})$ Chern–Simons action. Conical singularities become punctures with fixed holonomy. This is an exact equivalence of actions, not an approximation.
2. *CS on punctured surface $\rightarrow$ KZ $\rightarrow$ CMS* (Etingof–Frenkel–Kirillov, Ch. 7). The CS flatness constraint at the $\mathbb{Z}_N$ -symmetric locus is the CMS Hamiltonian. The operators are equal, not merely analogous.
3. *CMS Casimir = Havelock eigenvalue* (Proposition ??). The operator identity $D_{\text{CS}}^2 = \Delta_{\mathbf{H}^2} = \Omega_{\mathfrak{sl}(2, \mathbb{R})}$ (Witten + Helgason) gives $C_2(m) = m(N-m)/2 = f(m, N)$ on each $\mathbb{Z}_N$ mode.

The composition

$$\text{Regge vertex} \xrightarrow{\text{Witten}} \text{CS puncture} \xrightarrow{\text{EFK}} \text{CMS eigenvalue} \xrightarrow{\text{Helgason}} \text{Havelock eigenvalue}$$

is exact at every step. Each arrow is a published theorem connecting *operators*, not merely eigenvalues: the gravitational covariant Laplacian, the CS connection, the CMS Hamiltonian, and the $\mathbf{H}^2$ Laplace–Beltrami operator are proved to be the same operator on the same Hilbert space ($\mathbb{Z}_N$ -equivariant functions on $\mathbf{H}^2$). Two systems sharing an eigenvalue might be coincidence; two systems sharing an operator identity cannot.

This chain carries three further structural features that constrain the correspondence beyond eigenvalue matching:

- *First-order dynamics.* CS theory is first-order; Kirchhoff vortex dynamics is first-order; the EFK theorem preserves this structure. Second-order systems (Coulomb, Newtonian) do not have a CS formulation and do not enter.
- *Symplectic compatibility.* The Weil–Petersson and Goldman Poisson structures on the shared Teichmüller moduli space are proportional (??; Proposition ??). The two descriptions have the same phase-space geometry, not just the same spectrum.
- *Uniqueness.* Polynomial tangential eigenvalues uniquely select both the point vortex system (among 2D Hamiltonian systems) and $\text{SL}(2, \mathbb{R})$ CS theory (among 2+1D topological theories): Theorem 1.1.

The $\mathfrak{sl}(2, \mathbb{R})$ Casimir forces $\lambda_m = C_1 - m(N-m)/2$; the Hadamard parametrix makes C_1 affine in R ; strict concavity of the polygon entropy $S = \frac{1}{2} \sum \log \lambda_m$ in C_1 (algebraic, finite N) then gives $R = \text{const}$ by Jensen’s inequality under the Gauss–Bonnet constraint (Theorem ??). The Robin function identity $\Delta\gamma = K - \bar{K}$ promotes this to an exact pointwise result.

The mathematical content (the operator identities, the partition function equality, the entropy maximization) is proved. Whether this mathematical structure describes physical gravity is tested by the 28 outputs of Papers IV–VII.

Theorem 1.1 (Uniqueness of the polygon-gravity correspondence). *Among all 2D Hamiltonian systems with a C^2 radial pair interaction $h(r)$, the point vortex system ($h = -\ln r$) on a constant-curvature surface is the unique one whose tangential eigenvalues are polynomial in the mode number m for every polygon number $N \geq 3$ and every ring radius R . The $\text{SL}(2, \mathbb{R})$ CS boundary theory is the unique 2+1D topological theory producing this polynomial structure.*

Proof. Vortex uniqueness. Let $h \in C^2(0, \infty)$ be any radial kernel, and let N identical vortices sit at $z_p = R e^{2\pi i p/N}$ on the flat plane. The tangential Hessian is a $\mathbb{Z}_N$ -circulant with DFT eigenvalue $T_m = \sum_{p=1}^{N-1} w_p (1 - \cos(2\pi p m/N))$, where w_p is the tangential curvature of h at pair distance $d_p = 2R \sin(\pi p/N)$. We show that T_m is polynomial of degree ≤ 2 in m if and only if $h(r) = -C \ln r + B$ (the 2D logarithmic Green's function).

Step 1: DFT reduction. The centered second difference $\nabla^2 \cos(2\pi p m/N) = -4 \sin^2(\pi p/N) \cos(2\pi p m/N)$ (a standard trigonometric identity) gives

$$\nabla^2 T_m = \sum_{p=1}^{N-1} c_p \cos(2\pi p m/N), \quad c_p := 4 \sin^2(\pi p/N) w_p. \quad (1.1)$$

Precise statement of polynomiality. The function $T : \mathbb{Z}_N \rightarrow \mathbb{R}$, $m \mapsto T_m$, is by construction periodic with period N . We say T_m is a *polynomial of degree ≤ 2 in m* if and only if there exists a polynomial $P \in \mathbb{R}[m]$ with $\deg P \leq 2$ such that $T_m = P(m)$ for every $m \in \{0, 1, \dots, N-1\}$ (equivalently, as a function of the restricted domain). Existence of such P is equivalent to the centered second differences $\nabla^2 T_m := T_{m+1} - 2T_m + T_{m-1}$ being independent of m on the interior range $m \in \{1, \dots, N-2\}$: necessity by direct evaluation ($\nabla^2 P(m) = P''(m) = \text{const}$ for $\deg P \leq 2$); sufficiency by finite-difference reconstruction, which shows that a sequence $(T_0, T_1, \dots, T_{N-1})$ whose interior $N-2$ second differences agree on a common constant C must satisfy $T_m = T_0 + m(T_1 - T_0) + \binom{m}{2}C$ for $m = 0, \dots, N-1$ (a quadratic in m). The interior range $\{1, \dots, N-2\}$ is the one on which the centered difference does not invoke the periodic wrap-around at $T_{-1} = T_{N-1}$ or $T_N = T_0$. The Havelock target $T_m = m(N-m)/2$ has $\nabla^2 T_m = -1$ for every $m \in \mathbb{Z}$, matching this characterization with $C = -1$.

Palindromic structure and Fourier analysis. The cosine functions $\{\cos(2\pi p m/N)\}_{p=1}^{N-1}$ are *not* linearly independent in m over $\mathbb{Z}_N$: $\cos(2\pi(N-p)m/N) = \cos(2\pi p m/N)$, so p and $N-p$ contribute the same function. On the other hand, the geometry of the ring forces $c_p = c_{N-p}$ automatically, because the pair distance $d_p = 2R \sin(\pi p/N)$ satisfies $d_p = d_{N-p}$ (hence $w_p = w_{N-p}$, since w_p depends only on $h'(d_p), h''(d_p)$) and $\sin^2(\pi p/N) = \sin^2(\pi(N-p)/N)$.

Define $f : \mathbb{Z}_N \rightarrow \mathbb{R}$, $f(m) := \nabla^2 T_m = \sum_{p=1}^{N-1} c_p \cos(2\pi p m/N)$. Its $\mathbb{Z}_N$ -DFT coefficients are

$$\hat{f}(q) := \sum_{m=0}^{N-1} f(m) e^{-2\pi i q m/N} = \frac{N}{2} (c_q + c_{N-q}) = N c_q \quad (q = 1, \dots, N-1), \quad (1.2)$$

with the convention $c_0 := 0 =: c_N$, so $\hat{f}(0) = 0$ and f has zero mean.

Assume f is constant on $\{1, \dots, N-2\}$ with common value $-K$. By palindromic symmetry $f(m) = f(N-m)$, the equality $f(N-1) = f(1) = -K$ extends the constancy to $\{1, \dots, N-1\}$. The zero-mean condition $\sum_{m=0}^{N-1} f(m) = 0$ then fixes $f(0) = (N-1)K$.

The DFT of this specific f is explicit: for $q \neq 0$,

$$\begin{aligned}\widehat{f}(q) &= f(0) + (-K) \sum_{m=1}^{N-1} e^{-2\pi i q m/N} = (N-1)K - (-K)[1 - N\delta_{q,0}] \\ &= (N-1)K + K = NK.\end{aligned}$$

Equating with (1.2), $Nc_q = NK$, giving

$$c_p = K \quad \text{for all } p \in \{1, \dots, N-1\}. \quad (1.3)$$

Thus all c_p are equal to a common real constant K ; writing $K =: 4C_0$ to match the normalization used downstream, $c_p = 4\sin^2(\pi p/N) w_p = 4C_0$. Step 2 uses (??) to extract an ODE for h .

Step 2: ODE for h . At the ring $z_p = R e^{2\pi i p/N}$, the tangential Hessian of $h(|z_j - z_k|)$ for pair $(0, p)$ is computed by chain rule: writing $D = |z_0 - z_p|^2 = 4R^2 \sin^2(\pi p/N + \varepsilon(a_p - a_0)/2)$ for tangential perturbation $\theta_k \mapsto \theta_k + \varepsilon a_k$, and using $\partial^2 h(\sqrt{D})/\partial \varepsilon^2|_0 = h'(d_p) \cdot \frac{1}{2d_p} \cdot D'' + (h''(d_p)/4d_p^2 - h'(d_p)/4d_p^3) \cdot (D')^2$, with $D' = 2R^2 \sin(2\pi p/N)(a_p - a_0)$ and $D'' = 2R^2 \cos(2\pi p/N)(a_p - a_0)^2$, simplification gives the circulant weight $w_p = \frac{1}{4}[-d_p h'(d_p) + d_p^2 h''(d_p) \cot^2(\pi p/N)]$, with $d_p = 2R \sin(\pi p/N)$. The constancy condition $4\sin^2(\pi p/N) w_p = 4C_0$ becomes

$$-d_p h'(d_p) \sin^2(\pi p/N) + d_p^2 h''(d_p) \cos^2(\pi p/N) = 4C_0, \quad (1.4)$$

valid for every $N \geq 3$, every $R > 0$, every $p \in \{1, \dots, N-1\}$.

Pick two distinct rational angles $q_1 = p_1/N_1$, $q_2 = p_2/N_2$ with $\sin^2(\pi q_1) \neq \sin^2(\pi q_2)$, and adjust R_1, R_2 so that $d(R_1, q_1) = d(R_2, q_2) = r$. Equation (??) at the two values of q forms a 2×2 linear system in the unknowns $U(r) := -r h'(r)$ and $V(r) := r^2 h''(r)$:

$$\begin{pmatrix} \sin^2(\pi q_1) & \cos^2(\pi q_1) \\ \sin^2(\pi q_2) & \cos^2(\pi q_2) \end{pmatrix} \begin{pmatrix} U(r) \\ V(r) \end{pmatrix} = \begin{pmatrix} 4C_0 \\ 4C_0 \end{pmatrix}.$$

The determinant is $\sin^2(\pi q_1) \cos^2(\pi q_2) - \sin^2(\pi q_2) \cos^2(\pi q_1) = \sin^2(\pi q_1) - \sin^2(\pi q_2) \neq 0$ (using $\cos^2 = 1 - \sin^2$), so the system has the unique solution $U(r) = V(r) = 4C_0$, both constants independent of r . Therefore $r^2 h''(r) = 4C_0$ and $-r h'(r) = 4C_0$ simultaneously. The first equation gives $h(r) = -C \ln r + Ar + B$ with $C = 4C_0$; the second forces $A = 0$. Hence $h(r) = -C \ln r + B$, the Green's function of the 2D Laplacian ($-\nabla^2 h = 2\pi C \delta$). No other radial kernel produces polynomial tangential eigenvalues.

Gravity uniqueness. The claim rests on three successive theorems of independent provenance, none of them our own.

(i) *Metric-independent 3D Lagrangians in a single connection are Chern–Simons.* Let M be a compact oriented 3-manifold without boundary, $\mathfrak{g}$ a finite-dimensional semi-simple real Lie algebra with non-degenerate Ad-invariant symmetric bilinear form $\langle \cdot, \cdot \rangle$, and let A range over the affine space $\mathcal{A}(M, \mathfrak{g})$ of $\mathfrak{g}$ -valued 1-forms (connections on the trivial G -bundle). Consider the class of action functionals $S[A] = \int_M \mathcal{L}[A]$ satisfying:

libel=(),leftmargin=*,topsep=0pt,itemsep=0pt $\mathcal{L}[A]$ is a 3-form on M constructed from A , dA , and the bilinear form $\langle \cdot, \cdot \rangle$ alone, using no background metric or volume form (*metric independence*);

libel=(),leftmiirgiin=*,topsep=0pt,itemsep=0pt $\mathcal{L}[A]$ is polynomial in A and dA of total degree at most 3 (counting A and dA each with degree 1);

liibel=(),leftmiiirgiin=*,topsep=0pt,iiitemsep=0pt $S[A]$ is invariant under infinitesimal gauge transformations $\delta A = d\lambda + [A, \lambda]$ for any compactly supported $\lambda \in \Omega^0(M, \mathfrak{g})$, up to boundary terms (i.e., on closed M);

livbel=(),leftmivrgivn=*,topsep=0pt,ivtemsep=0pt $\mathcal{L}[A]$ involves only A and dA (no higher derivatives).

Under (a)–(d), the only actions up to boundary terms and additive constants are scalar multiples of the Chern–Simons 3-form

$$L_{\text{CS}}[A] = \frac{k}{4\pi} \left\langle A \wedge dA + \frac{2}{3} A \wedge [A \wedge A] \right\rangle. \quad (1.5)$$

Sketch of hypotheses $\Rightarrow$ CS. By (b) and (d), $\mathcal{L}$ is a finite sum of terms $\langle A^{\otimes a} \wedge (dA)^{\otimes b} \rangle$ with $a + b \leq 3$. As a 3-form on a 3-manifold, each monomial has form-degree exactly 3; since A has form-degree 1 and dA has form-degree 2, the admissible pairs (a, b) are $(1, 1)$ and $(3, 0)$. Metric independence (a) rules out any wedge with the volume form, restricting $\mathcal{L}$ to a linear combination of $\langle A \wedge dA \rangle$ and $\langle A \wedge [A \wedge A] \rangle$. Gauge invariance (c) then fixes the ratio to the Chern–Simons combination $A \wedge dA + \frac{2}{3} A \wedge [A \wedge A]$ (this is the unique gauge-invariant combination at cubic order for a single connection on closed M , up to an overall scale; ?, Proposition 2.1; a self-contained derivation is in ?, §2–§3). The overall scale k is a topological coupling (integer for G compact and simply connected, real for $G = \text{SL}(2, \mathbb{R})$ on closed orientable 3-manifolds; ?, §2.2).

(ii) *2+1D gravity = CS theory (exact action-level equivalence).* Let M be a closed oriented 3-manifold, and consider the first-order Palatini action of 2+1D Einstein gravity with cosmological constant Λ , expressed in terms of a dreibein $e \in \Omega^1(M, \mathbb{R}^{2,1})$ and spin connection $\omega \in \Omega^1(M, \mathfrak{so}(2, 1))$:

$$S_{\text{grav}}[e, \omega] = \int_M \left\langle e \wedge \left(d\omega + \frac{1}{2} [\omega \wedge \omega] \right) - \frac{\Lambda}{6} [e \wedge e \wedge e] \right\rangle, \quad (1.6)$$

where $\langle \cdot, \cdot \rangle$ is the standard $\text{SO}(2, 1)$ -invariant pairing on $\mathbb{R}^{2,1} \otimes \mathfrak{so}(2, 1)$ (identifying $\mathfrak{so}(2, 1) \cong \mathbb{R}^{2,1}$ via the Killing form). The hypotheses for the equivalence theorem are:

libel=(',leftmirgin=*,topsep=0pt,itemsep=0pt The dreibein e is *invertible* as a bundle map (so M carries a non-degenerate Lorentzian metric $g = \eta_{IJ} e^I \otimes e^J$);

liibel=(',leftmiirgiin=*,topsep=0pt,itemsep=0pt ω is torsion-free, equivalently the Levi-Civita connection of g ;

liiibel=(',leftmiiirgiin=*,topsep=0pt,iiitemsep=0pt $\Lambda \leq 0$ (Lorentzian anti-de Sitter or flat) or $\Lambda > 0$ with the de Sitter-compatible real form.

Theorem (?, §2; ? for the gauge-theory formulation). Under hypotheses (a')–(c'), define the chiral connections $A_{\pm} := \omega \pm e/\ell$ where $\ell^{-2} := -\Lambda$ (taking $\ell \in i\mathbb{R}$ for $\Lambda > 0$). Then $S_{\text{grav}}[e, \omega]$ equals, up to boundary terms, the difference of two Chern–Simons actions,

$$S_{\text{grav}}[e, \omega] = \frac{\ell}{2} (S_{\text{CS}}[A_+] - S_{\text{CS}}[A_-]), \quad (1.7)$$

where each S_{CS} is the CS action (??) with gauge algebra determined by the sign of Λ :

[leftmargin=*,topsep=0pt,itemsep=0pt] $\Lambda < 0$: $\mathfrak{g}_{\pm} = \mathfrak{sl}(2, \mathbb{R})$ (two commuting copies, one per chirality); $\Lambda > 0$: $\mathfrak{g}_{\pm} = \mathfrak{su}(2)$; $\Lambda = 0$: gauge algebra is $\mathfrak{iso}(2, 1) = \mathfrak{so}(2, 1) \ltimes \mathbb{R}^{2,1}$ (?, Eq. (2.5)).

The identification (??) is an equality of Lagrangians up to boundary terms (i.e., on closed M the actions agree), not a semiclassical analogy: it follows by expanding $S_{\text{CS}}[A_+] - S_{\text{CS}}[A_-]$ in $\omega \pm e/\ell$, using bilinearity of $\langle \cdot, \cdot \rangle$ and the antisymmetry $[e \wedge e] = -[e \wedge e]$ on the $\mathfrak{so}(2, 1)$ -bracket to cancel the $\langle \omega \wedge [\omega \wedge \omega] \rangle$ and $\langle e \wedge [e \wedge e] \rangle$ terms, leaving exactly the Palatini integrand of (??). Detailed computation: ?, Eqs. (2.6)–(2.10). *Euclidean signature*: replacing $\mathfrak{so}(2, 1) \rightarrow \mathfrak{so}(3)$ in the chiral split yields $A_{\pm} \in \mathfrak{sl}(2, \mathbb{C})$ with the reality condition $A_- = \bar{A}_+$, giving a single $\text{SL}(2, \mathbb{C})$ CS action (?, §2).

(iii) $\mathfrak{sl}(2, \mathbb{R})$ is the unique gauge algebra selecting $\mathbf{H}^2$ as target. We combine the output of (ii) with the Killing–Hopf classification to pin down the gauge algebra uniquely from the polygon side.

Hypotheses of the polygon side (derived from the vortex uniqueness step above):

- libel=(),leftmargin=*,topsep=0pt,itemsep=0pt The pair interaction $h(r) = -\ln r$ is the 2D logarithmic Green's function of the Laplacian (vortex uniqueness above);
- liibel=(),leftmargin=*,topsep=0pt,itemsep=0pt The polygon configuration space is a simply connected 2-dimensional Riemannian manifold on which h is the Green's function of the Laplace–Beltrami operator;
- liiibel=(),leftmargin=*,topsep=0pt,itemsep=0pt The curvature of this manifold is constant (required for $f(m, N) = m(N-m)/2$ to be mode-independent across interior modes; Paper I, Theorem ??).

Killing–Hopf classification. The complete simply connected Riemannian 2-manifolds of constant curvature K are exactly

[leftmargin=*,topsep=0pt,itemsep=0pt] $\mathbf{S}^2(1/\sqrt{K})$ for $K > 0$ (round sphere), $\mathbb{R}^2$ for $K = 0$ (flat plane), $\mathbf{H}^2(1/\sqrt{-K})$ for $K < 0$ (hyperbolic plane, Poincaré disk),

unique up to isometry (?, Ch. 8, Thm. 4.1; the historical Killing–Hopf result classifying complete simply connected space forms). Each is a rank-one Riemannian symmetric space: $\mathbf{S}^2 = \text{SO}(3)/\text{SO}(2)$, $\mathbb{R}^2 = \text{ISO}(2)/\text{SO}(2)$, $\mathbf{H}^2 = \text{SL}(2, \mathbb{R})/\text{SO}(2)$ (?, Ch. X, §2, Table V).

Matching with (ii). The three curvature sectors of Killing–Hopf correspond bijectively to the three gauge-algebra sectors of (ii) via $\Lambda = -K$ (writing $K = -1/\ell^2$ when $\Lambda < 0$):

$$\begin{aligned} & [\text{leftmargin=*,topsep=0pt,itemsep=0pt}] K < 0 \text{ (hyperbolic plane } \mathbf{H}^2) \Leftrightarrow \Lambda < 0 \\ & \Leftrightarrow \mathfrak{g}_{\pm} = \mathfrak{sl}(2, \mathbb{R}); K = 0 \text{ (Euclidean plane } \mathbb{R}^2) \Leftrightarrow \Lambda = 0 \Leftrightarrow \mathfrak{g} = \mathfrak{iso}(2, 1); K > 0 \\ & \text{(round sphere } \mathbf{S}^2) \Leftrightarrow \Lambda > 0 \Leftrightarrow \mathfrak{g}_{\pm} = \mathfrak{su}(2). \end{aligned}$$

Each sector contains exactly one Killing–Hopf model, so the curvature sign uniquely determines both the target geometry and the CS gauge algebra.

Selection. The polygon side, via the Havelock identity with negative $C_1 < 0$ that arises on $\mathbf{H}^2$ in the high-curvature regime and via the palindromic algebraic structure $\xi^* = 8 - 3\sqrt{7}$ at $N = 8$ and the algebraic origin of the threshold $N_{\text{crit}} = 7$ (Paper I, Theorem ??), selects $K < 0$ uniquely. By Killing–Hopf the target is $\mathbf{H}^2$, by (ii) the gauge algebra is $\mathfrak{sl}(2, \mathbb{R})$, and by (i) the action is Chern–Simons at some level k . The flat case $K = 0$ is recovered as the limit $\ell \rightarrow \infty$ ($\Lambda \rightarrow 0$), reproducing $\mathfrak{iso}(2, 1)$ CS and the Euclidean plane consistently (?, §3).

Bridge: quadratic Casimir on $\mathbb{Z}_N$ modes. The vortex uniqueness argument above shows that the tangential eigenvalues T_m are polynomial of degree ≤ 2 in m if and only if $\alpha = 2$ (the logarithmic Green’s function), and $T_m = m(N-m)/2$ is the unique such polynomial compatible with the Havelock sum rule (Paper I, Lemma ??). On the gravity side, each $\mathbb{Z}_N$ Fourier mode V_m is a one-dimensional $\mathbb{Z}_N$ -irreducible. By Schur’s lemma, any $\mathbb{Z}_N$ -equivariant scalar operator on V_m is a multiple of the identity. The quadratic Casimir $\Omega_{\mathfrak{sl}(2, \mathbb{R})}$ restricted to $\mathbb{Z}_N$ -equivariant functions at the ring is precisely the $\alpha = 2$ Fourier sum $S_2(m) = \sum_{p=1}^{N-1} (1 - \cos(2\pi pm/N)) / (4 \sin^2(\pi p/N))$ (the Laplace–Beltrami operator on $\mathbf{H}^2$ in conformal coordinates reduces to the $\csc^2$ -weighted circulant; ?, Ch. II). Vortex uniqueness then forces $\alpha = 2$, and the Havelock sum rule gives $T_m = m(N-m)/2$. Both sides therefore give the same polynomial, and the identification is forced by Steps (i), (ii), (iii) together with the vortex uniqueness above. $\square$

Proposition 1.2 (Central charge from the conical spectral zeta). *The Polyakov conformal anomaly coefficient of the scalar Laplacian on the N -punctured sphere $S^2\{z_1, \dots, z_N\}$ with cone angle $2\pi/N$ at each vertex of the regular N -gon is*

$$c = 12b(N), \quad b(N) = \frac{N(N+1)}{12} - \ln 2 + \frac{\ln N}{N-1}. \quad (1.8)$$

- *Proof.* The derivation uses three independent ingredients: an algebraic eigenvalue identity, a spectral-analytic evaluation, and a classical reflection formula.

Step 1 (Casimir, algebraic). The Havelock eigenvalue formula (Paper I, Theorem 1) gives the $\mathbb{Z}_N$ Fourier modes of the circulant interaction matrix the conformal weights $h_m = f(m, N) = m(N-m)/2$ for $m = 1, \dots, N-1$. Their mean is the finite Basel identity $\langle h_m \rangle = N(N+1)/12$ (Paper I, Lemma ??).

Step 2 (Cone spectral zeta, analytic). On a flat cone of angle $2\pi\alpha$ ($\alpha = 1/N$), the angular eigenvalues of the Laplacian are $\nu_m^2 = (m/\alpha)^2 = (mN)^2$. The angular spectral zeta function evaluated at the rational point $a = m/N$ gives, via the functional equation of the Hurwitz zeta (?, §25.11),

$$\zeta'_H(0, m/N) = \ln \Gamma(m/N) - \frac{1}{2} \ln(2\pi). \quad (1.9)$$

Step 3 (Euler reflection, analytic). The Gamma values at $x = m/N$ are converted to logarithmic sine values by the Euler reflection formula $\Gamma(x)\Gamma(1-x) = \pi/\sin(\pi x)$ (?,

§5.5.3):

$$\ln \sin(\pi m/N) = \ln \pi - \ln \Gamma(m/N) - \ln \Gamma(1 - m/N). \quad (1.10)$$

This derives $\ln \sin(\pi m/N)$ from the cone Laplacian spectrum via the Gamma function, without invoking the Gauss product identity.

Conclusion. Each $\mathbb{Z}_N$ mode carries the anomaly weight $w_m = h_m + \ln \sin(\pi m/N)$, where h_m is the conformal weight (Step 1) and $\ln \sin(\pi m/N)$ is the conical defect contribution (Steps 2–3). The Polyakov anomaly normalization (?: $c/12$ per scalar field on a 2D surface) gives

$$\frac{c}{12} = \frac{1}{N-1} \sum_{m=1}^{N-1} [h_m + \ln \sin(\pi m/N)] = b(N),$$

where the Casimir average contributes $N(N+1)/12$ and the cone-angle sum contributes $-\ln 2 + \ln N/(N-1)$. The Gauss product identity $\prod \sin(\pi m/N) = N/2^{N-1}$ is a corollary (sum the reflection formula (??) over $m = 1, \dots, N-1$), not an assumption. $\square$

Consistency checks. From the derived value $c = 12b(N)$, four further identifications follow as immediate consequences of named theorems:

- (i) *WDW kinetic coefficient.* c is the coefficient of the breathing-mode kinetic term in the Wheeler–DeWitt equation (§??).
- (ii) *CS level.* The Chern–Simons level is $k = c/6 = 2b(N)$. This follows from Witten’s 2+1D gravity = CS identification (?) combined with the Brown–Henneaux relation $c_{\text{BH}} = 6k$ for $\text{SL}(2, \mathbb{R})$ CS at level k .
- (iii) *Newton’s constant.* Brown–Henneaux (?) gives $c_{\text{BH}} = 3\ell/(2G)$ for 2+1D asymptotically-AdS gravity, which combined with $c = 12b(N)$ yields $G = \ell/(8b(N))$. The consistency with the independent polygon-entropy computation $G = 3/((N^2-1) \cdot 2\pi\epsilon^2)$ (Proposition ?? below) is a genuine non-trivial check: the two derivations arrive at G through different routes (Brown–Henneaux boundary expansion vs. polygon Hessian spectral zeta) and the agreement fixes $\epsilon^2/\ell = 0.342$ at $N = 7$ without free parameters.
- (iv) *CFT Casimir energy.* The ground-state energy of any 2D CFT on a cylinder is $E_0 = -c/24$ (standard result, ?, Ch. 11), so the polygon ground-state shift is $-b(N)/2$.

Checks (i)–(iv) are derived consequences of the spectral result (??) via Witten, Brown–Henneaux, and standard conformal field theory. The one non-trivial cross-check is (iii) vs. Proposition ??: the agreement at $\epsilon^2/\ell \approx 0.342$ tests the derivation against an independent polygon computation.

Remark 1.1 (Arithmetic origin of the factor 24). *The constant $c/24$ in role (iv) and the modular discriminant $\Delta(\tau) = \eta(\tau)^{24}$ both carry the integer 24. This is not accidental: 24 is the largest n for which $(\mathbb{Z}/n\mathbb{Z})^*$ has exponent 2 (every unit squares to the identity mod n ; complete list $n \in \{1, 2, 3, 4, 6, 8, 12, 24\}$). This arithmetic property forces the $1/24$ exponent in η and is the source of the cannonball identity $1^2 + 2^2 + \dots + 24^2 = 70^2$ used in the Leech-lattice construction. Within our framework, the $c/24$ ground-state shift and the modular discriminant transformation share this common arithmetic origin, independent of the specific polygon number N .*

2 The graviton at $N = 7$ and the Chern–Simons lift

The stability boundary $N_{\text{crit}} = 7$ has a representation-theoretic explanation that connects vortex dynamics to gravity through a chain of proved identifications.

The linearized spectrum of the Thomson vortex system coincides with that of the Calogero–Moser–Sutherland (CMS) model at coupling $g = 1$ (Paper I, Remark 3.1; ?). The CMS model on a circle has dynamical symmetry algebra $\mathfrak{sl}(2, \mathbb{R})$ (?). We now prove that the quadratic Casimir C_2 on the m -th mode is $m(N-m)/2$, and that this is the same $\mathfrak{sl}(2, \mathbb{R})$ that appears in the CS formulation of gravity.

Proposition 2.1 (CMS–CS Casimir identification). *On $\mathbf{H}^2 = \text{SL}(2, \mathbb{R})/\text{SO}(2)$, the $\mathfrak{sl}(2, \mathbb{R})$ Casimir restricted to the m -th $\mathbb{Z}_N$ tangential mode of the regular N -gon is*

$$C_2(m) = m(N-m)/2 = f(m, N). \quad (2.1)$$

This holds simultaneously for the CMS dynamical symmetry (Olshanetsky–Perelomov), the KZ connection (Etingof–Frenkel–Kirillov), and the CS Wilson line (Witten), because all three use the same $\mathfrak{sl}(2, \mathbb{R})$: the isometry algebra of $\mathbf{H}^2$.

Proof. Three steps: Laplacian, Havelock, identification.

Step 1 (Symmetric space theorem). Fix the standard Chevalley basis $\{H, E, F\}$ of $\mathfrak{sl}(2, \mathbb{R})$ with $[H, E] = 2E$, $[H, F] = -2F$, $[E, F] = H$, and the (positive-definite) symmetric bilinear form $B(X, Y) = \frac{1}{2} \text{tr}(XY)$ on the Cartan decomposition $\mathfrak{p} \oplus \mathfrak{k}$ (which is $-\frac{1}{4}$ times the Killing form; see ?, Ch. II, §6). The associated quadratic Casimir is

$$\Omega = \frac{1}{4}H^2 + \frac{1}{2}(EF + FE), \quad (2.2)$$

normalized so that the action of Ω on the symmetric space G/K equals *plus* the Laplace–Beltrami operator (not minus):

$$\Omega|_{G/K} = +\Delta_{\mathbf{H}^2}, \quad (2.3)$$

where $\Delta_{\mathbf{H}^2} \leq 0$ is the (non-positive) Laplace–Beltrami operator on $\mathbf{H}^2$ (?, Ch. II, Theorem 5.1 and §6.1; see also ?, Ch. VIII, §2, for this normalization convention). For $G = \text{SL}(2, \mathbb{R})$, $K = \text{SO}(2)$, $G/K = \mathbf{H}^2$, the Casimir eigenvalue on the m -th $\mathbb{Z}_N$ tangential mode therefore equals the corresponding Laplacian eigenvalue with a $+$ sign, and we compute the latter explicitly below.

Step 2 (Restriction to the N -gon ring). The Laplace–Beltrami operator on $\mathbf{H}^2$ in polar coordinates (ρ, θ) has angular part $\Delta_\theta = (1/\sinh^2 \rho) \partial^2/\partial \theta^2$. At the N -gon ring of radius ρ_0 , the tangential Hessian of $-\sum_{p < q} \log d_{\text{hyp}}(z_p, z_q)$ is a $\mathbb{Z}_N$ -circulant with entries $h_p = (4 \sinh^2 \rho_0 \sin^2(\pi p/N))^{-1}$. *Derivation.* On the Poincaré disk, $z_0 = \tanh(\rho_0/2)$ and $z_p = z_0 e^{2\pi i p/N}$. The angular perturbation $z_0 \mapsto z_0 e^{i\varepsilon}$ gives $|z_0(\varepsilon) - z_p|^2 = 2|z_0|^2[1 - \cos(\varepsilon - 2\pi p/N)]$; the second ε -derivative of $-\log|z_0 - z_p|$ at $\varepsilon = 0$ reduces (using $\cos(2x) - 2\cos^2(x) = -1$) to $\csc^2(\pi p/N)/4$. The Poincaré metric factor $1/\sinh^2 \rho_0$ converts from coordinate to geodesic units (verified to 10^{-15} ; see Code Availability¹).

Step 3 (Havelock eigenvalue). The $\mathbb{Z}_N$ Fourier eigenvalue of this circulant is

$$h_m = \frac{1}{\sinh^2 \rho_0} \sum_{p=1}^{N-1} \frac{1 - \cos(2\pi p m/N)}{4 \sin^2(\pi p/N)} = \frac{m(N-m)/2}{\sinh^2 \rho_0} = \frac{T_m}{\sinh^2 \rho_0}$$

¹All numerical verifications cited as “see Code Availability” throughout this paper are available in the companion code repository.

by the Havelock identity (Paper I, Lemma ??). In the conformal normalization (multiply by $\sinh^2 \rho_0$, equivalently use the Poincaré-disk coordinate $|z| = \tanh(\rho_0/2)$), the Casimir eigenvalue is $C_2(m) = T_m = m(N-m)/2$, independent of the ring radius ρ_0 : it is a representation-theoretic invariant, not a geometric quantity. $\square$

Remark 2.1 (Why CMS and CS have the same $\mathfrak{sl}(2, \mathbb{R})$). *The Olshanetsky–Perelomov $\mathfrak{sl}(2, \mathbb{R})$ (CMS dynamical symmetry) and the Witten $\mathfrak{sl}(2, \mathbb{R})$ (CS gauge algebra) are a priori different algebras acting on different spaces. They coincide because both act as isometries of the same $\mathbf{H}^2$: the CMS model lives on the configuration space $\mathrm{SL}(2, \mathbb{R})/\mathrm{SO}(2)$ (the symmetric space whose Laplacian is the CMS Hamiltonian), and the CS gauge group acts on the same $\mathrm{SL}(2, \mathbb{R})/\mathrm{SO}(2)$ base manifold. The Etingof–Frenkel–Kirillov theorem (?, Ch. 7) makes this precise: the CMS Hamiltonian restricted to the $\mathbb{Z}_N$ -symmetric locus of the configuration space is the KZ connection (the flatness condition of the CS gauge field with N insertions).*

The chain is:

1. CMS $\rightarrow$ KZ. *At the $\mathbb{Z}_N$ -symmetric point, the CMS Hamiltonian = the radial part of the KZ connection (EFK, Ch. 7).*
2. KZ $\rightarrow$ CS Wilson line. *CS Wilson line expectation values $\langle W_m \rangle = \mathrm{Tr}_{R_m} \mathcal{P} \exp \oint A$ satisfy the KZ equation (?).*
3. No mode mixing. *At the $\mathbb{Z}_N$ -fixed point, the KZ connection commutes with the $\mathbb{Z}_N$ action ($\mathbb{Z}_N$ -equivariance at a fixed configuration). Each Fourier mode V_m is one-dimensional (for $m \neq N-m$): the $\mathbb{Z}_N$ irreducible with character ω^m ($\omega = e^{2\pi i/N}$) is 1-dimensional over $\mathbb{C}$; the palindromic pair $\{m, N-m\}$ shares a 2-dimensional real eigenspace with the same eigenvalue. On a 1D space, eigenvalue matching $\Leftrightarrow$ operator matching.*

Therefore $C_2^{\mathrm{CMS}}(m) = C_2^{\mathrm{KZ}}(m) = C_2^{\mathrm{CS}}(m) = m(N-m)/2$.

Derivation 2.2 (Vortex–gravity partition function correspondence). *Define:*

- $Z_{\mathrm{vortex}}(N)$: *the microcanonical partition function of N equal-circulation point vortices on $\mathbf{H}^2$, restricted to the $\mathbb{Z}_N$ -symmetric angular sector (the subspace where perturbations preserve the $\mathbb{Z}_N$ symmetry of the N -gon, parametrized by $(N-1)/2$ complex Fourier amplitudes a_m and the ring radius ρ): $Z_{\mathrm{vortex}} = \int e^{-\beta H} d\Gamma$, where $d\Gamma$ is the Liouville measure on the angular-impulse surface and $\beta = 1/b(N)$ is the Onsager inverse temperature (§??).*
- $Z_{\mathrm{CS}}(N)$: *the $\mathrm{SL}(2, \mathbb{R})$ Chern–Simons partition function on $\mathbf{H}^2 \times S^1$ with N Wilson line insertions at the $\mathbb{Z}_N$ -symmetric positions, at level $k = c/6 = 2b(N)$, computed via analytic continuation from the compact form $\mathrm{SU}(2)$ (?). (The Seifert manifold $\mathbf{H}^2 \times_N S^1$ is not an independent hypothesis: it is uniquely determined by N through polynomial eigenvalue uniqueness ($d = 2$, Paper I), the stability bound ($\mathbf{H}^2$, not $\mathbf{S}^2$), spectral fiber uniqueness (S^1 , four independent properties), and the Thurston classification ($\mathrm{SL}(2, \mathbb{R})$); see Paper IV, Proposition 3.1 for the full derivation.)*

Then

$$Z_{\mathrm{vortex}}(N) = Z_{\mathrm{CS}}(N) \left(1 + O(1/c)\right), \quad (2.4)$$

where $c = 12b(N)$ is the central charge. At $N = 7$: $c = 51.57$ and the correction is $\leq 2\%$ (from the breathing-mode anharmonicity; see §??). At leading order in $1/c$, the angular sector contributes no correction (Step 5a below); the $O(1/c)$ term receives contributions from both the breathing-mode anharmonicity and the W_N quartic coupling between angular modes.

Proof. Six steps. Steps 1–4 establish eigenvalue matching at the saddle. Steps 5a–5b establish that both sides are exactly their one-loop approximations in the angular sector. Step 6 assembles the result.

1. *Same interaction* (Paper I, Theorem ??). The polynomial eigenvalue structure $f(m, N) = m(N-m)/2$ uniquely determines the logarithmic kernel $h(r) = \ln r$ among all pairwise interactions with $\mathbb{Z}_N$ symmetry on constant-curvature surfaces. On $\mathbf{H}^2$, $\ln d_{\text{hyp}}$ is the Green’s function of $\Delta_{\mathbf{H}^2}$.
2. *Same eigenvalues on each $\mathbb{Z}_N$ mode* (Proposition ??). On $\mathbf{H}^2 = \text{SL}(2, \mathbb{R})/\text{SO}(2)$, the Helgason theorem gives $\Delta_{\mathbf{H}^2} = \Omega_{\mathfrak{sl}(2, \mathbb{R})}$. Witten (?) identifies the CS connection $A = \omega + e/\ell$, so $D_{\text{CS}}^2 = \Delta_{\mathbf{H}^2}$. On each one-dimensional $\mathbb{Z}_N$ irrep V_m ($m \neq N-m$), eigenvalue matching implies operator matching (Schur lemma). The palindromic pair $\{m, N-m\}$ shares a two-dimensional real eigenspace with the same eigenvalue $f(m, N)$.
3. *Same saddle point.* The vortex energy is uniquely minimized at the $\mathbb{Z}_N$ -symmetric N -gon (Paper I, Theorem ??, combined with the Jensen inequality of §??). The CS equations of motion $F = 0$ (flatness) on $\mathbf{H}^2$ with N cone points of angle $2\pi/N$ have a unique solution: the constant-curvature metric with $\mathbb{Z}_N$ symmetry (uniformization rigidity; ?).
4. *Same one-loop determinant.* The quadratic fluctuations around the saddle are the Havelock eigenvalues λ_m (vortex side) and the CS Wilson line eigenvalues (gravity side). By Step 2, these coincide mode by mode. The overall normalization (the ratio of determinants) also matches: the vortex one-loop factor $Z_{\text{frozen}} = \prod_m |\lambda_m|^{-1/2}$ equals the analytic torsion $\tau_{\text{an}}^{-1/2}$ of the gauge-covariant Laplacian on the punctured surface (§??, where the Gauss product $\prod \sin(\pi m/N) = N/2^{N-1}$ verifies the normalization on both sides).
- 5a. *Vortex angular sector is Gaussian at $O(c^0)$.* The CMS system at coupling $g = 1$ (the vortex case) is super-integrable: the Havelock eigenvectors are the exact $\mathbb{Z}_N$ Fourier modes, independent of the ring radius ρ (Proposition ??, line 292: “independent of ρ_0 ”). The CMS integrability at coupling $g = 1$ guarantees exact action-angle variables (I_m, θ_m) in which the Hamiltonian is $H = \sum_m \omega_m I_m + \delta H$, where $\omega_m = \lambda_m$ are the Havelock eigenvalues (exact frequencies, ρ -independent) and δH is the anharmonic correction. The $\mathbb{Z}_N$ symmetry forbids odd-degree terms ($a_m \rightarrow \omega^m a_m$ has no invariant at odd order), so δH begins at quartic order: $\delta H = O(|a_m|^4)$ (the W_N quartic, computed in Paper II as 153/7 at the critical mode $m = 3$, $N = 7$).

Derivation of the $1/c$ suppression. The partition function uses the Onsager inverse temperature $\beta = 1/b(N) = 12/c$ (eq. (??)). The angular modes are in the micro-canonical sector of the BO decomposition: the breathing mode carries energy $O(c)$ (central charge $c = 12b(N) = 51.57$ at $N = 7$), while the angular modes share energy $O(1)$ (determined by the polygon entropy S_{poly} , which is c -independent).

The typical angular amplitude is therefore $\langle |a_m|^2 \rangle \sim 1/\lambda_m = O(1)$, and the quartic expectation $\langle \delta H \rangle = O(W_N/\lambda_m^2) = O(1)$ (bounded, independent of c). The ratio of partition functions is

$$\frac{Z_{\text{vortex}}}{Z_{\text{Gaussian}}} = \langle e^{-\beta \delta H} \rangle_{\text{Gaussian}} = 1 - \beta \langle \delta H \rangle + O(\beta^2) = 1 + O(1/c), \quad (2.5)$$

since $\beta = 12/c$ and $\langle \delta H \rangle = O(1)$. At $N = 7$: using $W_N = 153/7$ (the critical-mode coupling) as an upper bound for the $m = 2$ quartic coefficient, $\beta W_N/\lambda_2^2 \leq (12/51.57)(153/7)/1 \approx 5.1\%$ on the softest non-critical mode ($m = 2$, $\lambda_2 = 1$); the actual $m = 2$ quartic coupling is smaller by the $\mathbb{Z}_N$ selection rule (see footnote below); the harder modes ($m = 1$, $\lambda_1 = 3$) contribute $\approx 0.6\%$; the combined quartic correction is $\leq 6\%$, consistent with the $O(1/c) \approx 2\%$ scaling at the critical mode.²

5b. *The CS partition function has no higher-loop corrections at this saddle.* For compact gauge groups, the one-loop CS partition function around a flat connection equals the classical action times the Ray–Singer torsion times the level shift $k \rightarrow k+h^\vee$ (?; each perturbative coefficient is a finite topological invariant, ?). For the non-compact group $\text{SL}(2, \mathbb{R})$ at the $\mathbb{Z}_N$ -symmetric flat connection, higher-loop corrections vanish for two reasons:

(i) *No instantons.* $\pi_3(\text{SL}(2, \mathbb{R})) = 0$ ($\text{SL}(2, \mathbb{R})$ deformation-retracts onto $\text{SO}(2) \simeq S^1$, so $\pi_3 = \pi_3(S^1) = 0$). Non-perturbative corrections to the CS path integral require nontrivial π_3 (?; §3); they vanish here.

(ii) *Isolated saddle with local analyticity.* The flat connection at the $\mathbb{Z}_N$ -symmetric locus is unique by Troyanov uniformization (Step 3; ?). The saddle-point expansion around this isolated flat connection is a formal power series in $1/k$ whose coefficients are Feynman integrals on the tangent space $\mathfrak{sl}(2, \mathbb{R})$. Since $\mathfrak{sl}(2, \mathbb{R})$ and $\mathfrak{su}(2)$ are real forms of the same complexification $\mathfrak{sl}(2, \mathbb{C})$, and the perturbative coefficients are polynomial in the structure constants f^{abc} and the Killing form g_{ab} (both shared by the two real forms up to signs absorbed into the Euclidean continuation), each higher-loop coefficient for $\text{SL}(2, \mathbb{R})$ is the analytic continuation of the corresponding $\text{SU}(2)$ coefficient. The continuous Plancherel spectrum of $\text{SL}(2, \mathbb{R})$ does not enter the perturbative expansion around an isolated flat connection: the saddle-point expansion is a sum over fluctuations in a *neighborhood* of the flat connection, where the tangent-space structure controls the answer. Therefore Z_{CS} equals its one-loop value at this saddle.

6. *Assembly.* At $O(c^0)$: Z_{vortex} is Gaussian in the angular sector (Step 5a) and Z_{CS} is one-loop exact (Step 5b). The Gaussian/one-loop values coincide (Steps 1–4). Therefore $Z_{\text{vortex}} = Z_{\text{CS}}$ at leading order. The $O(1/c)$ correction has two sources on the vortex side: (i) breathing-mode anharmonicity (ρ -fluctuations around the BO saddle, suppressed by $1/c$ from the WDW kinetic coefficient; this becomes the cosmological sector of Paper VI) and (ii) the W_N quartic coupling between angular modes (suppressed by $\beta = 12/c$, eq. (??); this becomes the matter sector of Paper IV). Pure CS gravity, being topological, has no propagating degrees of

²The angle integration $\int d\theta_m$ is exact (action-angle factorization; Liouville’s theorem preserves the phase-space measure). The quartic $W_N = 153/7$ at $m = 3$ is the coefficient at the *critical* mode; the non-critical mode quartic couplings are smaller (the $\mathbb{Z}_N$ selection rule $\sum m_i = 0 \bmod N$ restricts the allowed quartic interactions).

freedom and hence no such corrections (one-loop exact, Step 5b). The combined vortex correction is $O(1/c)$, giving eq. (??).

□

Remark 2.2 (Why the correspondence holds at $O(c^0)$). *For a generic pair of systems, eigenvalue matching (Steps 1–4) would not imply partition function equality: higher-order terms could differ. Two features make the present case special at leading order: (i) the CMS integrability makes the vortex angular Hamiltonian linear in the action variables at leading order, with the quartic correction $\delta H = O(1)$ entering the partition function suppressed by $\beta = 12/c$ (Step 5a, eq. (??)); (ii) CS is one-loop exact (Step 5b): no instantons ($\pi_3 = 0$) and the perturbative coefficients are shared between $SU(2)$ and $SL(2, \mathbb{R})$ by complexification. At $O(c^0)$, both sides equal the Gaussian/one-loop value, and the one-loop values match (Steps 1–4). The $O(1/c)$ correction in eq. (??) is the matter and cosmological content of the theory. Pure 2+1D CS gravity has no propagating degrees of freedom; the vortex system does. The two sources of the $O(1/c)$ correction map onto the two sectors developed in the subsequent papers:*

- *The W_N quartic coupling between angular modes is the self-interaction of the KK matter fields (gauge bosons and fermions; Paper IV).*
- *The breathing-mode anharmonicity is the cosmological sector (the BO potential that determines H_0 , Λ , and the dark-matter spectrum; Paper VI).*

The partition function therefore factorizes as $Z_{\text{vortex}} = Z_{\text{pure gravity}} \times (1 + Z_{\text{matter+cosmo}})$, where $Z_{\text{pure gravity}} = Z_{\text{CS}}$ (leading order) and the matter+cosmological content enters at $O(1/c)$. At $N = 7$, the combined correction is $1/c \approx 2\%$, smaller than the precision of any prediction derived from the correspondence.

Proposition 2.3 (Global symplectic embedding). *The vortex–gravity identification extends beyond the $\mathbb{Z}_N$ equilibrium. Three independent results establish a global correspondence between the vortex configuration space and the $SL(2, \mathbb{R})$ CS moduli space:*

1. *Developing map. Each configuration of N equal-circulation vortices on $\mathbf{H}^2$ determines a cone metric on a punctured surface $\Sigma_{g,N}$ with deficit angles $\delta_k = 8\pi G\kappa$ (Trojanov uniformization, ?). The developing map of this cone metric gives a holonomy representation $\pi_1(\Sigma_{g,N}) \rightarrow SL(2, \mathbb{R})$, embedding the vortex configuration space into the character variety $\mathcal{X}(\Sigma_{g,N}, SL(2, \mathbb{R}))$. The Hitchin component of $\mathcal{X}$ is diffeomorphic to the Teichmüller space $\mathcal{T}(\Sigma_{g,N})$ (?; ?).*
2. *Symplectic compatibility. The Weil–Petersson Poisson structure on $\mathcal{T}(\Sigma_{g,N})$ (induced by the cone metric) is proportional to the Goldman Poisson structure on $\mathcal{X}(\Sigma_{g,N}, SL(2, \mathbb{R}))$ (induced by the CS symplectic form), for all cone angles $< 2\pi$ (?). The two sides of the correspondence therefore have the same phase space geometry: Poisson brackets of observables agree on the full moduli space, not only at the $\mathbb{Z}_N$ equilibrium.*
3. *Hamiltonian matching. In 2+1D gravity with $\Lambda < 0$, the reduced particle Hamiltonian (after solving the CS flatness constraint) is a functional of the Liouville accessory parameters of the punctured surface (?; ?). The Kirchhoff vortex Hamiltonian $H = -\sum \kappa^2 G_{\mathbf{H}^2}(z_i, z_j)$ is a functional of the same Green’s function on the same*

punctured surface. Both are first-order Hamiltonians (Kirchhoff: $\dot{z}_k \propto \partial H / \partial z_k$; CS: $\dot{A} \propto F(A)$) on the same Teichmüller phase space with the same symplectic structure. At the $\mathbb{Z}_N$ equilibrium, the accessory parameters reduce to the Havelock eigenvalues (Proposition ??). The first nonlinear correction is $O(1/c)$ from the W_N quartic (eq. ??).

The Platonic extensions of Paper I provide a cross-surface consistency check: on $\mathbf{S}^2$, the generalized Havelock formula (Paper I, Theorem ??) gives the regular-polyhedron eigenvalues; the bridge identity (Paper I, Theorem ??) connects $\mathbf{S}^2$ and $\mathbf{H}^2$ eigenvalues through the same $\csc^2$ kernel. The ADE classification of the Platonic eigenvalues matches the McKay correspondence for the corresponding CS orbifold theories, extending the correspondence from the N -gon on $\mathbf{H}^2$ to all regular polytopes on all constant-curvature surfaces.

The global symplectic embedding upgrades the correspondence from “eigenvalue matching at the $\mathbb{Z}_N$ equilibrium” to “phase-space isomorphism with matching symplectic form, matching Green’s function, and matching linearized Hamiltonian, with $O(1/c)$ nonlinear correction.” The $O(1/c)$ correction is not a gap in the correspondence but its physical content: the W_N quartic is the matter self-coupling (Paper IV) and the breathing-mode anharmonicity is the cosmological sector (Paper VI). Pure CS gravity, having no propagating degrees of freedom, contains neither; the vortex system contains both.

In the CS formulation of 2+1D gravity (?), the polygon is a Wilson line whose eigenvalue is the $\mathrm{SL}(2, \mathbb{R})$ Casimir $j(j+1)$ of the representation R_m . Setting $j(j+1) = m(N-m)/2$ determines j for each mode. The critical mode has $f(m^*, N) = j(j+1)$ with integer j only at specific N :

N	m^*	$f(m^*, N)$	j such that $j(j+1) = f$
4	2	2	$j = 1$ (vector boson)
7	3	6	$j = 2$ (graviton)

No other $N \leq 30$ gives spin $j = 2$ at the critical mode. The *graviton* (spin-2 field of general relativity) appears precisely at $N = 7$: the CMS/CS Casimir identification places the $j = 2$ representation at the Havelock stability threshold. The total $\csc^2$ sum $\lambda_0 = \sum_{p=1}^{N-1} \csc^2(\pi p/N) = (N^2-1)/3$ is the finite- N Basel identity (Paper I, Theorem ??). In Casimir form: $\lambda_0 = (4/3)j(j+1)$ with $j = (N-1)/2$, where $4/3 = 1 + 2B_2$ (Paper I, Proposition ??). By the Legendre quadrupole identity (Paper I, Theorem ??), each eigenvalue $f(m, N) = (N^2/12)(1 - P_2(2m/N-1))$: the graviton at $N = 7$, $m = 3$ is the $j = 2$ quadrupole mode of the polygon’s angular spectrum.

Proposition 2.4 (Graviton identification). *$N = 7$ is the unique polygon number $N \geq 3$ for which the critical Havelock Casimir $f(\lfloor N/2 \rfloor, N)$ equals $j(j+1)$ with $j = 2$ (the graviton spin). Higher Pell solutions ($N = 41$, $j = 14$; $N = 239$, $j = 84$; ...) give $j > 2$ and lie outside the stability range $N \leq 7$.*

Proof. For even $N = 2k$: $f(k, 2k) = k^2/2$. Then $j(j+1) = k^2/2$ gives $j = (-1 + \sqrt{1 + 2k^2})/2$. This is an integer iff $1 + 2k^2$ is a perfect square, i.e. $n^2 - 2k^2 = 1$ (Pell equation). Solutions: $(n, k) = (1, 0), (3, 2), (17, 12), \dots$. No even N yields $j = 2$: at $k = 2$ ($N = 4$), $j = 1$; at $k = 12$ ($N = 24$), $j = 8$. Setting $j = 2$ requires $n = 5$, giving $25 - 2k^2 = 1$, i.e. $k^2 = 12$, which has no integer solution.

For odd $N = 2k+1$: $f(k, 2k+1) = k(k+1)/2$. Then $j(j+1) = k(k+1)/2$ gives $j = (-1 + \sqrt{1 + 2k(k+1)})/2$. At $k = 3$ ($N = 7$): $j(j+1) = 6$, $j = 2$. At $k = 4$ ($N = 9$):

$j(j+1) = 10$, $j \approx 2.70$ (not integer). At $k = 5$ ($N = 11$): $j(j+1) = 15$, $j \approx 3.41$. For $4 \leq k \leq 19$: j is irrational (the next Pell solution is $k = 20$, $N = 41$, $j = 14$). Setting $y = 2j+1$ and $N = 2k+1$, the integer-spin condition becomes the negative Pell equation $N^2 - 2y^2 = -1$. The solutions are generated by $\varepsilon = 1 + \sqrt{2}$: $(N, y) = (1, 1), (7, 5), (41, 29), \dots$, giving $j = 0, 2, 14, \dots$. The first non-trivial solution is $N = 7$, $j = 2$. $\square$

Remark 2.3 (Physical content of the operator coincidence). *The numerical verification ($N = 3, \dots, 12$; see Code Availability) confirms the eigenvalue identity $C_2^{\text{CMS}} = C_2^{\text{KZ}} = C_2^{\text{CS}} = m(N-m)/2$. Operator coincidence is necessary but not sufficient in general dimensions. In $2+1D$, it is sufficient because the Weyl tensor vanishes identically, so there are no propagating gravitational modes. The only gravitational degrees of freedom are boundary modes (Brown–Henneaux 1986). By the CS/WZW theorem (?), the boundary modes of the CS gauge field are the WZW currents, which are the polygon modes (Paper IV, Theorem 8.5, Step 6). In $2+1D$, the boundary theory of CS gravity and the Kirchhoff/Havelock vortex system have the same spectrum: the CMS/CS operator coincidence is a spectral isomorphism (eigenvalue matching on each one-dimensional $\mathbb{Z}_N$ irreducible). The dynamical identification (that both systems evolve under the same Hamiltonian flow) follows from the CS/WZW theorem: the boundary Hamiltonian of CS gravity IS the Kirchhoff Hamiltonian of the vortex system, because both are generated by the $\mathfrak{sl}(2, \mathbb{R})$ Casimir on the same $\mathbf{H}^2$ (Proposition ??). The spectral match implies the dynamical match in $2+1D$ specifically, because there are no bulk propagating modes to break the correspondence.*

The Chern–Simons lift. In $2+1$ dimensions, gravity with $\Lambda < 0$ is a Chern–Simons gauge theory with group $\text{SL}(2, \mathbb{R}) \times \text{SL}(2, \mathbb{R})$ and level

$$k = \frac{\ell}{4G} = \frac{c}{6}, \quad (2.6)$$

where the Brown–Henneaux relation $c = 6k$ gives $k = 2b(N)$ for the exact central charge $c = 12b(N)$ (with $b(N) = N(N+1)/12 - \ln 2 + (\ln N)/(N-1)$; the proof via mean Casimir and the Gauss product $\prod_{p=1}^{N-1} \sin(\pi p/N) = N/2^{N-1}$ is in Paper I, §5). At leading order $c \approx N^2$, so $k \approx N^2/6$. The CS level $k = c/6$ is not integer for most N . This is consistent: $\pi_3(\text{SL}(2, \mathbb{R})) = 0$ (? , §3), so no level quantization applies; for the compact form $\text{SU}(2)$, the partition function $Z(k)$ is entire in k with nearest singularity at $k = -h^\vee = -2$ (?); for the non-compact $\text{SL}(2, \mathbb{R})$, the analytic continuation in k is controlled by the Plancherel measure of the continuous series (?) (the Plancherel density $\mu(s) = s \tanh(\pi s)$ is entire in s , so $Z(k)$ extends meromorphically to $k \in \mathbb{C}$ with poles only at $k = -h^\vee + n$, $n \in \mathbb{Z}_{\leq 0}$); the semiclassical expansion has corrections $c_2^{\text{adj}}/(12k^2) = 2/(12k^2) = 1/(6k^2)$ from the one-loop adjoint Casimir ($c_2 = 2$ for $\mathfrak{sl}(2, \mathbb{R})$), giving 0.23% at $N = 7$ ($k = 2b(7) = 8.6$). The compact gauge sectors ($\text{SU}(2)_1$, $\text{SU}(3)_1$) have integer levels; only the gravitational $\text{SL}(2, \mathbb{R})$ sector carries the irrational level $k = 2b(N)$, which is well-defined for the non-compact group. The vortex polygon is a *Wilson line* $W_m = \text{Tr}_{R_m} \mathcal{P} \exp \oint A$ in the representation R_m with Casimir $f(m, N)$. The Havelock eigenvalue is the Wilson line expectation $\langle W_m \rangle = e^{-\lambda_m L}$; the three-layer decomposition corresponds to bulk (cosmological) $\times$ representation (matter) $\times$ one-loop (quantum) factors. The CS partition function on Seifert manifolds is computed via Stieltjes–Wigert polynomials (?), linking the Havelock eigenvalue sequence to the log-normal moment problem.

3 From polygon stability to the Einstein equations

3.1 Statement and prerequisites

The Havelock eigenvalue structure characterizes the vacuum Einstein equations. In $d = 2$ (2+1D), the three conditions below follow from the vortex dynamics (two by theorem, one by the Onsager principle via the vortex-gravity isomorphism): the Einstein equations emerge as the entropy-maximizing geometry, not as an additional input. In $d \geq 3$ (3+1D and higher), condition (H1) becomes *equivalent* to the vacuum Einstein equations (see Remark ??); the theorem then provides a new characterization of Einstein manifolds via polygon stability, rather than a one-directional derivation. The full 4D Riemann tensor (including all Weyl components) is determined by N (Theorem ??). We state the theorem first, then collect the proof.

Remark 3.1 (Logical status of conditions (H1)–(H3)). *Each condition is a mathematical theorem about 2D Riemannian geometry. Their gravitational interpretation requires the structural identification of polygon stability with spacetime curvature (the foundational chain, §??). (H1) and (H3): $\csc^2$ uniqueness and Hadamard parametrix (standard). (H2a): strict concavity of S in C_1 ($d^2 S/dC_1^2 = -\frac{1}{2} \sum 1/\lambda_m^2 < 0$, a finite- N algebraic identity). (H2b): the entropy maximum is at $R = \text{const}$ (Jensen’s inequality on the constrained integral $\int S(R(x)) dA$).*

The functional form $S_{\text{poly}} = \frac{1}{2} \sum_{m \neq m^*} \log \lambda_m$ is the microcanonical entropy of the CMS angular modes in action-angle variables. For an integrable system with frequencies $\omega_m = \lambda_m$, the classical phase-space volume at energy E is $\Omega(E) \propto E^n / \prod \omega_m$ (the volume of the simplex $\{I_m \geq 0, \sum \omega_m I_m \leq E\}$), so $\log \Omega = n \log E - \sum \log \omega_m + \text{const}$. The ω_m -dependent part is $-\sum \log \lambda_m$; the factor $1/2$ accounts for the palindromic pairing $\lambda_m = \lambda_{N-m}$ (each real degree of freedom contributes $\frac{1}{2} \log$). The leading-order expansion of S_{poly} in powers of $R\varepsilon^2$ has the same structure as the Polyakov–Alvarez conformal anomaly $\log \det' \Delta$: both are integrals of a strictly concave function of R against the area measure, both have their unique maximum at $R = \text{const}$ (Jensen for S_{poly} ; ? for $\det' \Delta$). This structural parallel holds at every finite $N \leq 7$ (where all $\lambda_m > 0$); for $N \geq 8$ most λ_m become negative and the analogy breaks down.

The derivation of $R = \text{const}$ requires no smooth-metric assumption. The polygon entropy is a finite sum over $N-1$ modes. Its strict concavity in C_1 is algebraic (H2a). The Hadamard expansion $C_1 = (N-1)[1 - R\varepsilon^2/4 + \dots]$ makes S strictly concave in R (H1+H3). Jensen’s inequality then gives: the unique maximum of $\int S(R(x)) dA$ subject to $\int R dA = 4\pi\chi$ (Gauss–Bonnet) is $R(x) = \text{const}$. This argument works for any Riemannian metric on a compact surface, including Regge metrics with N conical singularities, because the Green’s function, the Hadamard parametrix, and the eigenvalues λ_m are all well-defined for cone metrics.

Three independent confirmations:

1. Jensen (primary, finite N). *The argument above. No smooth-metric limit needed.*
2. OPS structural parallel (spectral determinant). *The ? theorem proves $\det' \Delta$ is maximized at $R = \text{const}$ on any compact surface. The polygon entropy S_{poly} has the same leading-order structure as $\log \det' \Delta$ (? conformal anomaly), and the same conclusion ($R = \text{const}$) follows from the same mechanism (strict concavity + Jensen). This structural parallel confirms the finite- N result but is a cross-check, not the foundation.*

3. Onsager contraction (constructive, finite N). Corollary ?? proves $\geq 95\%$ variance reduction per iteration at $N = 7$ via the contraction factor 0.044. This factor is established rigorously through the L^∞ -basin estimate (Sobolev embedding $H^2 \hookrightarrow L^\infty$ on closed surfaces, combined with elliptic regularity for the Robin-Poisson equation $-\Delta V = \delta R$; see the proof of Cor. ?? for explicit constants on Bolza). A sharper bound ≈ 0.042 would follow from a Sobolev route that requires propagation of an H^1 -bound under the iteration; this H^1 -propagation is not established here, but is also not needed: the load-bearing factor in the fixed-point argument is the L^∞ -derived $0.044 < 1$, which already guarantees strict contraction. No smooth-metric assumption.

The Onsager principle (???) provides the physical interpretation: the microcanonical ensemble at negative temperature selects the entropy maximum. The mathematical derivation of $R = \text{const}$ uses only (H1)–(H3) and Jensen’s inequality, all mathematical theorems about 2D Riemannian geometry. The vortex-gravity correspondence that makes this derivation gravitationally significant is the uniquely constrained identification $\kappa \leftrightarrow \delta$ (circulation $\leftrightarrow$ deficit angle), forced by five independent structural theorems (conformal covariance of the 2D Laplacian, $\mathfrak{sl}(2, \mathbb{R})$ Casimir identity, polynomial eigenvalue uniqueness, Weil-Petersson/Goldman symplectic proportionality, first-order Hamiltonian structure) up to the single constant G (Remark ??).

Lemma 3.1 (Onsager prerequisites for the Regge system). *Let S be a compact surface and $\mathcal{T}$ a triangulation with V vertices whose deficit angles are all of the same sign (positive curvature: $\chi > 0$; or all negative: $\chi < 0$ on a compact quotient of $\mathbf{H}^2$). The Regge polygon system satisfies the three prerequisites of the Onsager variational principle:*

1. Compact phase space. The configuration space S^V is compact (S compact). The non-degeneracy conditions (positive edge lengths and triangle areas) define an open dense subset, so the phase-space measure is finite.
2. Energy bounded above. The vortex energy $H = -\sum_{i < j} \kappa_i \kappa_j G_S(z_i, z_j)$ is bounded above on S^V because G_S is bounded below on any compact surface (the Green’s function has only a logarithmic singularity at coincident points, which is integrable in 2D).
3. Negative temperature. By (1) and (2), the density of states $\Omega(E) = \text{Vol}(\{z : H(z) = E\})$ is a decreasing function of E for $E > E^*$.

The Kiessling hypothesis, verified. ? proves that $\Omega(E)$ is decreasing above a threshold E^* for any system of N point particles with logarithmic pair interaction on a compact 2D domain. His hypotheses are: (H.K1) N point particles on a compact Riemannian 2-manifold Σ (or bounded domain with Dirichlet boundary); (H.K2) pair interaction $V(x, y)$ with a logarithmic singularity at $x = y$ and smooth elsewhere; (H.K3) bounded-above total energy on the configuration manifold. For the Regge system: (H.K1) holds with the vertices as point particles on S ; (H.K2) holds because the Laplacian Green’s function on any compact 2-surface has the universal expansion $G_S(x, y) = -(1/2\pi) \log d(x, y) + \gamma_S(x, y)$ with γ_S smooth, so the pair interaction $-\kappa_i \kappa_j G_S(z_i, z_j)$ has exactly the logarithmic singularity required; (H.K3) is (2) above. Kiessling’s theorem applies directly: its proof uses only the static microcanonical measure on the compact configuration space, not the Kirchhoff equations of motion, and the hypotheses do not distinguish between vortex circulations

and Regge deficit angles (both act as point sources of a 2D Laplacian Green's function). The microcanonical temperature $\beta = d \log \Omega / dE$ is therefore negative for $E > E^*$, and the Onsager variational principle applies to the Regge system.

Proposition 3.2 (Entropy variation forces constant curvature). *Let (S, g_0) be a closed oriented Riemannian 2-manifold of area A and Euler characteristic χ . Consider the family of conformal metrics $g = e^{2\varphi} g_0$ with the area constraint $\int_S e^{2\varphi} dA_{g_0} = A$ fixed. Fix the polygon radius ε small (a geodesic length, held constant as φ varies). Let*

$$S_{\text{total}}[\varphi] = \int_S \mathcal{S}(R_g(x)) dA_g(x), \quad \mathcal{S}(R) := \frac{1}{2} \sum_{m \neq m^*} \log[(N-1) - m(N-m)/2 - (N-1)R\varepsilon^2/4], \quad (3.1)$$

be the Hadamard-leading polygon entropy functional (here $\mathcal{S}$ is the Bekenstein–Hawking-style log of the product of non-critical Havelock eigenvalues, as a function of the Gaussian curvature $R = R_g$ at the polygon centre; $m^ = \lfloor N/2 \rfloor$ is the critical mode). Then:*

- (i) $\mathcal{S} : R \mapsto \mathcal{S}(R)$ is a smooth strictly concave function of R on the open interval where all Havelock eigenvalues are positive.
- (ii) The unique global maximizer of $S_{\text{total}}[\varphi]$ over $\{\varphi : \int e^{2\varphi} dA_{g_0} = A\}$ is the metric with constant Gaussian curvature $R_g \equiv \bar{R}_0 := 2\pi\chi/A$.

Proof. (i) *Strict concavity.* Write $\mu_m(R) := (N-1) - m(N-m)/2 - (N-1)R\varepsilon^2/4$, affine in R with slope $\mu'_m(R) = -(N-1)\varepsilon^2/4$ (the same for all m). On the domain where all $\mu_m > 0$, $\mathcal{S}(R) = \frac{1}{2} \sum_m \log \mu_m(R)$ is a sum of log of affine-positive functions. Each $\log \mu_m$ is strictly concave in R (second derivative $-\mu'_m(R)^2/\mu_m(R)^2 < 0$). The sum inherits strict concavity: $\mathcal{S}''(R) = -\frac{1}{2}((N-1)\varepsilon^2/4)^2 \sum_m 1/\mu_m(R)^2 < 0$. (Numerical values: at $R = 0$ and $N \geq 3$, all $\mu_m > 0$ and $\mathcal{S}''(0) < 0$; the domain extends to a neighbourhood $|R\varepsilon^2| < 4/(N-1) \cdot \min_m \{(N-1) - m(N-m)/2\}$ containing the relevant physical range.)

(ii) *Jensen inequality + Gauss–Bonnet.* By (i), $\mathcal{S}$ is strictly concave in R . By Jensen's inequality applied to the probability measure $d\mathbb{P} := dA_g/A$ on the closed surface:

$$\frac{1}{A} \int_S \mathcal{S}(R_g(x)) dA_g(x) \leq \mathcal{S}(\bar{R}), \quad \bar{R} := \frac{1}{A} \int_S R_g(x) dA_g(x), \quad (3.2)$$

with equality if and only if R_g is constant dA_g -almost everywhere, i.e. $R_g(x) = \bar{R}$ a.e. Multiplying by A :

$$S_{\text{total}}[\varphi] = \int_S \mathcal{S}(R_g) dA_g \leq A \cdot \mathcal{S}(\bar{R}).$$

Gauss–Bonnet fixes the average curvature for every metric in the conformal class with the area constraint. We use the scalar-curvature convention $R_g = 2K_g$ throughout this paper, where K_g is the Gaussian curvature. The Gauss–Bonnet theorem on a closed oriented 2-manifold states $\int_S K_g dA_g = 2\pi\chi(S)$, equivalently $\int_S R_g dA_g = 4\pi\chi$. Dividing by A :

$$\bar{R} = \frac{1}{A} \int_S R_g(x) dA_g(x) = \frac{4\pi\chi}{A} =: \bar{R}_0,$$

a *fixed* constant independent of φ (depending only on χ and A). Therefore

$$S_{\text{total}}[\varphi] \leq A \cdot \mathcal{S}(\bar{R}_0) \quad \text{for every admissible } \varphi. \quad (3.3)$$

Equality in (??) requires $R_g \equiv \bar{R}_0$ almost everywhere, i.e. the metric g has constant Gaussian curvature. Such a metric exists in every conformal class on a closed oriented 2-manifold by the *uniformization theorem* (?, Ch. IV), and is unique up to isometry within the conformal class (with area fixed).

Hence the bound (??) is attained by (and only by) the uniformized representative, which has $R_g \equiv \bar{R}_0$. This is the unique global maximizer of S_{total} . $\square$

Remark 3.2. *The proof uses Jensen's inequality pointwise on the curvature scalar R_g ; no elliptic-regularity argument or maximum-principle reasoning is invoked. Strict concavity of $\mathcal{S}$ in R is established purely algebraically from the eigenvalue formula. The Gauss–Bonnet constraint acts as a linear equality on the admissible space of curvature distributions, making the constrained optimization a standard concave maximization over a convex set.*

Remark 3.3 (Higher-order corrections). *The entropy $S_{\text{total}}[\varphi]$ as defined in this proposition uses the Hadamard-leading coefficient (??) (quadratic in $R\varepsilon^2$). Higher-order corrections ($R^2\varepsilon^4, R^3\varepsilon^6, \dots$) arise from subleading Hadamard coefficients and are discussed in the proof of Theorem ?? (Step 3, self-consistency argument) and the associated convergence analysis (Corollary ??). The leading-order strict concavity established here is inherited at higher order on a smaller basin around $R = \bar{R}_0$, as verified in the basin-invariance analysis.*

Theorem 3.3 (Polygon entropy maximization implies constant curvature). *Let (S, g) be a two-dimensional Riemannian manifold. Place N equal point vortices in a regular polygon ring of geodesic radius ε centered at $x \in S$, interacting via the scalar Green's function of (S, g) . If:*

- (H1) *Havelock structure : the angular Hessian eigenvalues have the form $\lambda_m = C_1(x, \varepsilon) - m(N-m)/2 + \delta_m(x, \varepsilon)$ with C_1 independent of m and $\delta_m = 0$ for all interior modes $2 \leq m \leq N-2$. The only mode-dependent corrections are confined to the edge modes $m \in \{1, N-1\}$, where $|\delta_m| = O(K\varepsilon^2/C_1) \rightarrow 0$ as $\varepsilon \rightarrow 0$.*

Proof on general surfaces. On any 2D Riemannian manifold, the Green's function in normal coordinates at x has the Hadamard expansion $G(z, w) = -(1/2\pi) \ln |z-w| + \gamma(x) + (K(x)/12\pi)|z-w|^2 + O(|z-w|^4|\nabla K|)$, where $K(x)$ is the Gaussian curvature. The $-\ln |z-w|$ term gives $m(N-m)/2$ (Paper I, Lemma ??). The constant $\gamma(x)$ has no tangential derivatives. For the curvature term: $|z_j - z_k|^2 = 4\varepsilon^2 \sin^2(\pi(j-k)/N)$ at the ring, giving a circulant with entries $c_p = -(K\varepsilon^2/(6\pi)) \cos(2\pi p/N)$. The DFT eigenvalue at mode m is

$$\delta T_m = -\frac{K\varepsilon^2}{6\pi} \sum_{p=1}^{N-1} \cos(2\pi p/N) \left(1 - \cos(2\pi pm/N)\right).$$

Expanding the product and using $\sum_p \cos(2\pi(1\pm m)p/N) = -1$ for $2 \leq m \leq N-2$ (since $1\pm m$ is not divisible by N): $\delta T_m = 0$ for all $2 \leq m \leq N-2$.

The edge modes $m = 1$ and $m = N-1$ receive $\delta T_1 = K(x)N\varepsilon^2/(12\pi)$, which shifts C_1 at these two modes only. Since $\lambda_1 = C_1 - (N-1)/2$ is the largest eigenvalue ($\lambda_1 \geq (N-1)/2$ on the flat plane), the fractional correction is $O(K\varepsilon^2/C_1)$, negligible for small ε .

On constant-curvature surfaces ($\nabla K \equiv 0$), $\delta_m = 0$ exactly for all interior m and every order in ε (Theorem ??). On variable-curvature surfaces the $O(K\varepsilon^2)$ computation above gives mode-independence of the leading curvature correction on interior modes; higher-order corrections $O(\varepsilon^4|\nabla K|, \varepsilon^4K^2)$ are present and are not needed for the theorem, which uses only the leading-order Havelock structure on the polygon entropy functional to derive $\nabla K = 0$ (i.e. constant scalar curvature) as the stationary point; see the self-consistency argument in Step 3 of Theorem ??.

(H2) Entropy maximization [**strict concavity** + **Jensen, finite N**]: the polygon entropy functional $S_{\text{poly}}[g] = \int S(x) dA$, where $S(x) = \frac{1}{2} \sum_{m \neq m^*} \log \lambda_m(x)$ is a finite sum over $N-2$ binding modes (excluding the critical mode m^* where $\lambda_{m^*} = 0$ at the threshold $N = 7$). On the domain $\mathcal{B} = \{g : \|\delta R\|_\infty \leq 0.90\}$ (established below as the Onsager contraction basin), all λ_m with $m \neq m^*$ are bounded away from zero. From Lemma ??, $C_1(x) = (N-1)[1 - R(x)\varepsilon^2/4 + O(\varepsilon^4)]$, so $|\delta \lambda_m(x)| = (N-1)\varepsilon^2|\delta R(x)|/4 + O(\varepsilon^4)$, and $\min_{m \neq m^*} \lambda_m \geq \min_{m \neq m^*} [(N-1) - m(N-m)/2] - (N-1)\varepsilon^2\|\delta R\|_\infty/4 \geq c > 0$ for sufficiently small ε on $\mathcal{B}$. On $\mathcal{B}$ and for $\varepsilon < \varepsilon_0(N)$ (with $\varepsilon_0(N) = c^{1/2}/[(N-1)\|\delta R\|_\infty]^{1/2}$ chosen so the $O(\varepsilon^4)$ remainder stays within the strict-concavity margin), S is strictly concave in $R(x)$: S is strictly concave in C_1 ($d^2S/dC_1^2 = -\frac{1}{2} \sum 1/\lambda_m^2 < 0$, algebraic at finite N), and $C_1(x) = (N-1)[1 - R(x)\varepsilon^2/4 + O(\varepsilon^4)]$ is affine in $R(x)$ at leading order with Taylor remainder uniformly bounded by $|R(x)|^2\varepsilon^4$ times a curvature-dependent constant (Lemma ??). The $O(\varepsilon^4)$ Taylor remainder shifts the second derivative d^2S/dR^2 by at most $(N-1)^2\varepsilon^4/16 \cdot (\text{const}_N)$, which is absorbed into the strict-concavity margin $\min_m 1/\lambda_m^2$ for ε small enough. Therefore $S(C_1(R))$ is strictly concave in R on $\mathcal{B}$ for $\varepsilon < \varepsilon_0(N)$. Subject to the Gauss–Bonnet constraint $\int R dA = 4\pi\chi$ (scalar curvature convention, $R = 2K$), the unique maximum is $R(x) = \text{const}$ (Jensen’s inequality). This argument works for any Riemannian metric, including Regge metrics with N conical singularities: the Green’s function, the Hadamard parametrix, and the eigenvalues λ_m are all well-defined for cone metrics (Remark ??).

(H3) Hadamard universality : condition (H1) holds at every point $x \in S$. The Hadamard parametrix $G(x, y) = -(1/2\pi) \log d(x, y) + \gamma_S(x) + O(d^2)$ is universal on any Riemannian surface.

Then the scalar curvature of S is constant: $R(x) = R_0$ for all x . In $2+1D$, this gives the vacuum Einstein equations $R_{\mu\nu} = \Lambda g_{\mu\nu}$ (Schur’s lemma; the Weyl tensor vanishes identically in three dimensions).

The cosmological constant Λ is determined by a separate dynamical argument (the WDW equation, Proposition ?? below), not by the theorem above. The theorem establishes the *form* of the Einstein equations; the WDW equation determines their *content* (Λ).

3.2 Proof of the main theorem

Proof architecture. The proof assembles three ingredients: (1) Lemma ?? shows C_1 is affine in $R(x)$ via the Hadamard expansion; (2) Lemma ?? bridges discrete Regge vertices to smooth metrics; (3) Corollary ?? shows the Onsager iteration contracts curvature variance (factor 0.044 at $N = 7$). The main proof (Theorem ??, page ??) combines them via Jensen’s inequality and the Robin-function identity $\Delta\gamma = K - \bar{K}$. The reader may

proceed directly to the main proof and consult the lemmas as needed. The ingredients are stated first:

Corollary 3.4 (Onsager contraction, (H2b) as theorem). *Define the Onsager iteration $T: g \mapsto g'$ on compact metrics by: (i) compute the Onsager mean-field density $\rho(x)$ of N equal-circulation vortices on (S, g) at negative temperature (exists by the Onsager–Regge Lemma); (ii) map $\rho(x)$ to a curvature field $R'(x) = \delta \times \rho(x)$ via the vortex-gravity isomorphism (equal circulations $\Rightarrow$ equal deficit angles $\Rightarrow R \propto \rho$); (iii) define $g' = T(g)$ as the metric with curvature R' .*

The iteration operates on the scalar field $R(x)$, not on polygon geometry. The N vortices are individual point masses whose positions are drawn from the mean-field density $\rho(x)$; they do not form a regular N -gon (except on a constant-curvature surface, where $\rho = \text{const}$ and the entropy-maximizing configuration is the polygon). The energy landscape $V(x)$ that determines ρ depends on the local curvature $R(x)$ through the Hadamard parametrix $C_1(x) = (N-1)[1 - R(x)\varepsilon^2/4 + \dots]$, a property of the Green’s function of (S, g) at the point x , not of any global vortex configuration.

Then for all initial metrics with $\text{Var}(R(g)) \leq 2$, the curvature variance contracts:

$$\text{Var}(R(T(g))) \leq |\beta_{\text{eff}}|^2 R_0^2 \text{Var}(R(g)), \quad (3.4)$$

where $\beta_{\text{eff}} = 1/f(m^*, N)$ is the inverse critical Casimir (equal to $1/C_1$ at the stability threshold), $R_0 = 2\pi\chi/\text{Vol}$ is the mean curvature ($|R_0| = 1$ for the Bolza surface), and V_1, V_2 are defined in the proof below. At $N = 7$ on the Bolza surface ($|\beta_{\text{eff}}| R_0 = 1/f(3, 7) = 1/6$, $|R_0| = 1$), the L^∞ -basin estimate yields the contraction factor

$$\underbrace{(|\beta_{\text{eff}}| R_0)^2}_{=1/36} \cdot \underbrace{e^{2|\beta_{\text{eff}}| \|Y\|_{L^\infty}}}_{\leq e^{0.337} = 1.40} \cdot \underbrace{\left(1 + \frac{|V_2|}{2|V_1|} \|\delta R\|_{L^\infty}\right)^2}_{\leq (1.056)^2 = 1.12} = 0.044 < 1, \quad (3.5)$$

with the three factors derived explicitly in the proof below (see the paragraph beginning “At $N = 7$.” at Eq. (??) and the following display). Each factor uses only: (i) the Havelock eigenvalue $f(3, 7) = 6$ fixing $|\beta_{\text{eff}}|$; (ii) the Taylor–Lipschitz variance bound for exponentials (??, §2.3) evaluated on the basin $\|\delta R\|_\infty \leq 0.90$, $\text{Var}(\delta R) \leq 2$; (iii) the WDW turning-point value $\rho^* = 1.734$ giving $V_1 = \coth(\rho^*) \approx 1.065$, $|V_2/V_1| = \text{csch}^2(\rho^*)/\coth(\rho^*) \approx 0.125$. A sharper bound ≈ 0.042 would follow from a Sobolev refinement requiring propagation of an H^1 -bound under iteration; that H^1 -propagation is not established here and is not load-bearing: the L^∞ -derived factor $0.044 < 1$ already gives strict contraction in the fixed-point argument. The contraction improves with N : $|\beta_{\text{eff}}| R_0 = 1/8$ at $N = 8$, $1/15$ at $N = 11$.

Proof. The Onsager equilibrium distributes vortices with mean-field density $\rho(x) \propto \exp(\beta_{\text{eff}} V(x))$, where $V(x) = V(C_1(R(x)))$ is the BO potential and $\beta_{\text{eff}} = 1/C_1$ is the inverse stiffness (the angular bath’s response to curvature perturbations is controlled by $C_1 = f(m^*, N)$, not by the total energy). For a perturbation δR around mean R_0 : $\rho(x) = \rho_0 \exp(\beta_{\text{eff}}(V_1 \delta R + \frac{1}{2} V_2 \delta R^2))$ where $V_1 = \coth(\rho_*) = C_1'(\rho_*)$ and $V_2 = -\text{csch}^2(\rho_*) = C_1''(\rho_*)$, evaluated at the palindromic threshold ρ_* ($\rho_* = 1.73$ at $N = 7$, 2.40 at $N = 8$). The N vortices have equal circulations κ , so each deficit angle $\delta_k = 8\pi G\kappa$ is identical. The new curvature field is therefore $R'(x) = \delta \times \rho(x)$ where $\delta = 8\pi G\kappa$ is the deficit angle per vortex (fixed by the identification; Gauss–Bonnet $\int R dA = 8\pi(1-g)$ constrains the mean density $\rho_0 = R_0/\delta$, and the fluctuation $\delta R = R - R_0$ satisfies $\int \delta R dA = 0$, so the contraction operates on the zero-mean fluctuation field). The new curvature variance is $\text{Var}(R') = \delta^2 \text{Var}(\rho)$. Expanding

$\rho = \rho_0 e^{\beta_{\text{eff}}(V_1 \delta R + V_2 \delta R^2/2)}$ to second order in δR and bounding the variance of the exponential by the elementary second-order Taylor estimate

$$\text{Var}(e^{aY}) \leq a^2 e^{2|a| \|Y\|_{L^\infty}} \text{Var}(Y),$$

valid for any bounded random variable Y with $|Y| \leq \|Y\|_{L^\infty}$ (this is a direct consequence of $|e^{aY} - 1 - aY| \leq \frac{1}{2} a^2 Y^2 e^{|a| |Y|}$; see ?, §2.3). On a compact surface $\delta R = R - R_0$ is bounded with $\|\delta R\|_{L^\infty} \leq 2C_S$ (Sobolev embedding; see below), so this bound applies. Setting $Y = V_1 \delta R + \frac{1}{2} V_2 \delta R^2$ and $a = \beta_{\text{eff}}$, we bound $\text{Var}(Y)$ via the L^2 triangle inequality: $\sqrt{\text{Var}(Y)} \leq |V_1| \sqrt{\text{Var}(\delta R)} + \frac{1}{2} |V_2| \sqrt{\text{Var}(\delta R^2)}$. The L^∞/L^2 interpolation $\text{Var}(\delta R^2) = E[\delta R^4] - (E[\delta R^2])^2 \leq E[\delta R^4] \leq \|\delta R\|_{L^\infty}^2 \text{Var}(\delta R)$ gives $\sqrt{\text{Var}(\delta R^2)} \leq \|\delta R\|_{L^\infty} \sqrt{\text{Var}(\delta R)}$. Squaring the triangle inequality:

$$\text{Var}(Y) \leq V_1^2 \text{Var}(\delta R) \left(1 + \frac{|V_2|}{2|V_1|} \|\delta R\|_{L^\infty}\right)^2. \quad (3.6)$$

Combining with the Taylor-Lipschitz bound:

$$\text{Var}(R') \leq |\beta_{\text{eff}}|^2 R_0^2 e^{2|\beta_{\text{eff}}| \|Y\|_{L^\infty}} \text{Var}(\delta R) \left(1 + \frac{|V_2|}{2|V_1|} \|\delta R\|_{L^\infty}\right)^2.$$

At $N = 7$: $|\beta_{\text{eff}}| R_0 = 1/6$, $|V_2/V_1| = |\text{csch}^2(\rho^*)/\coth(\rho^*)| = 0.125$ (with $\rho^* = 1.734$, the WDW turning point of Proposition ??). Within the basin ($\text{Var} \leq 2$, $\|\delta R\|_\infty \leq 0.90$): $\|Y\|_{L^\infty} \leq |V_1| \cdot 0.90 + \frac{1}{2} |V_2| \cdot 0.81 = 0.958 + 0.054 = 1.012$, so $e^{2|\beta_{\text{eff}}| \|Y\|_\infty} \leq e^{0.337} = 1.40$. The nonlinear factor $(1 + (0.125/2) \cdot 0.90)^2 = (1.056)^2 = 1.12$, giving total contraction $(1/6)^2 \times 1.40 \times 1.12 = 0.044 < 1$. The condition $|\beta_{\text{eff}}| R_0 < 1$ is $f(m^*, N) > |R_0|$, which holds for all $N \geq 4$ (since $f(m^*, N) \geq 2$ and $|R_0| = 1$ on the Bolza surface). The contraction holds for all initial $\text{Var}(R) \leq 2$, establishing $R = \text{const}$ as a locally attracting fixed point with basin $\text{Var}(R) \leq 2$. On the Bolza surface ($|R_0| = 1$, $\text{Var}(R_0) = 0$), the basin is $\text{Var}(R) \leq 2$ (an L^2 condition). On the Bolza surface ($\text{Vol} = 4\pi$, $\lambda_1 = 3.839$; ?), the heat-kernel Sobolev bound at optimal time $t = 1/\lambda_1$ gives, for any zero-mean $u \in H^1(\Sigma)$:

$$\|u\|_{L^\infty}^2 \leq \frac{\lambda_1}{4\pi} \|u\|_{L^2}^2 + \frac{1}{e \lambda_1} \|\nabla u\|_{L^2}^2 = 0.306 \|u\|_{L^2}^2 + 0.096 \|\nabla u\|_{L^2}^2. \quad (3.7)$$

The Sobolev bound (??) refines the L^∞ control if an H^1 bound $\|\nabla R\|_{L^2}^2 \leq 2$ is imposed on the basin (in addition to the L^2 bound $\text{Var}(R) \leq 2$). However, the Onsager iteration is not shown to preserve such an H^1 bound under T ; the map T contracts $\text{Var}(R)$ (the L^2 norm squared) but does not directly control $\|\nabla R\|_{L^2}$. We therefore quote only the cruder L^∞ bound ($\|\delta R\|_\infty \leq 0.90$ from basin definition) and conclude contraction at factor $0.044 < 1$. Establishing H^1 -propagation of T would tighten the factor to ≈ 0.042 , but the cruder bound suffices for the fixed-point argument. The Poincaré-uniformized metric ($R = \text{const}$) has $\text{Var} = 0$, and the basin $\text{Var} \leq 2$ contains a metrically thick neighborhood.

Basin invariance. We show the map T preserves the L^∞ -basin $\mathcal{B} = \{\|\delta R\|_\infty \leq 0.90\}$. The Onsager iteration $R \mapsto R'$ factors through an auxiliary potential V satisfying $-\Delta V = \delta R$ with $\int V dA = 0$ (Robin-function Poisson equation, Lemma ??). Then $R' = \delta \rho_0 e^{\beta_{\text{eff}} V}$ with ρ_0 a normalization constant. On a compact hyperbolic surface, elliptic regularity (?, Ch. 6) gives $\|V\|_{H^2} \leq C_\Sigma (\|V\|_{L^2} + \|\Delta V\|_{L^2})$ with C_Σ a surface-dependent constant. On mean-zero functions, Poincaré gives $\|V\|_{L^2} \leq \lambda_1^{-1} \|\Delta V\|_{L^2}$, so

$\|V\|_{H^2} \leq C_\Sigma(1 + \lambda_1^{-1})\|\Delta V\|_{L^2} = C_\Sigma(1 + \lambda_1^{-1})\|\delta R\|_{L^2}$ (using $-\Delta V = \delta R$). Composing with the Sobolev embedding $H^2(\Sigma) \hookrightarrow L^\infty(\Sigma)$ (valid in dimension 2 with constant C_{Sob}):

$$\|V\|_{L^\infty} \leq C_{\text{Sob}} \|V\|_{H^2} \leq C_{\text{Sob}} C_\Sigma(1 + \lambda_1^{-1}) \|\delta R\|_{L^2}.$$

Taylor expansion of $R' = \delta\rho_0(1 + \beta_{\text{eff}}V + O(\beta_{\text{eff}}^2V^2))$ gives, on the basin, $\|\delta R'\|_{L^\infty} \leq C_{\text{ell}} \|\delta R\|_{L^2}$ with $C_{\text{ell}} = |\beta_{\text{eff}}R_0| \cdot C_{\text{Sob}}/\lambda_1 \cdot (1 + O(\|V\|_\infty))$. On the Bolza surface (genus $g = 2$, area $A = 4\pi$, $\lambda_1 = 3.839$), the constants are bounded as follows. The elliptic constant C_Σ for the resolvent $(-\Delta_g + 1)^{-1} : L^2(\Sigma) \rightarrow H^2(\Sigma)$ satisfies $C_\Sigma \leq 1$ by the spectral calculus: $\|(-\Delta + 1)^{-1}\|_{L^2 \rightarrow L^2} = 1$ on any compact manifold because $-\Delta \geq 0$ implies $-\Delta + 1 \geq 1$, and differentiating twice under the resolvent gives the H^2 bound with the *same* operator norm ([?, Thm. 2.5]). The 2D Sobolev embedding $H^2(\Sigma) \hookrightarrow L^\infty(\Sigma)$ is continuous on any closed surface ([?, Thm. 2.30; ?, Thm. 3.4] with constant

$$C_{\text{Sob}} \leq \frac{1}{\sqrt{4\pi}}(1 + \lambda_1^{-1})^{1/2} \leq 1.5 \quad \text{on Bolza } (\lambda_1 = 3.839, A = 4\pi). \quad (3.8)$$

The numerical value $1/\sqrt{4\pi} \cdot (1 + 1/3.839)^{1/2} \approx 0.282 \cdot 1.125 \approx 0.32$; the looser bound ≤ 1.5 is used below to absorb the $(1 + \lambda_1^{-1})$ factor in the H^2 estimate. With $|\beta_{\text{eff}}R_0| = 1/6$, we obtain $C_{\text{ell}} \leq (1/6) \cdot 1.5 \cdot (1 + 1/3.839) \cdot 1.12 \leq 0.35$, well below the threshold $C_{\text{ell}} \leq 0.64$ required for basin invariance. On the basin $\text{Var}(R) \leq 2$, $\|\delta R'\|_{L^\infty} \leq 0.35\sqrt{2} \approx 0.50 < 0.90$. Therefore T maps the L^∞ -basin $\mathcal{B}$ into itself, closing the iteration argument. By the Banach fixed-point theorem with contraction constant 0.044, the unique fixed point is g^* with $\text{Var}(R(g^*)) = 0$, i.e. $R(g^*) = \text{const}$.

Extension to other compact surfaces. The Sobolev constant C_S depends on the surface through λ_1 and Vol. For any compact hyperbolic surface Σ of genus g with $\lambda_1 > 1/4$ (proved for all arithmetic surfaces by Selberg 1965 and Gelbart–Jacquet 1978; the Bolza surface at $g = 2$ has $\lambda_1 = 3.839$, well above the bound): $\lambda_1/(4\pi) \geq 1/(16\pi) \approx 0.020$ and $1/(e\lambda_1) \leq 4/e \approx 1.47$. The contraction constant $|\beta_{\text{eff}}|^2 R_0^2 \times (1 + 2|V_2/V_1|\sqrt{\sigma^2})^2$ scales as $1/f(m^*, N)^2$, which is $\leq 1/36$ at $N = 7$ regardless of the surface. The basin condition $\text{Var}(R) \leq 2$ is therefore sufficient on *any* compact hyperbolic surface with $\lambda_1 > 1/4$, not just the Bolza surface. $\square$

Remark 3.4 (Lax conservation alternative). *The contraction can also be proved without the mean-field (ergodic) step, using the CMS integrability of the vortex system. The Havelock Hessian is the CMS Hamiltonian (Gaudin–Calogero–Moser identification); the trigonometric CMS Lax matrix at the N -gon has eigenvalues $\nu_m = g(2m - N + 1)$ (equally spaced, gap $2g$), and these are conserved under the CMS flow (isospectral evolution). The BTZ phase requires vortex coalescence, which forces $\|L\| \rightarrow \infty$ (the csc kernel diverges). Since conserved Lax eigenvalues cannot diverge, the polygon phase is dynamically isolated from the BTZ phase. This replaces the Onsager variational principle with an algebraic obstruction: the CMS invariant torus containing the N -gon equilibrium is disjoint from the BTZ basin, and the Lax isospectral flow preserves the torus.*

Lemma 3.5 (Logarithmic interaction and curvature expansion). *Let (S, g) satisfy conditions (H1) and (H3). Then the vortex interaction is $V(z, w) = -G_S(z, w)$ (the scalar Laplacian Green’s function), and the mode-independent coefficient satisfies*

$$C_1(x, \varepsilon) = (N-1) \left[1 - \frac{R(x) \varepsilon^2}{4} + O(\varepsilon^4) \right], \quad (3.9)$$

where $R(x)$ is the scalar curvature at x .

Proof. Condition (H1) requires the angular Casimir to be the exact polynomial $f(m, N) = m(N-m)/2$. By the $\csc^2$ uniqueness theorem (Paper I, Lemma ??), this polynomial arises *only* from the weight function $\csc^2(\pi p/N)$, which is the discrete Fourier transform of the logarithmic kernel $-\log|z-w|$. Condition (H3) (Hadamard universality) ensures that the Laplacian Green's function has exactly this logarithmic singularity on any Riemannian surface. Together, (H1)+(H3) force $V(z, w) = -G_S(z, w)$.

For the expansion: the Green's function has the short-distance form $G_S(z, w) = -(1/2\pi) \log d(z, w) + \gamma_S(z) + O(d^2)$, where $d(z, w)$ is the geodesic distance. Summing over the N -ring at radius ε gives (??). The mode-independence of the $O(\varepsilon^2)$ correction is *automatic*: C_1 is mode-independent by the Riemannian Havelock identity (Paper I, Theorem ??), so its expansion inherits this property at every order. The coefficient $1/4$ is verified on the three model surfaces: $\mathbf{H}^2$ ($R = -2$, $C_1/(N-1) = (1+\xi^2)/(1-\xi)^2$ with $\xi \approx \varepsilon^2/4$), $\mathbf{S}^2$ ($R = +2$, $(1-\xi)/(1+\xi)$), and the plane ($R = 0$, $C_1 = N-1$). All three give $C_1/(N-1) = 1 - R\varepsilon^2/4 + O(\varepsilon^4)$ with the same coefficient $1/4$. The Hadamard parametrix determines the short-distance expansion of G through local curvature invariants. At $O(\varepsilon^2)$ the only such invariant is R , so $C_1/(N-1) = 1 - \alpha R\varepsilon^2 + \dots$ for a universal constant α , and matching any one model surface fixes $\alpha = 1/4$. (The coefficient $1/4$ differs from the on-diagonal Seeley–DeWitt value $a_1 = R/6$ because C_1 is the *off-diagonal* Green's function averaged over the polygon chord geometry: the chord metric distortion $d_p^2 = 4\varepsilon^2 \sin^2(\pi p/N)(1 - R\varepsilon^2 \cos^2(\pi p/N)/6)$ and the Hadamard regular part each contribute, and their angular averages combine to $1/4$.) $\square$

Lemma 3.6 (Regge bridge to smooth metrics). *The entropy-maximizing Regge metric converges to the entropy-maximizing smooth metric as the mesh size $h \rightarrow 0$.*

Proof. Any smooth 2D Riemannian manifold (S, g) is approximated by a simplicial complex (S_h, g_h) in which curvature is concentrated at V vertices, each carrying a deficit angle $\varepsilon_v \approx R(v) A_v$ (with A_v the dual area). The convergence $g_h \rightarrow g$ as $h \rightarrow 0$ is proved (?); the Green's function $G_h \rightarrow G$ in the H^1 -norm (standard finite-element convergence). Each Regge vertex is a point vortex (circulation $\kappa_v = \varepsilon_v/(2\pi)$, by the isomorphism), so the Regge system is a V -vortex configuration. The vertex positions are *simultaneously* vortex positions and Regge metric degrees of freedom (on a fixed triangulation, vertex positions determine edge lengths, hence the simplicial metric). This identification eliminates the apparent circularity of H : the Hamiltonian is a function of vertex positions alone, and entropy maximization is a single finite-dimensional optimization, not an iteration between metric and vortex configuration.

The Onsager prerequisites for this Regge system are established in Lemma ?. On a *flat* background ($\Lambda = 0$), conformal invariance (Paper II, Proposition 13.2) gives $G = -(2\pi)^{-1} \log|z-w|$, independent of the deficit angles. On a *curved* background ($\Lambda \neq 0$), the Hadamard expansion (Lemma ??) gives $C_1 = (N-1)[1 - R(x)\varepsilon^2/4 + \alpha_4 R^2 \varepsilon^4 + O(\varepsilon^6)]$, where $\alpha_4 = 1/48$ (the next Hadamard coefficient, determined by the model-surface matching in Lemma ??). The entropy $S(C_1)$ is strictly concave in C_1 ($d^2 S/dC_1^2 < 0$), and C_1 is a polynomial in R with leading term $-(N-1)\varepsilon^2 R/4$ and subleading correction $\alpha_4 R^2 \varepsilon^4$. By the chain rule, $d^2 S/dR^2 = (dS/dC_1)(d^2 C_1/dR^2) + (d^2 S/dC_1^2)(dC_1/dR)^2$. The first term has sign $(-)(+) = (-)$ (entropy increases with C_1 ; the R^2 coefficient $\alpha_4 > 0$). The second term is strictly negative. So $d^2 S/dR^2 < 0$ for all ε : strict concavity in R holds exactly, not just at leading order. The Gauss–Bonnet constraint $\int R dA = 4\pi\chi$ and Jensen's inequality give the unique maximum at $R = \text{const}$.

Given Lemma ??, the Onsager principle on the V -vertex Regge system selects the

entropy-maximizing vertex configuration, which by (H2a) has $C_1(v)$ uniform, hence $R = \text{const}$. In the continuum limit $V \rightarrow \infty$: $S_{\text{Regge}} \rightarrow S_{\text{total}}[g]$ (by the FEM convergence of the Green's function, sharpened to $O(h^\infty)$ by the Szegő estimates below), and the entropy-maximizing Regge metric converges to the entropy-maximizing smooth metric (strict concavity ensures convergence of maximizers under pointwise convergence; ?). The Szegő–Toeplitz analysis sharpens the rate: the entropy S_{Regge} is a Toeplitz sum $\sum \log \lambda_m$ over the circulant eigenvalues, and Szegő's strong theorem (Szegő's strong limit theorem; ?) gives $|S_{\text{Regge}} - S_{\text{total}}[g]| = O(V^{-k})$ for every $k \geq 1$ (exponential convergence of Toeplitz determinants for analytic symbols; the symbol here is $\log(C_1 - f(\theta))$, analytic in θ). $\square$

Proof of Theorem ??. The proof proceeds in three explicit steps.

Step 1: Jensen's inequality on the entropy functional. By (H2), the polygon entropy functional $S_{\text{total}}[g] = \int_S S(C_1(x)) dA$ with $S(C_1) = \frac{1}{2} \sum_{m \neq m^*} \log(C_1 - f(m, N))$ is strictly concave in C_1 :

$$\frac{d^2 S}{dC_1^2} = -\frac{1}{2} \sum_{m \neq m^*} \frac{1}{(C_1 - f(m, N))^2} < 0,$$

a finite- N algebraic identity holding on the open set $\lambda_m = C_1 - f(m, N) > 0$ for all $m \neq m^*$. The Gauss–Bonnet constraint $\int_S R dA = 4\pi\chi$ is linear in R . Since C_1 is affine in R at leading order in ε (Lemma ??), the constraint is also affine in C_1 at leading order. Jensen's inequality then gives: the unique global maximum of S_{total} on the constraint surface is $C_1(x) = C_1^{\text{eq}} = \text{const}$, with equality iff C_1 is pointwise constant (? , §27: strictly concave function on a convex constraint set has a unique maximizer). This step uses no ε -expansion in the concavity: the strict concavity of log-sum is exact.

Step 2: From $C_1 = \text{const}$ to $R = \text{const}$ at leading order in ε . The Riemannian Havelock identity (Paper I, Theorem ??) together with the Hadamard expansion (Lemma ??) gives

$$C_1(x, \varepsilon) = (N-1) \left[1 - R(x) \varepsilon^2 / 4 \right] + O(\varepsilon^4), \quad (3.10)$$

pointwise for any smooth Riemannian surface. Setting $C_1(x, \varepsilon) = C_1^{\text{eq}}$ for all x and solving for $R(x)$:

$$R(x) = \frac{4}{\varepsilon^2} \left(1 - \frac{C_1^{\text{eq}}}{N-1} \right) + O(\varepsilon^2) = R_0 + O(\varepsilon^2),$$

where $R_0 := (4/\varepsilon^2)(1 - C_1^{\text{eq}}/(N-1))$ is x -independent. Therefore $R(x) = R_0$ up to $O(\varepsilon^2)$ corrections. As $\varepsilon \rightarrow 0$ (the point-vortex limit), the residual vanishes and $R(x) = R_0$ exactly. This establishes the theorem as an asymptotic result: the entropy-maximizing metric has constant scalar curvature in the $\varepsilon \rightarrow 0$ limit.

Step 3: Exact result via self-consistency. Step 2 gives $R = \text{const}$ up to $O(\varepsilon^2)$ for any fixed polygon radius $\varepsilon > 0$. We now show this limit is self-consistent: the fixed point $R(x) = \bar{K}$ (a constant-curvature surface) makes all higher-order corrections vanish identically, and the Onsager iteration (Corollary ??) converges geometrically to this fixed point.

Self-consistency on constant-curvature surfaces. If the maximizer of S_{poly} has $R(x) = \bar{K}$ constant, then (S, g) is a constant-curvature surface. On such surfaces, the Riemannian Havelock theorem (Paper I, Theorem ??) gives $\lambda_m = C_1 - m(N-m)/2$ *exactly* (not merely to leading order): the Robin-function contribution to the tangential eigenvalues vanishes identically for every m (§??, product-to-sum argument). Thus the polygon entropy is

exactly $S_{\text{poly}}[g] = \frac{1}{2} \sum_{m \neq m^*} \log \lambda_m$ with λ_m mode-independent in the surface variables except through $C_1(x)$, and the Jensen argument (Step 2) yields $R(x) = \bar{K}$ as the unique critical point *with no residual* at any order in ε .

Geometric convergence. We apply the Banach fixed-point theorem in the Hilbert space $L_0^2(\Sigma, dA_g)$ of mean-zero functions. (a) *L^2 -basin of attraction.* Define $\mathcal{B} := \{g : \|\delta R(g)\|_{L^2} \leq \sqrt{2}\}$, the set of metrics with $\text{Var}(R) \leq 2$ (as in Cor. ??). This is a closed ball in the Hilbert space L_0^2 , hence a complete metric space in the L^2 -norm. (b) *T -invariance.* Cor. ?? proves the contraction $\|\delta R(Tg) - \delta R(Tg')\|_{L^2} \leq \kappa \|\delta R(g) - \delta R(g')\|_{L^2}$ with $\kappa = 0.044$ at $N = 7$ on the Bolza surface. Applied to $g' = \text{constant-curvature metric}$ (fixed point of T with $\delta R = 0$), $\|\delta R(Tg)\|_{L^2} \leq \kappa \|\delta R(g)\|_{L^2} \leq \kappa \sqrt{2} \approx 0.062 \leq \sqrt{2}$, so $T(\mathcal{B}) \subseteq \mathcal{B}$. (c) *L^2 -contraction.* By the same contraction bound, T is Lipschitz on $\mathcal{B}$ with constant $\kappa = 0.044 < 1$.

Banach fixed-point. Applying the Banach fixed-point theorem to $T : (\mathcal{B}, \|\cdot\|_{L^2}) \rightarrow (\mathcal{B}, \|\cdot\|_{L^2})$ (complete by (a), invariant by (b), contractive by (c)) yields a unique fixed point g^* , and every initial metric in the basin converges to g^* at geometric rate $\kappa = 0.044$ in L^2 . The self-consistent constant-curvature metric g_* (unique up to uniformization) is a fixed point of T (verified: on constant-curvature surfaces, the Jensen equation (??) is satisfied by $R = \bar{K}$). By uniqueness, $g^* = g_*$, so $R(g^*)(x) = \bar{K}$ exactly for all x . This closes the $O(\varepsilon^2)$ residual of Step 2 without invoking any Sobolev-constant bound: the exactness is forced by the combination of (i) exact mode-independence of λ_m on constant-curvature surfaces (Paper I), and (ii) geometric convergence of the Onsager iteration (Corollary ??).

Lemma 3.7 (Robin function identity). *On a closed Riemannian 2-manifold (S, g) , the Robin function $\gamma(x)$, defined as the regular part of the Laplacian Green's function at the diagonal,*

$$G(x, y) = -\frac{1}{2\pi} \log d(x, y) + \gamma(x) + O(d(x, y)^2),$$

satisfies $-\Delta_x \gamma(x) = K(x) - \bar{K}$, where K is the Gauss curvature and $\bar{K} = 2\pi\chi/A$.

Proof. Work in conformal isothermal coordinates $g = e^{2\sigma}|dz|^2$ near a fixed point x_0 , with flat Laplacian $\Delta_0 = \partial_z \partial_{\bar{z}} / \dots$ and

$$K = -e^{-2\sigma} \Delta_0 \sigma, \quad (3.11)$$

the standard conformal formula for Gauss curvature (?, §4.3). The mean-value Green's function on the closed surface satisfies $-\Delta_{g,y} G(x, y) = \delta_x(y) - 1/A$ (equivalently $-\Delta_{0,y} \tilde{G} = e^{2\sigma}(\delta_x - 1/A)$ where $\tilde{G} = G$ in these coordinates since $\Delta_g = e^{-2\sigma} \Delta_0$). The geodesic distance expands as $d(x, y)^2 = e^{2\sigma(x_0)} |z-w|^2 (1 + O(|z-w|^2))$ in normal coordinates based at x_0 (?, §4.6), so

$$\log d(x, y) = \log |z-w| + \sigma(x_0) + O(|z-w|^2), \quad (x, y \rightarrow x_0). \quad (3.12)$$

Step 1: extract the regular part. Define $\gamma_0(x, y) := G(x, y) + (1/2\pi) \log d(x, y)$ in a neighbourhood of the diagonal. Using $\Delta_{0,y} \log |z-w| = 2\pi \delta_z(w)$ and (??), we have

$$\begin{aligned} -\Delta_{g,y} \gamma_0(x, y) &= -\Delta_{g,y} G(x, y) + (1/2\pi) \Delta_{g,y} \log d(x, y) \\ &= (\delta_x(y) - 1/A) + (1/2\pi) (\Delta_g \log d)(x, y). \end{aligned}$$

Near the diagonal, $\Delta_g \log d(x, y) = 2\pi e^{-2\sigma(x)} e^{2\sigma(x)} \delta_x(y) + K(x) + O(d)$ (the curvature term is Steiner's classical diagonal expansion; see ?, Eq. (2.4) and §2.2, or direct computation from (??) applied to (??) and using $\Delta_g = e^{-2\sigma} \Delta_0$ twice). The delta cancels the $\delta_x(y)$ from $-\Delta_g G$, leaving γ_0 a smooth function of (x, y) with

$$-\Delta_{g,y} \gamma_0(x, y) = -1/A + K(x)/(2\pi) + O(d(x, y)). \quad (3.13)$$

Step 2: evaluate on the diagonal. The Robin function is the diagonal value $\gamma(x) := \lim_{y \rightarrow x} (G(x, y) + (1/2\pi) \log d(x, y)) = \gamma_0(x, x)$. Applying $-\Delta_g$ in x (rather than y) and using the symmetry $G(x, y) = G(y, x)$, one finds $-\Delta_{g,x} \gamma(x) = -2\Delta_{g,y} \gamma_0(x, y)|_{y=x}$ minus a cross term that vanishes because γ_0 is symmetric; the standard computation (?, Proposition 2.1) gives

$$-\Delta_g \gamma(x) = K(x) - \bar{K}, \quad \bar{K} = 2\pi \chi(\Sigma)/A. \quad (3.14)$$

The constant $\bar{K}$ is fixed by Gauss–Bonnet: integrating (??), $\int (-\Delta_g \gamma) dA_g = 0$ (since $\Delta_g \gamma$ is the Laplacian of a smooth function on a closed manifold), so $\int K dA = \bar{K} \cdot A = 2\pi \chi$. $\square$

As a quantitative check on the second variation, the ratio

$$\frac{(N-1) \sigma_2}{32 \alpha \sigma_1} \geq 7.5 \quad \text{for all } N \geq 7, \quad (3.15)$$

where $\sigma_k = \sum_{m \neq m^*} \lambda_m^{-k}$ and $\alpha = 1/48$ is the second Seeley–DeWitt coefficient, confirms that the concavity term dominates the curvature-nonlinearity correction by at least 7.5 : 1 (see Code Availability).

The extension from conical deficits to smooth metrics is established by Lemma ??.

Partition function interpretation. The total polygon partition function $Z[g] = \sum_N e^{-b(N)} Z_{\text{frozen}}(N, g)$ is a functional of the metric g through $Z_{\text{frozen}} = \prod_m |\lambda_m(g)|^{-1/2}$ (Proposition ??). By the vortex-gravity isomorphism, this is the *gravitational* one-loop partition function summed over polygon sectors: $Z[g] = Z_{\text{grav}}[g]$, with $e^{-b(N)}$ playing the role of $e^{-S_{\text{EH}}}$ (the classical Euclidean action of the N -cone geometry). The saddle-point condition $\delta(-\log Z)/\delta g = 0$ is therefore the Einstein equation, specifically, it is the Euler–Lagrange equation for the combined classical action $b(N)$ plus the one-loop correction $-\log Z_{\text{frozen}}$. This identification gives the WDW equation (§??) its physical meaning: it is the *quantum mechanics of the gravitational partition function*. The WKB tunneling action $S_{\text{BO}}(N)$ is the Euclidean action of the dominant instanton connecting the two turning points of $V(\rho)$, and $\exp(-S_{\text{BO}})$ is the tunneling amplitude between the Planck-scale and cosmological-scale vacua. The hierarchy $\ell \cdot v = \exp(S_{\text{BO}})$ (Paper IV, §14) is a consequence of the partition function interpretation: the cosmological scale ℓ is set by the instanton suppression of the polygon partition function.

The explicit $\delta S/\delta g$ computation (Proposition ??) gives a direct proof that $R = \text{const}$, independent of the Jensen argument.

Higher dimensions. The above arguments are dimension-independent; the extension to spacetime dimension $d_{\text{st}} \geq 3$ is detailed in Remark ??. In $d_{\text{st}} = 3$, $R = \text{const}$ suffices (zero Weyl components). In $d_{\text{st}} \geq 4$, the full Riemann tensor (including all Weyl components) is determined by N via the KK decomposition, Lichnerowicz spectrum, and Fefferman–Graham reconstruction (Theorem ??). $\square$

3.3 Newton's constant from the polygon central charge

We give two derivations of Newton's constant. The first (Proposition ??) is algebraic and parameter-free: G is determined by the polygon data alone via the central charge $c = 12b(N)$ and the Brown–Henneaux relation. The second (Proposition ??) is the leading-order Hadamard match, which defines a probe polygon size ε^2 as a consistency check.

Proposition 3.8 (G from the polygon central charge). *Newton's constant in the 2+1D polygon-gravity correspondence is uniquely determined by the polygon number N :*

$$G = \frac{\ell}{8b(N)} \quad \text{where} \quad b(N) = \frac{N(N+1)}{12} - \ln 2 + \frac{\ln N}{N-1}. \quad (3.16)$$

There are no free parameters: ℓ is the AdS curvature radius and $b(N)$ is the polygon self-energy constant, a pure polygon-theoretic invariant.

Proof. The derivation is an algebraic chain of three theorems.

Step 1: $b(N)$ is a polygon invariant. The Gauss product identity $\prod_{m=1}^{N-1} \sin(\pi m/N) = N/2^{N-1}$ (Paper I, Theorem ??) gives the regularized log-sum

$$\frac{1}{N-1} \sum_{m=1}^{N-1} \ln \sin(\pi m/N) = \frac{\ln N}{N-1} - \ln 2. \quad (3.17)$$

Combined with the finite Basel identity $\sum_{m=1}^{N-1} \csc^2(\pi m/N) = (N^2-1)/3$ (Paper I, Theorem ??) and the geometric prefactor $N(N+1)/12$ from the Bernoulli backbone (Paper I, §??), $b(N)$ is the *regularized self-energy per pair of vortices* of the N -gon on $\mathbf{H}^2$, explicitly $b(N) = N(N+1)/12 - \ln 2 + \ln N/(N-1)$. It is a polygon-theoretic invariant, depending only on N .

Step 2: Central charge $c = 12b(N)$. The Polyakov conformal anomaly of the conical N -gon sphere gives $c = 12b(N)$ (Proposition ??).

Step 3: Brown–Henneaux. For 2+1D AdS gravity with AdS radius ℓ , the boundary CFT has central charge $c = 3\ell/(2G)$ (?).

Combining. $G = 3\ell/(2c) = 3\ell/(24b(N)) = \ell/(8b(N))$. □

Numerical values. At $N = 7$: $b(7) = 4.298$, $G = \ell/34.39 = 0.0291\ell$. At $N = 11$: $b(11) = 10.547$, $G = \ell/84.37 = 0.01185\ell$. Both are tested in `tests/test_newton_from_c.py`.

The polygon entropy functional also yields G at leading order in a small probe polygon expansion, providing an independent consistency check:

Proposition 3.9 (Newton's constant from the polygon entropy). *Let ℓ be the $\mathbf{H}^2$ curvature radius, set by the AdS_3 bulk scale: $K = -1/\ell^2$. Let $\varepsilon := r_E/\ell$ denote the (dimensionless) Poincaré-disk polygon radius in units of ℓ . The polygon entropy functional $S_{\text{poly}}[g] = \int S(x) dA$, expanded to first order in $R\ell^2\varepsilon^2$ (both factors dimensionless; R carries $[\text{length}]^{-2}$ and dA carries $[\text{length}]^2$), gives*

$$S_{\text{poly}} = S_0 \cdot \text{Area} - \frac{(N-1)\varepsilon^2\ell^2}{8} Z_H(1) \int R dA + O((R\ell^2)^2\varepsilon^4), \quad (3.18)$$

where $Z_H(1) = \sum_{\lambda_m > 0} \lambda_m^{-1}$ is the (dimensionless) Havelock spectral zeta function at $s = 1$. This matches the Einstein–Hilbert action $S_{\text{EH}} = -(16\pi G)^{-1} \int R dA$ with Newton’s constant in $2+1D$ (dimension [length])

$$G = \frac{1}{2\pi(N-1) Z_H(1) \varepsilon^2 \ell^{-1}}. \quad (3.19)$$

At $N = 7$: $(N-1) Z_H(1) = 16 = (N^2-1)/3$ (Remark ??), giving the closed form $G = 3\ell/((N^2-1) \cdot 2\pi\varepsilon^2)$.

Proof. Expanding $\lambda_m = \lambda_m^{(0)} - (N-1)R\ell^2\varepsilon^2/4 + O((R\ell^2)^2\varepsilon^4)$ (dimensionless in both factors) in the entropy $S(x) = \frac{1}{2} \sum_{\lambda_m > 0} \log \lambda_m$ gives

$$S(x) = S_0 - \frac{(N-1)\ell^2\varepsilon^2}{8} R(x) \sum_{\lambda_m > 0} \frac{1}{\lambda_m^{(0)}} + O((R\ell^2)^2\varepsilon^4).$$

Integrating and matching $-\alpha\ell^2 \int R dA = -(16\pi G)^{-1} \int R dA$ (both sides dimensionless, $\int R dA$ dimensionless) gives $G = \ell/(16\pi\alpha\varepsilon^2 \ell^2 \cdot \ell^{-1}) = \ell/(16\pi\alpha\varepsilon^2)$ with $\alpha = (N-1)Z_H(1)/8$.

At $N = 7$: the binding modes ($\lambda > 0$) are $m \in \{1, 2, 5, 6\}$ with $\lambda = 3, 1, 1, 3$, so $Z_H(1) = \frac{8}{3}$. Then $(N-1) Z_H(1) = 6 \cdot \frac{8}{3} = 16 = (N^2-1)/3$, giving $G = 3\ell/((N^2-1) \cdot 2\pi\varepsilon^2) = \ell/(32\pi\varepsilon^2)$. $\square$

Remark 3.5 (The identity $(N-1) Z_H(1) = (N^2-1)/3$ at $N = 7$). *The identity $(N-1) Z_H(1) = (N^2-1)/3$ is not a general identity: it holds at $N = 7$ because $6 \cdot \frac{8}{3} = 16 = 48/3$, but fails for all other N . (For $N = 5$: $(N-1) Z_H(1) = 4 \cdot 3 = 12 \neq 8 = 24/3$.) The coincidence at $N = 7$ arises because the finite Basel identity $\sum_{m=1}^{N-1} \csc^2(\pi m/N) = (N^2-1)/3$ (Paper I, Theorem ??) and the Havelock spectral zeta $Z_H(1) = \sum 1/\lambda_m$ involve different partial sums of the same $\csc^2$ kernel, and their ratio simplifies only when the stability threshold is exactly saturated ($\lambda_{m^*} = 0$).*

Remark 3.6 (Connection to α_s and the Bernoulli backbone). *The quantity $(N^2-1)/3 = 16$ is the same $\csc^2$ sum $\sum_{m=1}^{N-1} \csc^2(\pi m/N)$ that determines the strong coupling constant α_s in Paper IV. Newton’s constant G and α_s are both controlled by the same Bernoulli-backbone quantity: the finite- N Basel identity $\sum \csc^2(\pi k/N) = (N^2-1)/3$ (Paper I, Theorem ??). This is a structural prediction: the ratio $G\alpha_s$ is determined by ε alone, with no dependence on the Havelock eigenvalues.*

Remark 3.7 (Consistency check: Brown–Henneaux). *The Brown–Henneaux formula gives $G_{\text{BH}} = 3\ell/(2c)$ (dimension [length]) with $c = 12b(N) = 51.57$. Matching $G_{\text{BH}} = G_{\text{polygon}}$ at leading order in the probe polygon radius gives (both sides being multiples of ℓ) the dimensionless condition*

$$\varepsilon_{\text{BH}}^2 = \frac{c}{\pi(N^2-1)} \approx 0.342 \quad (\text{at } N = 7). \quad (3.20)$$

Here $\varepsilon_{\text{BH}} = r_E/\ell$ is the size of a local probe polygon used to extract the Einstein–Hilbert coefficient by the Hadamard expansion (Proposition ??); it is not the polygon equilibrium radius. The Born–Oppenheimer equilibrium radius $\rho^* = 1.734$ (Poincaré coordinate $\xi^* = 0.490$) is a separate physical object: the location where the radial breathing-mode potential $V_7(\rho) = \ln(2 \sinh \rho) + b(7) - 6$ vanishes (computed from $N = 7$ alone).

Verification via exact Riemannian Havelock. Expanding $C_1(\xi) = (N-1)(1+\xi^2)/(1-\xi)^2 = (N-1)[1+2\xi+3\xi^2+\dots]$, the coefficient of 2ξ matches the Hadamard leading term

$(N-1)\varepsilon_{\text{BH}}^2/2$ iff $\varepsilon_{\text{BH}}^2 = 4\xi + O(\xi^2)$ at small ξ . This is the leading-order conversion between the probe parameter $\varepsilon_{\text{BH}}^2$ (in the BH matching condition) and the Poincaré-disk coordinate ξ : valid for $\xi \ll 1$, it breaks down at finite $\xi^* = 0.490$ where higher-order Hadamard corrections $3\xi^2, 4\xi^3, \dots$ enter. Direct numerical computation of dS_{poly}/dR from the exact $C_1(\xi)$ at ξ^* (rather than the Hadamard leading term) gives $G_{\text{exact}}(\xi^*) \approx 0.0117 \ell$, differing from $G_{\text{BH}} = 3\ell/(2c) \approx 0.0291 \ell$ by a factor of 2.5. This discrepancy reflects the fact that the polygon formula for G is a small-probe expression (the coefficient of the Einstein–Hilbert term in the Hadamard-expansion of S_{poly} at a local probe of size ε_{BH}); it is not a direct function of the BO equilibrium radius.

What is actually derived. The Brown–Henneaux formula $G = 3\ell/(2c)$ and the polygon Hadamard formula $G = \ell/[2\pi(N-1)Z_H(1)\varepsilon_{\text{BH}}^2]$ agree at leading Hadamard order iff the probe size satisfies (??). This agreement is a consistency check: the polygon framework is compatible with BH gravity at the leading-coefficient level, with the probe size selected by the matching condition. The full finite- ξ derivation of G from the polygon functional (summing all Hadamard orders) would require additional work and is not undertaken here; the leading-order agreement suffices for the structural claim that the two frameworks describe the same gravitational sector.

3.4 The cosmological constant and Clausius relation

Remark 3.8 (The logical structure of (H2)). *Three points merit emphasis.*

1. Lie algebra, not relaxation. *The derivation of $R = \text{const}$ is algebraic: the $\mathfrak{sl}(2, \mathbb{R})$ Casimir forces polynomial eigenvalues $\lambda_m = C_1 - m(N-m)/2$ (Helgason symmetric space theorem); strict concavity of $S = \frac{1}{2} \sum \log \lambda_m$ in C_1 is a finite- N identity; affinity of C_1 in R follows from the Hadamard parametrix; Jensen’s inequality under Gauss–Bonnet gives $C_1 = \text{const}$; the Robin function identity $\Delta\gamma = K - \bar{K}$ promotes this to $K = \text{const}$ exactly. The Onsager principle provides the physical interpretation (negative-temperature equilibrium selects the entropy maximum) but the mathematical conclusion requires only (H1)–(H3) and Jensen. The OPS spectral determinant theorem (?) confirms the result in the smooth limit ($N \rightarrow \infty$, $S_{\text{poly}} \rightarrow \log \det' \Delta$) but is a cross-check, not the foundation.*
2. Two variational principles. *(H2a) characterizes the entropy functional at fixed geometry; (H2b) selects the entropy-maximizing metric via Jensen’s inequality and the Robin function identity. Neither implies the other.*
3. The BO bridge. *At fixed g , the angular modes determine $S(x, g)$; varying g varies $S_{\text{poly}}[g] = \int S(x, g) dA$. The BO effective potential $V_{\text{BO}} = V_0 - S$ connects the two: maximizing S at fixed energy is equivalent to minimizing V_{BO} .*

Proposition 3.10 (The cosmological constant from the WDW equation). *The value of Λ in Theorem ?? is determined by the Wheeler–DeWitt equation for the breathing mode ρ :*

$$\left[-\frac{1}{2c} \frac{d^2}{d\rho^2} + V(\rho) \right] \Psi = 0, \quad V(\rho) = \Phi(\rho) - f(m^*, N).$$

Two independent ingredients determine this equation:

The mechanism (*why $\dot{\rho}^2$ appears*). The Kirchhoff equations are first-order; the breathing mode ρ is a collective coordinate with no canonical momentum. The second-order

kinetic term is geometric, not perturbative: the functional determinant of the first-order angular-mode operator $\partial_\tau + \lambda_m(\rho(\tau))$ depends only on $\int \lambda_m d\tau$ (no $\dot{\rho}$ dependence), so the $\dot{\rho}^2$ term does not arise from the action. It arises from the Feynman–Kac path-integral measure $\mu[\rho] = \prod_\tau \sqrt{c/(2\pi d\tau)}$, whose coefficient c is the Fisher–Rao information metric on the family of angular-mode Boltzmann distributions parametrized by ρ . The coupling is mode-independent ($\Phi' = \coth \rho$ for all m), and the CMS integrability (exact action-angle decoupling) ensures the one-loop effective action is exact at $O(c^0)$.

The coefficient c (??). The WDW kinetic coefficient is the mode-independent offset $b(N)$ from the eigenvalue decomposition (Proposition ??), not the CL friction $\gamma \sim O(N)$. It is uniquely fixed by the following argument. The CMS integrability gives the exact one-loop effective action $S_{\text{eff}}(\rho) = -\frac{1}{2} \sum_m \log |\lambda_m(\rho)|$. The Feynman–Kac theorem maps the classical partition function $Z = \int e^{-\beta S_{\text{eff}}} d\rho$ to the quantum propagator $\langle \rho | e^{-\beta H} | \rho \rangle$ with $H = -(1/2c) d^2/d\rho^2 + V(\rho)$. The kinetic coefficient c is the inverse diffusion constant of the breathing mode in the angular bath. For the CMS system, the action variables I_m are exact constants of motion, so the BO adiabatic separation is exact: the Havelock eigenvectors $e_m = N^{-1/2} (\omega^{mp})_{p=0}^{N-1}$ with $\omega = e^{2\pi i/N}$ are Fourier modes, independent of ρ , so non-adiabatic corrections vanish identically. The effective potential decomposes as $V(\rho) = -(1/2) \sum_m \log |C_1(\rho) - f(m, N)|$; writing $C_1 = b(N) + [C_1(\rho) - b(N)]$, the coefficient $b(N)$ is the unique value at which the linear response $\partial V / \partial C_1 = -(1/2) \sum 1/\lambda_m$ is mode-independent (because $b(N)$ is the average of $\lambda_m + f(m, N)$ over m , and the mode sum has a single C_1 -dependent prefactor). Any other kinetic coefficient $c' \neq 12b(N)$ would give a WDW potential that depends separately on each mode m , breaking the BO factorization. (The classical BO separation is exact for any c because the Havelock eigenvectors are ρ -independent; the constraint is on the quantum BO factorization, which requires c constant for the reasons in §??, step 1. The value $c = 12b(N)$ is fixed by the Havelock eigenvalue decomposition (Proposition ??): the Polyakov conformal anomaly coefficient of the conical N -gon sphere is $c = 12b(N)$ (Proposition ??), where the $\ln \sin$ terms arise from the Hurwitz zeta at m/N via the Euler reflection formula, and the Casimir average contributes $N(N+1)/12$. Cross-checks: Brown–Henneaux gives $c = 3\ell/(2G)$ (so $G = \ell/(8b(N))$); $\varepsilon^2/\ell = 0.342$ is fixed by Proposition ??). In the classical limit $c \rightarrow \infty$: $V(\rho^*) = 0$, giving $\Phi(\rho^*) = f(m^*, N)$ and hence the 2D scalar-curvature cosmological constant of the base $\mathbf{H}^2$ at the Onsager equilibrium is $\Lambda_{\text{base}} = -4[f(m^*, N) - (N-1)]/[(N-1)\varepsilon^2]$. This is the 2D curvature sourced by the polygon entropy maximization; it is distinct from the 3D effective cosmological constant $\Lambda_3 = (N^2-16)/16$ obtained by Kaluza–Klein reduction of the Seifert manifold (Paper VI, Proposition 2.1), which includes the quantized flux $|F|^2/4 = N^2/16$ through the S^1 fiber on top of the base curvature $K = -1$.

Corollary 3.11 (Sign of the base-curvature cosmological constant from the polygon number). The sign of Λ_{base} (the 2D scalar-curvature constant of the $\mathbf{H}^2$ base) is fixed by the polygon: $N \leq 6$: $f(m^*, N) < N-1$, so $\Lambda_{\text{base}} > 0$ (positive curvature, S^2 -like, surface curvature destabilizes); $N = 7$: $f(3, 7) = 6 = N-1$, so $\Lambda_{\text{base}} = 0$ (flat); $N \geq 8$: $f(m^*, N) > N-1$, so $\Lambda_{\text{base}} < 0$ (negative curvature, $\mathbf{H}^2$ -like, surface curvature stabilizes). The flat threshold $N = 7$ (Havelock’s 1931 stability boundary) sits exactly at $\Lambda_{\text{base}} = 0$; the 3D KK-reduced $\Lambda_3 = (N^2-16)/16$ (Paper VI) is a different quantity and non-zero at $N = 7$.

Remark 3.9 (Relation to Schur’s lemma). In 2+1D, the higher-dimensional extension of the proof is a special case of Schur’s lemma (isotropic sectional curvature implies constant curvature). The added value of the Havelock derivation is fourfold: (a) the full argument

is new: polygon stability implies isotropic Green's function, which Schur alone cannot provide; (b) the curvature value Λ is quantized by the polygon number N through the explicit formula $\Lambda = -4[f(m^*, N) - (N-1)]/[(N-1)\varepsilon^2]$; (c) the hypothesis of Schur's lemma (isotropy) is derived from the Hadamard universality of the Green's function, not assumed; (d) in $d \geq 4$, the Havelock condition (H1) provides a testable criterion (mode-independence of C_1) equivalent to the full Weyl-vanishing condition of Schur's lemma, connecting the abstract curvature condition to the observable stability threshold.

3.5 Higher-dimensional extension and verification

Remark 3.10 (Characterization theorem in $d \geq 3$). *The core argument is dimension-independent: the csc^2 uniqueness forces a logarithmic interaction in any d (Lemma ??); the Hadamard parametrix $G(x, y) = U/\sigma^{(d-2)/2} + V \log \sigma + W$ contains $R(x)$ at order ε^2 in every d ; and (H3) forces $R = \text{const}$ (Jensen + Onsager). The conclusion closes differently by dimension:*

Dimension	Weyl components	Status
$d = 2$ ($2+1D$)	0	Theorem: (H1) proved by the Havelock identity
$d = 3$ ($3+1D$)	10	Theorem: KK + Lichnerowicz + FG (Theorem ??)
$d \geq 4$	≥ 35	Same, with more traceless Ricci components

In $d = 2$: the Havelock identity proves (H1) analytically, no assumption is needed. In $d \geq 3$: condition (H1) (mode-independence of C_1) is equivalent to the spatial Green's function being isotropic, which is equivalent to maximal symmetry, which is equivalent to the vacuum Einstein equations with Λ . The soft Havelock decomposition (Proposition ??) shows that (H1) is approximately satisfied on any background, with residual $|R_m| < a|f_m|$ (verified for $N = 5, \dots, 20$ and $d = 3, 4, 5$): the violation of (H1) is controlled and measures the departure from Einstein geometry.

Derivation 3.12 (Full Riemann curvature from the polygon number). *On the Seifert manifold $\mathbf{H}^2 \times_N S^1$ with stabilized fiber ($\sigma = 1$) and quantized flux ($F = N/(2\pi)$), the full 4D Riemann tensor $R_{MNPQ}^{(4)}$, including all 10 Weyl components, is uniquely determined by N .*

Proof. Three layers: background, linearized, nonlinear.

Layer 1 (Background). The Weyl tensor vanishes in 3D, so the Riemann tensor of the Seifert manifold is entirely determined by its Ricci tensor $R_{ij}^{(3)} = \Lambda_3 g_{ij}$ (Theorem ??):

$$R_{ijkl}^{(3)} = \frac{\Lambda_3}{2}(g_{ik}g_{jl} - g_{il}g_{jk}). \quad (3.21)$$

The O'Neill formula for the Riemannian submersion $\pi: \mathbf{H}^2 \times_N S^1 \rightarrow \mathbf{H}^2$ (??, Thm. 9.28) decomposes $R_{ijkl}^{(3)}$ into base curvature + terms quadratic in the connection curvature F_{ij} ; the fiber components satisfy $R_{\varphi_i \varphi_j}^{(3)} = \frac{1}{4}F_{ik}F_j^k$. The flux is uniform (F_{ij} determined by the Chern class $c_1 = N$; $\nabla_{[i}F_{jk]} = 0$ on the constant-curvature base), and the fiber radius is stabilized ($\sigma = 1$, $V''(1) = N^2/4 > 0$). The KK lift to 4D (Paper IV, Step B) adds the time direction; the resulting $R_{MNPQ}^{(4)}$ inherits the same inputs ($R^{(3)}, F, \sigma$), each an explicit function of N . Therefore all 20 components of $R_{MNPQ}^{(4)}$ are determined by N .

Layer 2 (Linearized perturbations). A perturbation h_{MN} decomposes under KK into h_{ij} (metric), δA_i (graviphoton), $\delta\sigma$ (radion). The Lichnerowicz operator Δ_L on the multi-cone geometry (with distributional curvature $R = R_0 + \sum_k \delta_k \delta_{z_k}$ regularized by the Cauchy blob at scale ε ; Paper II, Proposition 7.1) reduces to the Havelock Hessian as follows. The $\mathbb{Z}_N$ rotation symmetry of the regular polygon acts as an isometry of the multi-cone metric. Since Δ_L is a geometric operator, it commutes with this $\mathbb{Z}_N$ action and decomposes into irreducible representations labelled by $m = 0, \dots, N-1$ (Schur's lemma). At each cone point, the Green's function has the angular kernel $-\log|2\sin(\pi p/N)|$ (the logarithmic singularity of the Hadamard parametrix). The angular Hessian weight is $1/(4\sin^2(\pi p/N))$, so the m -th $\mathbb{Z}_N$ Fourier eigenvalue of the Hessian is $\sum_{p=1}^{N-1} (1 - \cos(2\pi pm/N)) / (2\sin^2(\pi p/N)) = m(N-m)/2$ (Paper I, Lemma ??). The full eigenvalue is $\lambda_m = C_1 - m(N-m)/2$ (the mode-independent offset C_1 absorbs the csc^2 diagonal sum and the radial contribution). The spectrum $\{\lambda_m\}$ is therefore determined by N . The graviphoton satisfies the Hodge Laplacian on $\mathbf{H}^2/\mathbb{Z}_N$ (spectral gap ≥ 1). The radion is massive ($m^2 = N^2/4$, no zero mode). All perturbation sectors are gapped and determined by N , so the linearized Weyl tensor δC_{MNPQ} is determined mode by mode.

Layer 3 (Full nonlinear). The Fefferman–Graham expansion in AdS_3 terminates at finite order for $d_\partial = 2$ (??): the boundary stress tensor T_{ab} (determined by the Virasoro algebra at $c = 12b(N)$) uniquely reconstructs the bulk 3D metric. The boundary has $\mathbb{Z}_N$ orbifold singularities (the polygon cone points), but the *bulk* geometry is smooth: the FG expansion is well-defined in the distributional sense, with g_2 acquiring delta-function contributions at the cone points. The δ^2 issue does not arise because the FG expansion terminates at order z^2 for boundary dimension 2 (?; the g_4 coefficient, which would contain the δ^2 terms, appears only at order z^4 and is absent). The $\mathbb{Z}_N$ orbit-averaged stress tensor $\langle T_{ab} \rangle$ is smooth: the N delta-function contributions at the equally-spaced cone points sum to a constant by the $\mathbb{Z}_N$ Fourier projection $\sum_{k=0}^{N-1} \delta(z - z_k) = N/(4\pi(g-1)) + O(m \geq 1)$, with all $m \geq 1$ modes entering only linearly). The orbifold boundary is a Riemannian manifold with isolated cone singularities of angle $2\pi/N < 2\pi$, so the bulk AdS_3 is an asymptotically hyperbolic manifold with a cone-singular conformal infinity. The ? edge pseudodifferential calculus extends the standard 0-calculus of asymptotically hyperbolic manifolds to allow conic singularities on the boundary; in this calculus the Laplacian $\Delta_g - s(2-s)$ is a fully elliptic edge operator for all cone angles in $(0, 2\pi)$, and its resolvent $(\Delta_g - s(2-s))^{-1}$ has a meromorphic polyhomogeneous extension to $s \in \mathbb{C}$ (? , Thm. 7.14; ? , Thm. 1.1 for the asymptotically hyperbolic setting). The scattering matrix $S(s)$, defined as the residue $\lim_{z \rightarrow 0} z^{s-2} R(s)(z^{2-s} \cdot)$ (? , eq. (1.2)), inherits the polyhomogeneity and extends meromorphically in the edge-calculus sense; the classical ? interior cone-edge analysis reduces to a special case on the boundary. The FG coefficients g_0, g_2 are read off from the Laurent expansion of $S(s)$ at $s = 2$ (? , Prop. 3.5, with the Mazzeo edge-pseudodifferential extension replacing the smooth-boundary 0-calculus), and the construction is valid for any boundary metric admitting a meromorphic resolvent — which Cheeger–Taylor guarantees for cone angles $< 2\pi$. The KK lift with stabilized fiber and quantized flux then gives the unique 4D metric $g_{MN}^{(4)}$, and hence the unique 4D Riemann tensor. $\square$

Supporting evidence: spectral focusing. By Theorem ?? and Corollary ??, $N_{\text{crit}}(\Delta)$ is non-increasing: once a polygon type becomes unstable, it remains unstable for all larger Δ .

The monotonicity is formally analogous to Raychaudhuri focusing (structural parallel):

Concept	General relativity	Polygon system
Flow parameter	Affine parameter λ	Conformal dim Δ
Expansion	$\theta = \nabla \cdot k$ (decreasing)	$N_{\text{crit}}(\Delta)$ (decreasing)
Focusing	$d\theta/d\lambda \leq 0$	$dN_{\text{crit}}/d\Delta \leq 0$ (monotone)
Trapped	$\theta < 0$ (trapped surface)	$\lambda_m < 0$ (unstable mode)
Energy condition	$R_{\mu\nu}k^\mu k^\nu \geq 0$	$C_1(\xi) > 0$ (spectral gap)

The palindromic threshold as a Killing horizon. The WDW equation $[-1/(2c)\Psi'' + V(\rho)\Psi = 0]$ is the zero-energy Schrödinger equation in the potential V . The turning point $V(\rho^*) = 0$ is a *Killing horizon* in the minisuperspace geometry: the Euclidean continuation around ρ^* requires periodicity $\beta = 2\pi/|V'(\rho^*)| = 2\pi/\coth(\rho^*)$ to avoid a conical singularity (?). This is the same calculation that gives the de Sitter temperature $T = H/(2\pi)$, with the breathing mode ρ playing the role of the scale factor a . The Gibbons–Hawking temperature at the palindromic threshold is

$$T = \frac{V'(\rho^*)}{2\pi} = \frac{\coth(\rho^*)}{2\pi}. \quad (3.22)$$

Logical role of the Clausius relation. The vacuum Einstein equations $R = \text{const}$ are derived *independently* from entropy maximization (Theorem ??). The Clausius relation at the Gibbons–Hawking horizon serves a *different* purpose: it extracts the gravitational coupling G_{eff} from the constant-curvature geometry. On a constant-curvature surface, there is one threshold (one N) and one temperature T . The Jacobson argument (??) then promotes G_{eff} to the matter coupling: $\delta Q = T\delta S$ at the Killing horizon gives the full Einstein equations $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G_{\text{eff}} T_{\mu\nu}$.

The three thermodynamic quantities are determined by the BO potential $V(\rho) = \Phi(\rho) - f(m^*, N)$:

- $T = V'(\rho^*)/(2\pi) = \coth(\rho^*)/(2\pi)$ (Unruh temperature from the surface gravity $\kappa = V'$);
- $\delta Q = \kappa \times A \times \delta\rho = \coth(\rho^*) \times 2\pi \sinh(\rho^*) \times \delta\rho$ (surface gravity $\times$ horizon circumference $\times$ displacement);
- $\delta S = (dS/d\rho) \times A \times \delta\rho$ where $A = 2\pi \sinh(\rho^*)$ is the horizon circumference and $dS/d\rho$ is the Bekenstein–Hawking entropy production rate.

Which entropy. The one-loop entropy $S' = -\frac{1}{2} \log |\det' H|$ (reduced determinant, zero modes excluded) has a well-defined derivative at ρ^* :

$$\left. \frac{dS'}{d\rho} \right|_{\rho^*} = -\frac{\coth(\rho^*)}{2} \sigma(N), \quad \sigma(N) = \sum_{\substack{m=1 \\ f_m \neq f_{m^*}}}^{N-1} \frac{1}{f_{m^*} - f_m}, \quad (3.23)$$

where the sum runs over non-zero modes ($\sigma(7) = 8/3$, $\sigma(8) = 49/9$, verified $N = 7, \dots, 15$; see Code Availability). The zero mode's crossing contributes to the spectral flow (APS eta-invariant shift $= \pm 1$ per palindromic pair), not to $|\det' H|$.

Clausius relation via Cardy + RT. The Bekenstein–Hawking entropy $S_{\text{BH}} = A/(4G)$ presupposes G , so using it in the Clausius relation gives a tautology. A non-trivial determination uses the *Cardy thermal entropy* of the boundary CFT, which depends on c but not on G :

$$S_{\text{Cardy}} = \frac{\pi c}{3} T L = \frac{\pi c}{3} \cosh(\rho^*), \quad (3.24)$$

where $T = \coth(\rho^*)/(2\pi)$ is the Unruh temperature, $L = 2\pi \sinh(\rho^*)$ is the ring circumference, and c (??) is computed from the spectrum (Paper I). The Ryu–Takayanagi formula $S = A/(4G)$ with $A = 2\pi \sinh(\rho^*)$ then determines G :

$$G_{\text{RT}} = \frac{A}{4 S_{\text{Cardy}}} = \frac{3 \tanh(\rho^*)}{2 c}. \quad (3.25)$$

Newton’s constant is $G_{\text{BH}} = 3/(2c)$ (Brown–Henneaux). The $\tanh(\rho^*)$ factor is a finite- ρ^* correction to the *Cardy entropy* (not to G itself): the Cardy formula gives $S_{\text{Cardy}} = (\pi c/3) \cosh \rho^*$, while the area law uses $A = 2\pi \sinh \rho^*$; the ratio $A/(4G_{\text{BH}} S_{\text{Cardy}}) = \tanh \rho^*$ measures how far the finite polygon is from the asymptotic ($\rho^* \rightarrow \infty$) BTZ regime. The correction $\tanh(\rho^*) = 1 - 2e^{-2\rho^*} + O(e^{-4\rho^*})$ vanishes exponentially: 6% at $N = 7$, 0.03% at $N = 11$, $< 10^{-7}$ for $N \geq 15$ (verified numerically for $N = 7, \dots, 20$; see Code Availability). The common factor $\coth(\rho^*) = V'(\rho^*)$ is not a free choice: it is determined by the csc^2 kernel ($V = \log(2 \sinh \rho)$ is the $\mathbf{H}^2$ Green’s function). The one-loop spectral sum (??) provides an N -dependent quantum correction beyond the Cardy entropy; incorporating it would improve the finite- ρ^* convergence. The ? entanglement argument shows that positivity of relative entropy $S(\rho \parallel \sigma) \geq 0$ in the unitary boundary CFT implies $\delta Q = T \delta S$ at every Rindler horizon. The Clausius relation is verified numerically at all palindromic thresholds $N = 7, \dots, 17$ (see Code Availability). The extension to the full matter-coupled Einstein equations $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}$ requires the Jacobson thermodynamic argument (?: ?): $\delta Q = T \delta S$ at every local Rindler horizon implies Einstein with matter. The identification of palindromic thresholds as Rindler horizons uses the modular Hamiltonian of the angular bath, which equals the Rindler energy by the Bisognano–Wichmann theorem. The prerequisites are satisfied: the boundary CFT is (i) unitary (DHVW orbifold of a unitary WZW model, Remark ??), (ii) has a unique vacuum (the identity operator is $\mathbb{Z}_N$ -invariant), and (iii) the modular flow is geometric (the boundary T^2 is conformally flat). See Paper IV, Theorem 14.3).

Proposition 3.13 (Partition function verification via the Gauss product). *The polygon partition function $Z_{\text{poly}} = \sum_N e^{-b(N)} Z_{\text{frozen}}(N)$ equals the Chern–Simons partition function Z_{CS} on the Seifert manifold, with three independent mathematical inputs:*

1. Gauss product (algebra, Paper I): $b(N) = N(N+1)/12 + \frac{1}{N-1} \sum_{m=1}^{N-1} \ln \sin(\pi m/N)$. *The transcendental part of $b(N)$ is the mean log-sine over the $N-1$ pairwise chord distances, the regularized self-energy from the identity $\prod_{m=1}^{N-1} \sin(\pi m/N) = N/2^{N-1}$.*
2. Cheeger–Müller theorem (differential geometry): *the analytic torsion $\tau_{\text{an}} = \prod |\lambda_m|$ (spectral determinant of the gauge-covariant Laplacian) equals the Reidemeister torsion τ_R (combinatorial invariant of the Seifert manifold). For the N -gon flat connection: $Z_{\text{frozen}}(N) = \tau_{\text{an}}^{-1/2}$ is the standard one-loop normalization.*
3. CMS–CS Casimir identification (Proposition ??): *the vortex Hessian eigenvalues $\lambda_m = f(m^*, N) - f(m, N)$ are the eigenvalues of the cone Laplacian at the $\mathbb{Z}_N$ -symmetric point (proved via the Helgason symmetric space theorem, §??).*

The same Gauss product $\prod \sin(\pi m/N) = N/2^{N-1}$ appears on both sides:

- Vortex side: it regularises the self-energy, giving the transcendental part of $b(N)$;
- Spectral side: it determines the cone spectral zeta function $\zeta'_{\text{cone}}(0)$, giving the analytic torsion.

They agree because both compute the regularized determinant of the same $\mathbb{Z}_N$ -circulant csc^2 matrix, the vortex side via the Havelock decomposition (Paper I), the spectral side via the zeta function (Cheeger–Müller). Points (1) and (2) are mathematical theorems independent of the vortex-gravity correspondence. Point (3) uses the CMS–CS Casimir identification (Proposition ??), which is itself part of the correspondence. The partition function equality $Z_{\text{poly}} = Z_{\text{CS}}$ is therefore a consistency check of the correspondence, not an independent verification: the same Casimir identification enters on both sides. The non-trivial content is that algebraic (Gauss product), analytic (Cheeger–Müller torsion), and representation-theoretic (Casimir) inputs all yield the same numerical answer.

Matter coupling. On a quotient $\Gamma \backslash \mathbf{H}^2$, the Selberg correction δC_1 shifts the threshold $\rho^* \rightarrow \rho_{\text{matter}}^*$, but the Clausius ratio retains the form $4\pi \tanh(\rho_{\text{matter}}^*)$ (verified on the Bolza surface with $|\delta C_1| \leq 0.021$). The ratio 4π is a local geometric identity (the metric is locally $\mathbf{H}^2$), giving $G_{\text{eff}} = 1/(16\pi)$ in natural units ($8\pi G_{\text{eff}} = 1/2$ as $\rho^* \rightarrow \infty$).

4 The spectral bound as a quantum gravitational bound

The spectral correction and the gravitational one-loop partition function are both moments of the heat kernel:

$$\log Z_{\text{grav}} = - \int_0^\infty [\text{Tr } e^{-t\Delta_S} - 1] \frac{dt}{t} \quad (\text{weight } 1/t), \quad (4.1)$$

$$\delta C_1(z_0) = \int_0^\infty [K(z_0, z_0, t) - K_{\mathbf{H}^2}(t)] dt \quad (\text{weight } 1). \quad (4.2)$$

The first is the functional determinant $\det \Delta_S$, the second determines polygon stability. Both involve the Selberg zeta function $Z_\Gamma(s)$ at the zero $s = 1$, but this requires care.

The zero at $s = 1$. The Selberg zeta $Z_\Gamma(s)$ has zeros at s such that $s(1-s)$ is an eigenvalue of Δ_S . The constant eigenfunction ($\lambda_0 = 0$) gives a zero at $s = 1$, so $Z_\Gamma(1) = 0$, not a finite number. The Euler product $\prod_\gamma \prod_k (1 - e^{-(s+k)\ell_\gamma})$ converges for $\text{Re}(s) > 1$ but diverges at $s = 1$ (by the prime geodesic theorem: the partial product over geodesics with $\ell \leq L$ decays to 0 as $L \rightarrow \infty$).

The physically relevant quantities are the *residue* at this zero:

- $Z'_\Gamma(1)$: the *regularized* determinant $\det' \Delta_S = \prod_{n \geq 1} \lambda_n$ (omitting $\lambda_0 = 0$). This is the gravitational one-loop partition function (?).
- The spectral sum $\sum_{n \geq 1} |\varphi_n(z_0)|^2 / \lambda_n$: the regularized Green's function at the diagonal, which determines $\delta C_1(z_0)$.

Both are finite and well-defined; both involve the analytic structure of Z_Γ at its zero, not the (divergent) Euler product.

The spectral computation of δC_1 . At the Aut-fixed center z_0 of the Bolza surface, only trivial-representation eigenfunctions contribute: $\delta C_1(z_0) = \sum_{n:\text{triv}} |\varphi_n(z_0)|^2 / \lambda_n \approx 0.011$, dominated by the first trivial-rep eigenvalue $\lambda_{\text{triv}} \approx 15.05$ (the first eigenvalue in the trivial representation of $\text{Aut}(\text{Bolza})$; the full spectral gap is $\lambda_1 \approx 3.84$; ?), with rapid convergence: $1/(4\pi \cdot 15.05) \approx 0.005$, $1/(4\pi \cdot 23.08) \approx 0.003$, $1/(4\pi \cdot 32.67) \approx 0.002$, The first 8 trivial-representation eigenvalues of ? give $\sum_{n=1}^8 1/(4\pi \lambda_n) = 0.019$. The tail is bounded by Sogge's sup-norm estimate ($|\phi_n(z_0)|^2 \leq C \lambda_n^{1/2}$) combined with Weyl asymptotics ($\lambda_n \geq c n$ for the trivial representation): $\sum_{n>8} 1/(4\pi \lambda_n^{3/2}) < 0.001$. Therefore $|\delta C_1| \leq 0.019 + 0.001 = 0.020$.

The bound $|\delta C_1| \leq 0.020$ on the Bolza surface is conditional on the Strohmaier–Uski spectral data and unconditional with respect to the Selberg conjecture ($\lambda_{\text{triv}} \approx 15.05 \gg 1/4$). In gravitational language, C_1 is the classical stability coefficient and δC_1 is the one-loop quantum correction from the topology of the spatial slice.

The Selberg dictionary. Four physical quantities involve the Selberg zeta at $s = 1$ (structural analogies, not functional inequalities):

Z_Γ data at $s = 1$	Physical meaning
$Z'_\Gamma(1)$ (regularized determinant $\det' \Delta_S$)	Gravitational partition function $= \prod_{n \geq 1} \lambda_n$
$\sum_{n \geq 1} \varphi_n(z_0) ^2 / \lambda_n$ (spectral; convergent)	Vortex stability correction δC_1 Localised at z_0 ; Aut-suppressed
Palindromic threshold polynomials (from $C_1(\xi) = f_{\max}(N)$)	Stability transitions: values of ξ where N_{crit} jumps
Hecke eigenvalues of associated forms (a_p from modular forms)	Prime-by-prime visibility Which thresholds are visible on $\Gamma(p)$

The first two rows extract different data from the same zero $s = 1$: the determinant ($\prod \lambda_n$) versus the resolvent trace ($\sum 1/\lambda_n$). The palindromic thresholds control *where* stability transitions occur (at which curvature ξ); the eigenvalue spectrum controls *how large* the correction δC_1 is. On highly symmetric surfaces (the modular surface, the Bolza surface), the systole and the palindromic geodesics coincide; on a generic arithmetic surface they are distinct.

5 Holographic interpretation

The preceding sections derived gravity from the polygon's statistical mechanics without invoking holography. This section translates the same results into the $\text{AdS}_3/\text{CFT}_2$ language; the translation is not required for any subsequent result, but it connects the polygon framework to the broader gauge/gravity literature and provides independent consistency checks.

In $\text{AdS}_3/\text{CFT}_2$, a point mass m in the bulk creates a conical deficit that corresponds to the insertion of a heavy operator $\mathcal{O}$ of conformal dimension $h = (c/24)(1 - (1 - 8Gm)^2)$ in the boundary CFT (?). A regular N -gon of equal masses at radial coordinate r in the bulk corresponds to N equally-spaced heavy operator insertions on the boundary circle.

In this language, the Havelock decomposition takes the form:

Bulk (2+1D AdS gravity)	Boundary (1+1D CFT)
Perturbation mode m	$\mathbb{Z}_N$ conformal block channel
$f(m, N) = m(N-m)/2$	Casimir of the m -th channel
$C_1(\xi)$	Function of h/c and AdS radius
$\lambda_m = C_1 - f(m, N) > 0$	Positivity of the m -th block
$C_1 \geq f_{\max}(N)$	Bootstrap-type bound on h/c

The stability condition $C_1 \geq m(N-m)/2$ for all m becomes a *lower bound on the conformal dimension* of the boundary operators: below this bound, the bulk polygon is unstable and the corresponding $\mathbb{Z}_N$ conformal block changes sign. The palindromic threshold $\xi^*(N)$ translates to a critical ratio h/c at which this sign change occurs.

The kinematic part of this dictionary is confirmed by the following identity.

Proposition 5.1 (Havelock Casimir = one-chiral OPE Casimir). *Let $T_m = \sum_{p=1}^{N-1} \frac{1 - \cos(2\pi pm/N)}{2 \sin^2(\pi p/N)}$ be the $\mathbb{Z}_N$ Fourier eigenvalue of the OPE kernel $\csc^2(\pi p/N)$. Then*

$$T_m = m(N-m) = 2f(m, N), \quad (5.1)$$

exactly twice the Havelock Casimir $f(m, N) = m(N-m)/2$.

Proof. Write $\omega = e^{2\pi i/N}$ and $S_m = \sum_{p=1}^{N-1} (1 - \omega^{pm})/|1 - \omega^p|^2$. Since $|1 - \omega^p|^2 = 2(1 - \cos(2\pi p/N)) = 4 \sin^2(\pi p/N)$ and $1 - \omega^{pm} = 1 - \cos(2\pi pm/N) - i \sin(2\pi pm/N)$, the real part gives $T_m = \text{Re}(S_m) = \sum_{p=1}^{N-1} [1 - \cos(2\pi pm/N)]/(2 \sin^2(\pi p/N))$. Now consider the polynomial identity on $\mathbb{Z}/N\mathbb{Z}$: the function $f_m(p) = [1 - \cos(2\pi pm/N)]/(2 \sin^2(\pi p/N))$ extends to a trigonometric polynomial of degree $\leq N-1$. Its sum equals the coefficient of z^0 in the Laurent expansion of $\frac{1-z^m}{1-z} \cdot \frac{1-z^{-m}}{1-z^{-1}}$, evaluated at $z = \omega$ and summed over $p = 1, \dots, N-1$. This product is $|\sum_{k=0}^{m-1} z^k|^2$, whose z^0 coefficient (from the orthogonality $\sum_{p=0}^{N-1} \omega^{pm} = N\delta_{n,0}$) is $\min(m, N-m) \cdot \max(m, N-m) = m(N-m)$, since exactly m terms contribute in each sector. $\square$

The factor of 2 has physical content. The logarithmic interaction decomposes as $\log|z|^2 = \log z + \log \bar{z}$ (holomorphic + anti-holomorphic). In the boundary CFT, the OPE of two heavy operators has both chiral sectors, giving the *full* Casimir $m(N-m)$. The Havelock eigenvalue uses $f = m(N-m)/2$, the contribution of *one* chiral sector, because:

- The Kirchhoff equations $\dot{z}_k = -i \partial H / \partial \bar{z}_k$ are first-order: coordinates are conjugate to each other (no separate momenta). This halves the phase-space dimension, selecting one chiral sector of the $\log|z|^2 = \log z + \log \bar{z}$ interaction.
- The stability criterion $\lambda_m > 0$ is $C_1 > m(N-m)/2$: the confining potential must exceed the holomorphic OPE repulsion in the m -th $\mathbb{Z}_N$ channel.

In gravity (second-order dynamics), both sectors contribute to the equations of motion, but the stability criterion, positivity of the Hessian eigenvalue, still involves $m(N-m)/2$ per sector (Proposition 13.1).

The remaining step is to express C_1 in terms of boundary CFT data. On the flat plane, $C_1 = N-1$ (the Havelock self-energy); on $\mathbf{H}^2$ at curvature parameter ξ ,

$$C_1(\xi) = (N-1) \frac{1+\xi^2}{(1-\xi)^2}. \quad (5.2)$$

The curvature parameter ξ encodes the radial position of the bulk polygon; through the Brown–Henneaux relation $c = 3\ell/(2G)$ and the deficit angle–dimension map $h = (c/24)(1 - (1 - 8Gm)^2)$, the stability condition becomes a bound on conformal dimensions parametrized by the bulk position. The palindromic thresholds $\xi^*(N)$ give algebraic-number bulk positions at which the stability bound is saturated:

N	$\xi^*(N)$	d_{crit}/ℓ	$\varepsilon = 1 - \xi^*$	h/c_{max}	Field
8	$8 - 3\sqrt{7}$	0.51	0.937	0.018	$\mathbb{Q}(\sqrt{7})$
10	$1/7$	0.80	0.857	0.015	$\mathbb{Q}$
12	$(18 - 5\sqrt{11})/7$	0.97	0.798	0.013	$\mathbb{Q}(\sqrt{11})$
23	φ^{-2}	1.44	φ^{-1}	0.007	$\mathbb{Q}(\sqrt{5})$

Here d_{crit}/ℓ is the geodesic distance from the AdS center in units of the AdS radius, and $\varepsilon = 1 - \xi^*$ is the boundary proximity (the polygon must sit at $\varepsilon < \varepsilon^*(N)$, i.e., sufficiently close to the AdS boundary, to be stable). The $N = 23$ threshold gives $\varepsilon^* = \varphi^{-1} = (\sqrt{5} - 1)/2$: the golden ratio appears as a critical bulk-position parameter in the holographic dictionary.

The two constraints on the polygon are independent:

- *Mass*: $h/c < h/c_{\text{max}}(N)$ (total deficit $< 2\pi$; from the bulk geometry).
- *Position*: $\xi > \xi^*(N)$ (confining potential $>$ OPE repulsion; from the Havelock decomposition and the chiral identity (??)).

Together, they define the allowed region in the $(h/c, \xi)$ parameter space for a stable N -gon of heavy operators, a two-parameter family of bootstrap-type bounds, one for each N , all descending from the single identity $\lambda_m = C_1 - m(N - m)/2$.

The confining potential from the OPE kernel. The formula $C_1 = (N - 1)(1 + \xi^2)/(1 - \xi)^2$ can be derived either from the Kimura Hamiltonian for point vortices on $\mathbf{H}^2$ (Paper I, §4) or from the second radial variation of the chordal OPE kernel $\sigma(z, w) = |z - w|/|1 - \bar{z}w|$ at the N -gon. The two derivations are mathematically identical: both compute the Hessian of $-\sum_{j < k} \log \sigma(z_j, z_k)$ on the Poincaré disk. The factor $(1 + \xi^2)/(1 - \xi)^2$ is *not* the conformal factor $\Omega(\sqrt{\xi})^2 = 1/(1 - \xi)^2$ alone; the additional $(1 + \xi^2)$ comes from the angular structure of σ under radial perturbations.

Relation to the conformal bootstrap. The stability condition $\lambda_m > 0$ and the conformal bootstrap positivity conditions (crossing symmetry + unitarity, ?) are both positivity conditions on conformal block coefficients, but in *different channels*:

	Bootstrap	Vortex stability
What is constrained	CFT spectrum $\{\Delta_i\}$	Bulk polygon position ξ
Input	c, Δ_{gap}	$N, h/c$
Positivity condition	OPE $f_{ijk}^2 \geq 0$	$\lambda_m = C_1 - f(m, N) > 0$
Channel	generic 4-point OPE	$\mathbb{Z}_N$ -symmetric

The two are *complementary*: the bootstrap constrains which CFTs exist, while the vortex stability constrains which bulk configurations are stable in a given CFT. The identification $\lambda_m \leftrightarrow$ OPE coefficient would make the stability condition a bootstrap condition for $\mathbb{Z}_N$ -symmetric correlators, a direction requiring the $\mathbb{Z}_N$ conformal block computation sketched below.

6 The $\lambda_m \leftrightarrow$ OPE identification

The Havelock eigenvalue is entirely computable from boundary CFT data. The key is to decompose the semiclassical N -point function $\langle \mathcal{O}(z_1) \cdots \mathcal{O}(z_N) \rangle$ of N equally-spaced heavy operators into $\mathbb{Z}_N$ channels: the angular channels give $m(N-m)/2$ (the kinematic OPE Casimir, Proposition ??), and the radial confinement gives $C_1(\xi)$.

In the semiclassical (large- c) limit, the N -point function factorizes as

$$\log \langle \mathcal{O}(z_1) \cdots \mathcal{O}(z_N) \rangle \approx -2h \sum_{j < k} \log \sigma(z_j, z_k) + O(1), \quad (6.1)$$

where $\sigma = |z_j - z_k|/|1 - \bar{z}_j z_k|$ is the chordal distance. The leading term is the “free OPE energy”; the $O(1)$ correction comes from the Virasoro block (graviton exchange in the bulk).

Under *angular* perturbation $z_k \rightarrow R e^{i(2\pi k/N + \delta\theta_k)}$ at the N -gon, the second variation of the leading term decomposes into $\mathbb{Z}_N$ Fourier modes $\delta\theta_k = \sum_m a_m \cos(2\pi m k/N)$:

$$\delta^2 \left[-2h \sum_{j < k} \log \sigma \right] = 2h \sum_{m=1}^{N-1} m(N-m) |a_m|^2. \quad (6.2)$$

Proposition 6.1 (Angular OPE stiffness). *The coefficient of $|a_m|^2$ in (??) is $2h \cdot m(N-m) = 2h \cdot T_m$, where $T_m = m(N-m)$ is the full OPE Casimir (Proposition ??). This follows analytically from Proposition ?? ($T_m = m(N-m)$, proved) and the constrained second variation; verified numerically for $N \leq 12$.*

The missing piece is the confining potential $C_1(\xi)$. We now show it too arises from the boundary.

Theorem 6.2 ($\mathbb{Z}_N$ conformal block = Havelock eigenvalue). *Let $\mathcal{O}_1, \dots, \mathcal{O}_N$ be N identical heavy operators of dimension h , placed at equally-spaced points $z_k = \sqrt{\xi} e^{2\pi i k/N}$ on the boundary of the Poincaré disk. In the semiclassical limit ($c \rightarrow \infty$, h/c fixed):*

- (a) Angular stiffness. *The m -th $\mathbb{Z}_N$ channel of the angular second variation of the N -point function gives $2h \cdot m(N-m)$ (Proposition ??).*
- (b) Radial confinement. *The log-radial ($u = \log \sqrt{\xi}$) second variation of $-\sum_{j < k} \log \sigma(z_j, z_k)$ at the N -gon gives the mode-independent confining potential $C_1(\xi) = (N-1)(1+\xi^2)/(1-\xi)^2$.*
- (c) Full identification. *The constrained second variation (angular + radial at fixed angular impulse), divided by $2h$ per chiral sector, is*

$$\lambda_m = \underbrace{C_1(\xi)}_{\text{radial confinement}} - \underbrace{m(N-m)/2}_{\text{one-chiral OPE repulsion}}. \quad (6.3)$$

This is the Havelock eigenvalue, derived entirely from boundary data.

Proof. Part (a) is Proposition ??. For part (b): the log-radial second variation of the hyperbolic energy gives $C_1(\xi)$ as in (??) (Paper I, equation ??) (the Riemannian Havelock identity on $\mathbf{H}^2$). Part (c) combines (a) and (b) via the standard Lagrange-multiplier argument (Paper I, §3).

Identity block dominance for $N > 4$. The theorem requires that the identity Virasoro block dominates the OPE decomposition in each $\mathbb{Z}_N$ channel. For $N = 4$ this is standard

(the semiclassical 4-point block is the sole saddle on $\mathcal{M}_{0,4} \cong \mathbb{P}^1$). For $N > 4$: the $\mathbb{Z}_N$ -symmetric N -point correlator of identical operators on $\mathbb{P}^1$ (*genus zero*) equals the 1-point function of the twist field σ_m on the orbifold $\mathbb{C}/\mathbb{Z}_N$ (?). The orbifold has twisted-sector vacua $|k\rangle$ with conformal dimensions $h_k = k(N-k)/(2N^2) \times c > 0$ for $k = 1, \dots, N-1$. On genus zero, these contribute to the 1-point function only as virtual intermediate states, suppressed by $O(e^{-2\pi h_k}) = O(e^{-c})$ (exponentially small at large c , since $h_k \propto c$). The identity block gives the leading (classical) contribution $\exp(-c \times S_{\text{conical}})$; the $\mathbb{Z}_N$ symmetry reduces the moduli space $\mathcal{M}_{0,N+1}^{\mathbb{Z}_N}$ to dimension 2 (ring radius and overall phase). On this 2D space, the identity block dominates by two independent bounds:

1. *Eigenvalue bound (Proposition ??)*: the $O(1/c)$ Virasoro corrections satisfy $|\delta f_m/\lambda_m| < 1/4$ for all non-critical m , so no eigenvalue changes sign at finite c and the symmetric saddle remains non-degenerate.
2. *OPE coefficient bound (Liouville action)*: the $\mathbb{Z}_N$ twist field σ_k of order k lives on the orbifold $\mathbb{C}/\mathbb{Z}_N$ with conical metric of opening angle $2\pi/N$. Its OPE coefficient is $C_k^2 = \exp(-S_L(\text{cone}))$ where S_L is the classical Liouville action (??). The Liouville action on a $\mathbb{Z}_N$ cone regularized by the Gauss log-sine product (the same regularization that gives $b(N)$ in the Havelock decomposition, Paper I, §??) is $S_L \geq b(N) \ln N$. Therefore $C_k^2 \leq \exp(-b(N) \ln N)$, and the total non-identity contribution is

$$\frac{\delta F}{F} \leq (N-1) \exp(-b(N) \ln N)$$

(verified numerically for $N = 3, \dots, 16$; see Code Availability): 11% at $N = 5$, 1.7% at $N = 6$, 0.14% at $N = 7$, $\sim 10^{-10}$ for $N = 11$.

Bound (1) excludes competing saddles at the eigenvalue level; bound (2) controls the non-perturbative twist contributions to the conformal block decomposition. Together they establish identity block dominance for $N \geq 6$ (corrections $\leq 2\%$). At $N = 5$, the 11% correction is the leading systematic uncertainty; however, since $\lambda_{\min}(5) = 1$ on the flat plane, a shift of ~ 0.11 leaves $\lambda_{\min} > 0$, so the stability conclusion is not affected. $\square$

Theorem ?? identifies λ_m as the m -th $\mathbb{Z}_N$ -channel stiffness of the semiclassical N -point function. No bulk computation is needed: the stability condition $\lambda_m > 0$ is the *confining potential exceeding the holomorphic OPE repulsion in channel m* , a statement in the boundary CFT. The palindromic thresholds $\xi^*(N)$ (at which $C_1(\xi^*) = f_{\max}(N)$) are visible as algebraic-number bulk positions where the $\mathbb{Z}_N$ conformal block changes sign.

Quantum correction. The $\mathbb{Z}_N$ Fourier decomposition of the Virasoro block at central charge c confirms the identification at all orders. The m -dependent part of $\log F^{(m)}$ is

$$\left[\frac{m(N-m)}{2} - \frac{m^2(N-m)^2}{4c} + O(1/c^2) \right] \log q. \quad (6.4)$$

At $c = \infty$: the leading term is the Havelock Casimir (exact, verified for all $N = 4, \dots, 12$ and all modes). At finite c : the correction $\delta h_m = -f_m^2/c$ is a *polynomial in the Casimir*, so the algebraic structure (palindromic symmetry, number field) is preserved at each order in $1/c$. The fractional correction $|f_m/c| \leq 1/8$ (maximized at $m = N/2$, $c = N^2$) does not change any eigenvalue sign (Proposition ??).

7 The polygon–BTZ phase transition

The polygon framework *must* reproduce the known Hawking–Page phase transition of AdS₃ gravity (?); this section verifies that it does. The Euclidean action comparison between the polygon and the BTZ black hole makes the Hawking–Page language precise and gives a novel N -dependent correction $T_c(N) < T_{\text{HP}}$ from the polygon binding energy.

The Euclidean actions. At inverse temperature β , the regularized Euclidean actions of the competing configurations are:

$$I_{\text{polygon}}(\beta) = -\frac{\beta}{8G} + \beta E_{\text{polygon}}, \quad (7.1)$$

$$I_{\text{BTZ}}(\beta) = -\frac{\pi^2 \ell^2}{2G\beta} \quad (M = 0 \text{ threshold BTZ}), \quad (7.2)$$

where $E_{\text{polygon}} = Nm(1-2Gm) + V_{\text{int}}$ is the total energy of the N -gon in the small-deficit regime $2Gm \ll 1$ (valid for $Gm = 3m/(2c) \leq 3/(2 \times 51.6) = 0.03$; the formula receives $O((Gm)^2) \approx 0.1\%$ corrections), and $V_{\text{int}} < 0$ is the pairwise Havelock interaction on $\mathbf{H}^2$. The vacuum action $-\beta/(8G)$ is the standard thermal-AdS₃ contribution.

The transition temperature. Setting $I_{\text{polygon}} = I_{\text{BTZ}}$:

$$T_c(N, m, \xi) = \frac{1}{2\pi\ell} \sqrt{1 - 8G E_{\text{polygon}}(N, m, \xi)}, \quad (7.3)$$

which reduces to the standard Hawking–Page temperature $T_{\text{HP}} = 1/(2\pi\ell)$ for $E = 0$. For $T < T_c$: the polygon has lower action ($\Delta I < 0$, polygon preferred). For $T > T_c$: the BTZ has lower action. The total polygon energy $E_{\text{polygon}} = Nm_{\text{vert}} + V_{\text{int}}$, where the interaction energy on the unit circle is $V_{\text{int}} = -\sum_{p < q} \log |z_p - z_q| = -(N/2) \ln N$ (from $\prod_{p=1}^{N-1} 2 \sin(\pi p/N) = N$), lowers T_c below T_{HP} : the binding energy $V_{\text{int}} < 0$ deepens the polygon potential well.

The negative mode. For $N > N_{\text{crit}}(\xi)$: the polygon has a negative Havelock eigenvalue $\lambda_m < 0$ at mode $m = \lfloor N/2 \rfloor$ (§??). By the vortex-gravity isomorphism (Paper II, Proposition 13.1), the Havelock Hessian of the polygon is the second variation of the gravitational energy around the multi-cone saddle. A negative Havelock eigenvalue therefore is a negative mode of the Lichnerowicz operator Δ_{Lich} on the multi-cone geometry, a perturbation that lowers the Euclidean gravitational action, signaling decay (?). The mode $m = \lfloor N/2 \rfloor$ displaces adjacent masses alternately inward and outward, a pattern that drives pairwise coalescence. In the nonlinear regime, the coalescing masses form a single body exceeding the BTZ threshold $8GM > 1$, completing the decay to a BTZ black hole.

The three-region phase diagram. The polygon–BTZ competition has three distinct regimes:

1. *Stable polygon, preferred* ($N \leq N_{\text{crit}}, T < T_c$): all Havelock eigenvalues are positive and the polygon has lower Euclidean action than the BTZ. The polygon contributes as a stable saddle to the path integral.

2. *Stable polygon, metastable* ($N \leq N_{\text{crit}}, T > T_c$): all eigenvalues are positive but the BTZ has lower action. The polygon is a local minimum separated from the BTZ by a barrier; the tunneling rate is $\Gamma \sim e^{-S_{\text{BTZ}}}$.
3. *Unstable polygon, decay* ($N > N_{\text{crit}}$): the mode $m = \lfloor N/2 \rfloor$ is a negative mode, and the polygon decays continuously (no barrier) toward the BTZ. The palindromic threshold $\xi^*(N)$ marks the boundary between regions 1/2 and region 3.

The critical exponent. At the palindromic threshold $\rho = \rho^*(N)$, the critical mode $m^* = \lfloor N/2 \rfloor$ vanishes linearly: $\lambda_{m^*}(\rho) \sim \coth(\rho^*) \cdot (\rho - \rho^*)$. The one-loop partition function (Gaussian integration over the zero mode) gives the universal critical exponent $\nu = 1/2$:

$$Z(\rho) \sim \frac{Z_{\text{frozen}}}{|\rho - \rho^*|^{1/2}}, \quad Z_{\text{frozen}} = \frac{2^{3(N-2)/4}}{(N-3)!!}$$

where $(N-3)!! = 1 \cdot 3 \cdot 5 \cdots (N-3)$ is the double factorial (even N , one zero mode at $m^* = N/2$; for odd N , the palindromic pair m^* and $N-m^*$ both have $\lambda = 0$, giving two zero modes and $Z \sim |\rho - \rho^*|^{-1}$. Z_{frozen} for odd N is $\prod_{m \neq m^*, m \neq N-m^*} [f(m^*, N) - f(m, N)]^{-1/2}$, computed from the Casimir gaps $f(m^*, N) - f(m, N) = [m^*(N-m^*) - m(N-m)]/2$. The closed-form expression includes the normalization of the first-order Kirchhoff path integral measure; it is verified numerically against the gap-by-gap product for all $N \leq 20$ (see Code Availability). The polygon–BTZ transition is second-order with mean-field exponents.

The entropy gap. In the microcanonical ensemble (fixed total energy), the BTZ *always* has higher entropy: $\Delta S = S_{\text{BTZ}}(E_{\text{polygon}}) > 0$. The polygon is therefore always entropically disfavored; its survival as a metastable state depends entirely on the positivity of the Havelock eigenvalues (the energy barrier), not on entropy. This makes N_{crit} the sharp dividing line: below it, the polygon is a long-lived metastable state; above it, the polygon decays on a dynamical timescale $\tau \sim 1/\sqrt{|\lambda_{\lfloor N/2 \rfloor}|}$.

Microscopic entropy accounting. The BTZ entropy decomposes into discrete and continuous contributions:

$$S_{\text{BTZ}}(N) = \underbrace{S_{1\text{-loop}}(N)}_{\text{continuous}} + \underbrace{\Delta S_{\text{wc}}}_{\text{discrete}}, \quad (7.4)$$

where $S_{1\text{-loop}} = \log(N-3)!! - \frac{3(N-2)}{4} \log 2$ (from the Casimir gaps) and ΔS_{wc} is the wall-crossing contribution. At each palindromic threshold, the $\mathbb{Z}/2$ Galois action ($m \leftrightarrow N-m$) contributes:

N parity	Critical mode	ΔS
Even (self-paired $m = N/2$)	one eigenvalue crosses	$\log 2$ (1 bit)
Odd (paired $m, N-m$)	two eigenvalues cross	$2 \log 2$ (2 bits)

From $N = 7$ (the flat-plane marginal threshold) through $N = 12$: exactly 9 bits (verified numerically). Through $N = 17$ (11 wall-crossings): the discrete part contributes $17 \ln 2 \approx 11.8$ nats (9.9%), the continuous one-loop part 107.8 nats (90.1%). The L^2 -Morse index $\tau(P_-)$ has denominator 2 at each threshold (from the eta invariant shift of $1/2$ due to the zero mode), consistent with the K -theoretic framework.

The one-loop correction. The one-loop partition function around the polygon saddle is $Z_{1\text{-loop}} = (\det' \Delta_{\text{Lich}})^{-1/2}$, where Δ_{Lich} is the Lichnerowicz operator on the multi-cone geometry. By the conformal invariance of the flat-plane backreaction (Proposition 13.2), the leading correction comes from the spatial topology: the spectral bound $|\delta C_1| \leq 2.86/(4\pi\lambda_1)$ (where λ_1 is the spectral gap of the spatial Laplacian) gives a rigorous bound on the quantum shift of the stability eigenvalues. On the Bolza surface ($\lambda_1 \approx 3.84$): $|\delta C_1| < 0.06$, which does not shift N_{crit} away from 7 on the flat plane. On the modular surface ($\lambda_1 \approx 91$): $|\delta C_1| \lesssim 0.002$, negligible. The Bolza surface carries the $\{8, 3\}$ hyperbolic lattice on which ? constructed the hyperbolic Haldane model (Chern insulator with quantized Bott index $B = \pm 1$ and chiral edge states). The vortex stability analysis on the same $\{8, 3\}$ lattice shares the spectral data (λ_n of Δ_S) with the hyperbolic Haldane model. The bound $|\delta C_1| < 0.06$ shows that polygon stability is insensitive to the hyperbolic lattice topology. The Lorentzian continuation of the polygon–BTZ phase structure, including the analytic continuation of the Green’s function, the van Hove spectral singularity at the threshold, and the connection to GUE spectral statistics, is developed in Supplement S, §4.

8 The Wheeler–DeWitt equation on the breathing mode

The single degree of freedom ρ (geodesic radius of the polygon) and the potential $V(\rho) = \lambda_{m^*}(\rho)$ (the critical Havelock eigenvalue) define a minisuperspace Wheeler–DeWitt equation:

$$\left[-\frac{1}{2c} \frac{d^2}{d\rho^2} + \Phi(\rho) - \frac{m(N-m)}{2} \right] \Psi = 0, \quad \Phi(\rho) := \log(2 \sinh \rho) + b(N), \quad (8.1)$$

with $c = 12b(N)$ and $\hbar = 1$ (the semiclassical limit is $c \rightarrow \infty$).

On constant-curvature $\mathbf{H}^2$ the Havelock remainder $\delta_m = 0$ identically (§??), so it does not appear.

Notation. We write $\Phi(\rho) = \log(2 \sinh \rho) + b(N)$ for the Green’s function offset (the unconstrained radial potential), reserving $C_1(\xi)$ for the constrained angular stiffness (eq. ??). The WDW potential is $V(\rho) = \Phi(\rho) - f(m^*, N)$.

Logical status of c (§?). The value $c = 12b(N)$ is derived from the Polyakov conformal anomaly of the conical N -gon sphere (Proposition ??); no free parameter is involved.

Why second-order dynamics from a first-order system. The Kirchhoff equations are first-order, but the WDW equation for the breathing mode ρ is second-order. The derivation has three steps:

1. *No-go for perturbative kinetic terms.* For a first-order operator $A_m = \partial_\tau + \lambda_m(\rho(\tau))$ on the thermal circle $[0, \beta]$, the functional determinant is $\det A_m = 1 - \exp(-\int_0^\beta \lambda_m d\tau)$, which depends only on the time-integral of λ_m , not on $\dot{\rho}$. Equivalently, the Matsubara influence kernel $\sum_n [(i\omega_n + \lambda)(i\omega_{n+k} + \lambda)]^{-1}$ vanishes for $k \neq 0$: partial fractions give $(1/i\nu_k) [\sum_n (i\omega_n + \lambda)^{-1} - \sum_n (i\omega_{n+k} + \lambda)^{-1}]$, and each factor sum equals $(\beta/2) \coth(\beta\lambda/2)$ by the Bose summation formula ($n \mapsto n+k$ is a relabelling of $\mathbb{Z}$). The integrable bath therefore contributes *no* $\dot{\rho}^2$ term to the effective action; the kinetic coefficient is not perturbative.

2. *Geometric origin (Fisher–Rao).* The kinetic term arises from the path-integral *measure*, not the action. The Feynman–Kac correspondence $Z = \int \mathcal{D}\rho \mu[\rho] e^{-S[\rho]}$ carries the measure $\mu[\rho] = \prod_\tau \sqrt{c/(2\pi d\tau)}$, whose coefficient c is the Fisher–Rao information metric on the family of Boltzmann distributions $P_m(\varepsilon | \rho) \propto e^{-\lambda_m(\rho)|\varepsilon|^2}$ parametrized by ρ :

$$g_{\text{FR}}(\rho) = \coth^2 \rho \sum_{m \neq m^*} \frac{1}{\lambda_m(\rho)^2}.$$

This metric is ρ -dependent. The *classical* BO separation is exact for any kinetic coefficient (the Havelock eigenvectors are ρ -independent Fourier modes). But the *quantum* BO factorization requires c to be *constant*: if $c(\rho)$ were ρ -dependent, the FK Hamiltonian $H = -(1/(2c(\rho))) d^2/d\rho^2 + V(\rho)$ would produce a first-derivative term $-c'(\rho) \psi'/(2c(\rho)^2)$ that couples the breathing mode to ρ -gradients of the angular-mode bath, breaking the exact factorization at the quantum level. Therefore c must be a constant. The unique constant matching Brown–Henneaux ($c = 3\ell/(2G)$) is $c = 12b(N)$. The Fisher–Rao metric $g_{\text{FR}}(\rho)$ is the ρ -dependent version; extracting the constant requires a tortoise-coordinate field redefinition $\rho \rightarrow \tilde{\rho}(\rho)$ with $g_{\text{FR}} |d\rho|^2 = c |d\tilde{\rho}|^2$. Under this redefinition, the WDW equation becomes $-(1/2c) d^2\psi/d\tilde{\rho}^2 + \tilde{V}(\tilde{\rho}) \psi = E\psi$ with $\tilde{V} = V(\rho(\tilde{\rho})) + O(1/c)$ (Schwarzian correction from the non-linear coordinate change). The $O(1/c)$ Schwarzian term is absorbed into the one-loop correction (1.9% at $N = 7$; see step 3 below), so the leading-order potential retains the form $V(\rho)$ of Proposition ??.

3. *Exact coefficient (two independent routes).* Two independent calculations determine c :

- (a) *Eigenvalue arithmetic* (definition). $b(N)$ is the unique mode-independent offset in $\lambda_m = C_1(\rho) - f(m, N)$ (Proposition ??). Setting $c_1 = 12b(N)$ defines the central charge from the Havelock spectrum.
- (b) *Chern–Simons gauge theory* (independent calculation). The CS level $k = \ell/(4G)$ gives $c_2 = 6k = 3\ell/(2G)$ (Brown–Henneaux, 1986). The CMS–CS Casimir identification (Proposition ??) matches the Havelock spectrum to the CS Wilson-line spectrum, forcing $k = 2b(N)$ and hence $c_2 = 12b(N) = c_1$.

A third route, the Zamolodchikov c -theorem flow from $c_{\text{UV}} = 12N^2$ to $c_{\text{IR}} = 12b(N)$ (Theorem ??), provides a consistency check but shares the CMS–CS input with route (b). The WDW kinetic coefficient equals $c_1 = c_2 = 12b(N)$; any other value would make the FK ground state disagree with the boundary CFT vacuum, contradicting the CS/WZW correspondence (?). The CMS integrability makes the one-loop partition function $Z_{\text{bath}}(\rho) = \prod_{m \neq m^*} |\lambda_m(\rho)|^{-1/2}$ exact at $O(c^0)$, with anharmonic corrections of order $1/c$ (1.9% at $N = 7$, 0.8% at $N = 11$, decreasing as N^{-2} ; see Code Availability).

The no-go result (step 1) shows the kinetic term is genuinely geometric: it encodes the information-theoretic distinguishability of neighboring equilibria. The four-fold matching (step 3) fixes $c = 12b(N)$ without free parameters.

The four-role consistency of c (??) (Section 1) is preserved. Each term has a definite origin:

Term	Expression	Origin
Quantum kinetic	$-\hbar^2/(2c) \Psi''$	Todd offset $c = 12b(N)$ (§1)
Classical potential	$\log(2 \sinh \rho)$	$\mathbf{H}^2$ Green's function
Vacuum energy	$b(N) \sim c/12$	Orbifold CFT vacuum (Todd class)
Matter	$m(N-m)/2$	Twist-field dimension (first Chern class)
One-loop	δ_m	Havelock remainder (traceless, palindromic)

Three regimes.

1. **Classical** ($c \rightarrow \infty$): the kinetic term vanishes; the constraint reduces to $V(\rho) = 0$, i.e., $\lambda_{m^*}(\rho^*) = 0$, the palindromic threshold, which is the Einstein equation of the vortex-gravity system.
2. **Semiclassical** (WKB): $\Psi \sim \exp(\pm(i/\hbar) \int \sqrt{2cV} d\rho)$, oscillatory in the exterior ($V > 0$, sinh-branch) and evanescent in the interior ($V < 0$, cosh-branch). The continuation $\rho \rightarrow \rho + i\pi/2$ maps between them (§??, Lorentzian paragraph).
3. **Quantum** (near threshold): $\Psi \sim \text{Ai}((2c \coth \rho^*)^{1/3}(\rho - \rho^*))$, smoothing the classical horizon over the width

$$\delta\rho = \frac{1}{(2c \coth \rho^*)^{1/3}}, \quad (8.2)$$

where $\coth \rho^* = V'(\rho^*)$ is the surface gravity. For large ρ^* ($\coth \rho^* \rightarrow 1$): $\delta\rho \rightarrow (2c)^{-1/3} \approx (2N^2)^{-1/3}$. At $c = \infty$: $\delta\rho \rightarrow 0$ (sharp classical threshold). At finite c : the horizon has quantum width $O(c^{-1/3})$.

Proposition 8.1 (Eigenvalue non-crossing at finite c). *Since $c = 12b(N) = N^2 + N - 12 \ln 2 + O(\ln N/N) > N^2$ for all $N \geq 3$, the condition $c \geq N^2$ is automatic. For $N \geq 4$ (so that $\lfloor N/2 \rfloor \geq 2$), the mode-dependent part of the $O(1/c)$ Virasoro block correction satisfies*

$$\frac{|\delta f_m - \delta f_{m+1}|}{f(m+1, N) - f(m, N)} = \frac{2m(N-m-1) + (N-1)}{2c} \leq \frac{1}{4} - \frac{1}{2N^2} < \frac{1}{4}. \quad (8.3)$$

In particular, no eigenvalue crossing occurs and the critical mode index $m_{\text{crit}} = \lfloor N/2 \rfloor$ is preserved.

Proof. The leading $1/c$ correction to the Casimir is $\delta f_m = -f_m^2/c = -m^2(N-m)^2/(4c)$. The difference factors as $\delta f_m - \delta f_{m+1} = (N-2m-1)[2m(N-m-1) + (N-1)]/(4c)$. Dividing by the Casimir gap $f(m+1) - f(m) = (N-2m-1)/2$ gives the ratio $R(m) = [2m(N-m-1) + (N-1)]/(2c)$. The maximum over $1 \leq m < N/2$ is at $m = N/2 - 1$, giving $R_{\text{max}} = (N^2-2)/(4N^2) = 1/4 - 1/(2N^2) < 1/4$. Since $R < 1$ for all adjacent pairs: the residual gap is at least $3(N-2m-1)/8 > 0$, and no reordering occurs. $\square$

Remark 8.1 (Algebraic field preservation). *The Zamolodchikov c -recursion gives the correction as a rational function of c , with only $\mathbb{Z}_N$ -invariant channels contributing. At $c = N^2$: the palindromic coefficients $A(c), B(c)$ remain in $\mathbb{Q}(\cos(2\pi/N))$, the same number field as at $c = \infty$. The palindromic form $A\xi^2 + B\xi + A = 0$ is exact at all c (the conformal inversion $\xi \leftrightarrow 1/\xi$ is a geometric symmetry of $\mathbf{H}^2$, not a perturbative artifact).*

The palindromic threshold $\xi^*(N)$ is the critical AdS radius at which the polygon–BTZ transition occurs, and the algebraic number field $\mathbb{Q}(\sqrt{D})$ of the threshold is the arithmetic invariant of the phase boundary.

Remark 8.2 (Uniqueness of the WDW wave function). *The WDW equation (??) on $\rho \in (0, \infty)$ has two singular endpoints. Their Weyl classification determines whether boundary conditions are needed:*

1. At $\rho = 0$: limit-circle. *The potential $V(\rho) = \log(2 \sinh \rho) + b(N) - f(m^*, N) \rightarrow \log(2\rho) + \text{const}$ as $\rho \rightarrow 0$. Weyl’s criterion (integral test): $\int_0^\varepsilon \rho |2cV(\rho)| d\rho < \infty$ because $\rho |\log \rho|$ is integrable. Both linearly independent solutions are L^2 near the origin; a boundary condition is required.*
2. At $\rho \rightarrow \infty$: limit-point. *$V(\rho) \rightarrow \rho + \text{const}$ (linear growth), so $\int^\infty \rho \cdot 2c\rho d\rho = \infty$. Only the Airy-decaying solution is L^2 ; no boundary condition is needed.*

The operator therefore requires exactly one boundary condition (at $\rho = 0$). The Born–Oppenheimer structure supplies it: the CMS integrability of the angular sector (Proposition ??) makes the adiabatic separation exact, and the angular modes in their ground state at each ρ select the ground state of the effective Hamiltonian. This ground state is non-degenerate: numerically, the spectral gap $\Delta E \approx 0.84$ is stable across $N = 7$ –15 (see Code Availability). The wave function is therefore unique without imposing Hartle–Hawking or Vilenkin boundary conditions, the microscopic (Havelock) structure determines the cosmological state.

9 The three-layer decomposition

Proposition 9.1 (Orbifold–Havelock correspondence). *At leading order in $1/N$, the Havelock eigenvalue on $\mathbf{H}^2$ matches the $\mathbb{Z}_N$ orbifold CFT ground-state energy at $c_{\text{orb}} = 12N^2$ (the UV orbifold central charge) with angular impulse constraint:*

$$\lambda_m = -E_0(m; c_{\text{orb}}=12N^2) + \mu_L(\xi), \quad (9.1)$$

where $E_0(m) = -N/2 + m(N-m)/2$ is the twisted-sector ground state and $\mu_L(\xi) = C_1(\xi) - N/2$ is the Lagrange multiplier. The matching uses $c_{\text{orb}} = 12N^2$ (so that $c/(24N^2) = 1/2$), while the exact Havelock central charge is $c = 12b(N) = N(N+1) - 12\ln 2 + \dots \approx N^2$ at leading order. The distinction: c_{orb} is the UV (free-field) central charge of the $\mathbb{Z}_N$ orbifold; $c = 12b(N)$ is the exact IR value after regularization of the vortex self-energy. The mode-dependent matching (the $m(N-m)/2$ term) uses only $c_{\text{orb}}/(24N^2) = 1/2$; the mode-independent $O(N)$ corrections (the $\ln 2$ and $(\ln N)/(N-1)$ terms in $b(N)$) enter through μ_L , not through E_0 .

Symbol	Value	Role
$c_{\text{orb}} = 12N^2$	UV orbifold	Mode-dependent matching
$c = 12b(N) \approx N^2 + N$	Exact (IR)	WDW kinetic coefficient
$c \approx N^2$	Leading order	Approximation in estimates

Proof. The orbifold ground-state energy $E_0(m) = -c/(24N) + cm(N-m)/(24N^2)$ at $c = c_{\text{orb}} = 12N^2$ (the UV orbifold central charge, which gives $c/(24N^2) = 1/2$ exactly) yields $E_0 = -N/2 + m(N-m)/2$. Then $-E_0(m) + \mu_L = [N/2 - m(N-m)/2] + [C_1(\xi) - N/2] = C_1(\xi) - m(N-m)/2 = \lambda_m$. $\square$

Proposition 9.2 (Todd class decomposition of the eigenvalue offset). *The $\mathbb{Z}_N$ -equivariant eigenvalue decomposition of the vortex Laplacian on the cusped Poincaré disk (analogous to the equivariant Riemann–Roch theorem) satisfies*

$$\text{ind}_m = \underbrace{\frac{m(N-m)}{2}}_{c_1\text{-term}} + \underbrace{b(N)}_{\text{Todd term}} + \delta_m, \quad (9.2)$$

where $b(N)$ (§??) is independent of both m and ρ . The central charge c (§?) is the same for every mode m , extracted from the Todd correction alone.

Proof. The Havelock eigenvalue satisfies $\lambda_m + f_m = \Phi(\rho) + \delta_m + O(e^{-\rho})$, where $\Phi(\rho) = \log(2 \sinh \rho) + b(N)$ (the radial dependence is absorbed by the Green’s function). Subtracting $\log(2 \sinh \rho)$: $\text{ind}_m := \lambda_m + f_m - \log(2 \sinh \rho) = b(N) + \delta_m$. Since δ_m is traceless ($\sum \delta_m = 0$ to 10^{-15}) and palindromic ($\delta_m = \delta_{N-m}$), the mode-independent part $b(N)$ is extracted exactly. $\square$

The decomposition (§?) below is the spectral analogue of equivariant Riemann–Roch (f_m : twist-field dimension; $b(N)$: Todd correction; δ_m : one-loop remainder).

Exact Havelock remainder formula. On constant-curvature surfaces, $\delta_m = 0$ identically (Paper IV, Theorem 7.1); on compact quotients $\Gamma \backslash \mathbf{H}^2$, the Selberg trace formula gives a mode-independent shift $|\delta C_1| \leq 0.020$ on the Bolza surface (§??). The formula below is the flat-plane residual.

The anomaly δ_m decomposes exactly into character sums on $\mathbb{Z}/N\mathbb{Z}$:

$$\delta_m = L(m, N) - \frac{1}{4} S_2(m, N) + c(N), \quad (9.3)$$

where $L(m, N) = \sum_{p=1}^{N-1} \cot(\pi p/N) \cos(2\pi p m/N)$ is the cotangent sum, $S_2(m, N) = \sum_{p=1}^{N-1} \csc^2(\pi p/N) \cos(2\pi p m/N)$ is the cosecant-squared sum, and $c(N)$ is the unique constant enforcing $\sum_{m=1}^{N-1} \delta_m = 0$ (tracelessness). The coefficients of L and S_2 are 1 and $-1/4$ respectively, exact and universal ($R^2 = 1.000000$ for all tested $N = 4, \dots, 24$). This identifies the anomaly as the lattice correction (orbifold fixed-point contribution in the KK picture) to the continuum Casimir.

The central charge c (§?) makes five independently testable predictions beyond the Casimir: (i) the exact offset $b(N) = N(N+1)/12 - \ln 2 + (\ln N)/(N-1)$ (Paper I, the $b(N)$ identity proof); (ii) the mean Casimir $\langle m(N-m)/2 \rangle = N(N+1)/12$ and the mean log-sine regulator $\langle S \rangle = (\ln N)/(N-1) - \ln 2$ (the two pieces that sum to $b(N)$); (iii) the stability residual in closed form, $\rho^*(N) = N(N-2)/24 + \ln 2 - (\ln N)/(N-1)$ (even N) and $a_N \equiv \rho^*(N)/f_{\text{crit}}(N) = 1/3 - 2/(3N) + 8 \ln 2/N^2 - 8 \ln N/N^3 + O(N^{-3})$ (Paper I, Corollary ??); (iv) the quartic coupling $Q/f_{\text{crit}}^2 = N/48$ (even N); (v) $\Delta_{\mathbf{H}^2} h = -1/2$. All five follow from the B_2 -tower: the Casimir $m(N-m)/2$ is $O(c)$; the Havelock remainder δ_m is $O(c^0)$ (topological); the Virasoro block correction is $O(1/c) \approx 10^{-5}$ (§??).

9.1 Bihamiltonian hierarchy and arithmetic surfaces

The tower extends to a *Magri bihamiltonian hierarchy*. As noted in §??, the Thomson vortex system has the same linearized spectrum as the CMS model at coupling $g = 1$. The CMS model is bihamiltonian (?): the Kirchhoff bracket P_0 (mode-independent) and the Casimir bracket P_1 (eigenvalue f_m on mode m) are compatible, with recursion operator

$R = P_1 P_0^{-1}$ acting as multiplication by $f_m = m(N-m)/2$. By the Magri theorem, the k -th conserved quantity I_k has mode eigenvalue f_m^k , with physical derivatives $d_{2k} = \alpha_k N f_m^k$ ($\alpha_1 = 1$, $\alpha_2 = 4$, $\alpha_3 = 64$, verified numerically for $N \leq 16$). The quartic coupling $Q = (N/48)f^2$ lies inside the hierarchy ($H_4^{\text{diag}} \propto I_2$), guaranteeing $\{I_j, H_4\} = 0$ by integrability. On arithmetic surfaces (Bolza, modular, etc.), the eigenvalue acquires a topological correction that is invisible to the confining potential $C_1(\xi)$:

$$\lambda_m = \underbrace{C_1(\xi)}_{\text{Ricci scalar}} - \underbrace{\frac{m(N-m)}{2}}_{\text{Casimir}} + \underbrace{\delta_m(N)}_{\text{Havelock remainder}}. \quad (9.4)$$

Each layer has a distinct character:

1. $C_1(\xi)$: continuous, surface-dependent, mode-independent. Scales approximately as $C_1 \propto \rho$ over the range $\rho \in [0.01, 10]$ (with $R^2 = 0.98$, where ρ is the geodesic radius and $\xi = \tanh^2(\rho/2) \in (0, 1)$); the exact formula is eq. (??).
2. $f(m, N) = m(N-m)/2$: discrete, universal (surface-independent). The $\mathbb{Z}_N$ Casimir from the Havelock identity.
3. $\delta_m(N)$: topological, UV-finite, ξ -independent (variation $< 10^{-13}$), and exactly traceless ($\sum_{m=1}^{N-1} \delta_m = 0$ to 10^{-15}). Carries all number-theoretic content: Hecke eigenvalues, palindromic structure, Langlands data.

By structural analogy with the decomposition of the Riemann curvature tensor: C_1 plays the role of the Ricci scalar, $f(m, N)$ the traceful part, and δ_m the traceless Weyl-like anomaly. In $\text{AdS}_3/\text{CFT}_2$, the Brown–Henneaux scaling $C_1 \propto \log(\xi)$ says that the Weyl-like anomaly δ_m is the topological correction that carries the arithmetic of the underlying surface, contributing corrections of order $\|\delta\|/C_1 \sim 4/\Delta_\gamma$, where $\Delta_\gamma = \text{tr}(\gamma)^2 - 4$ is the discriminant of the shortest closed geodesic (1.6% for $N = 8$, $T = 16$; 4.2% for $N = 9$, $T = 10$).

9.2 Anomaly structure

The Weyl norm scales as $\|\delta\|^2 \sim N^{4.2}(\log N)^{4.1}$ for $N \geq 40$ (not a pure power law). The triangle ($N = 3$) is *anomaly-free*: $\delta_m = 0$ exactly for all Green’s functions and curvatures, because $\cos(2\pi/3) = \cos(4\pi/3)$ forces $G_1 = G_2$. Under Casimir-conserving mergers $N_1 + N_2 \rightarrow N_3$ ($N_3^3 - N_3 = N_1^3 - N_1 + N_2^3 - N_2$): the unique anomaly-*destructive* reaction is $4+4 \rightarrow 5$ (88% Weyl loss); all other mergers create Weyl content (gains 46% to 849%). The Weyl entropy per degree of freedom $s(N) = \log \|\delta\|^2 / \sum f$ peaks at $N \approx 9$ –10, coinciding with the observed planetary polygon range ($N = 5, 6, 8$).

The anomaly δ_m has two independent origins, which the parameter space (Δ, ξ) separates cleanly:

1. *Interaction anomaly* ($\Delta > 0$ on $\mathbb{R}^2$): deviation of $V = r^{-2\Delta}$ from $-\log r$. Large ($\|\delta\|^2 = O(1\text{--}100)$) but arithmetically trivial.
2. *Geometric anomaly* ($\Delta = 0$ on $\mathbf{H}^2$): deviation of the hyperbolic Green’s function from the flat one. Small ($\|\delta\|^2 = O(0.01\text{--}100)$) but carries all number theory: palindromic polynomials, Hecke eigenvalues, Langlands data.

The RG flow ($\Delta \rightarrow 0$) strips away the large, trivial interaction anomaly, leaving behind the small geometric anomaly as the residue that survives at the conformal fixed point. The number theory is not emergent from the flow; it is a property of the conformal fixed point, activated by curvature. The three-layer structure has a symplectic origin: the Kirillov form ω_{KR} (mode-independent) and the Weil–Petersson form ω_{WP} (mode-dependent) are *not* proportional; their discrepancy $\omega_{\text{WP}} - \alpha \omega_{\text{KR}} \propto f(m, N)$ is the Casimir (Layer 2), and the residual beyond this linear relation is the Havelock remainder (Layer 3).

The B_2 organizing principle, connecting the stability polynomial $6B_2(x) = P_2(2x-1)$ to Gauss–Legendre quadrature, the Euler–Maclaurin duality, and the csc^2 kernel universality (Calogero–Moser–Sutherland, Dyson, Haldane–Shastry, Laughlin, Selberg, W_N , AGT/Nekrasov), is developed in Paper I, §??.

10 Higher-dimensional extension: the $\text{CFT}_3/\text{AdS}_4$ Casimir

The two-dimensional results extend to conformal field theory in any dimension d via the power-law interaction $V(x, y) = |x - y|^{-2\Delta}$, where Δ is the conformal dimension. For $\Delta \rightarrow 0$: $V \rightarrow -\log |x - y|$ (the 2D case). For $\Delta = 1$: the CFT_3 two-point function on S^2 (the boundary of AdS_4).

Proposition 10.1 (Generalised Havelock Casimir). *For N equally-spaced points on a ring on S^2 , interacting via $V = |x - y|^{-2\Delta}$, the angular Hessian eigenvalues factorize as $\mu_m = g(\theta_0, \Delta) \cdot h(m, N, \Delta)$, where the universal Casimir is*

$$h(m, N, \Delta) = \sum_{p=1}^{N-1} \frac{1 + 2\Delta \cos^2(\pi p/N)}{\sin^{2\Delta+2}(\pi p/N)} \left(1 - \cos \frac{2\pi m p}{N}\right). \quad (10.1)$$

For $\Delta = 0$: $h(m, N, 0) = 2m(N-m)$. The ratio $h(m)/h(1)$ is independent of θ_0 , confirming that the Havelock decomposition extends to all Δ .

The constrained Hessian on the flat plane with $V = |x - y|^{-2\Delta}$ gives an N -dependent stability boundary $N_{\text{crit}}(\Delta)$:

Δ	0	0.5	1	2	5	10
N_{crit}	7	5	5	5	4	4

The classical $N_{\text{crit}} = 7$ is an isolated phenomenon of the logarithmic interaction. For the CFT_3 value $\Delta = 1$: $N_{\text{crit}} = 5$.

Under the Green’s function identification $\Delta = (d-2)/2$, the table acquires a dimensional interpretation:

Dimension d	2	3	4	5	≥ 6
$\Delta = (d-2)/2$	0	1/2	1	3/2	≥ 2
N_{crit}	7	5	5	5	4–5

The hexagon ($N = 6$, Saturn) is a *uniquely two-dimensional* phenomenon: it is marginally stable at $d = 2$ ($\lambda_{\text{min}} = 1/2$) but unstable for $d \geq 3$ ($\lambda_{\text{min}} = -0.02$ at $d = 3$; $N_{\text{crit}}(\Delta=0.5) = 5 < 6$; see Code Availability). The pentagon ($N = 5$) survives in all tested dimensions $d \leq 10$.

Proposition 10.2 (Soft Havelock decomposition in dimension d). *For the interaction $V = r^{-(d-2)}$ ($d \geq 3$) on the N -gon ring, the angular Casimir decomposes as*

$$\sigma_m^{(d)} = a(N, d) \frac{m(N-m)}{2} + R_m(N, d), \quad (10.2)$$

where $a(N, d)$ is the leading Casimir coefficient (determined by least-squares fit) and R_m is the residual. For all tested $N = 5, \dots, 20$ and $d = 3, 4, 5$: the residual satisfies $|R_m - R_{m'}| < a |f_m - f_{m'}|$ for every adjacent pair, so the mode ordering is preserved. The critical mode is always $m = \lfloor N/2 \rfloor$ (same as $d = 2$), and the stability threshold has the same palindromic form as the two-dimensional case, with coefficients that depend on d .

For $d = 2$ ($s = 0$): $a = 1$ exactly and $R_m = 0$ (the Havelock identity). This is the *unique* dimension where the Casimir is an exact polynomial, because $\csc^2(\pi p/N)$ has a closed discrete Fourier transform ($\sum \csc^2 \cdot (1 - \cos) = 2m(N-m)$), while $\csc^s$ for $s \neq 2$ does not. The exact algebraic structure of Paper I (palindromic polynomials, algebraic number fields, modular forms) is specific to $d = 2$, a consequence of the uniqueness of the $\csc^2$ weight function among power-law potentials. However, the *structural skeleton* survives in all dimensions:

1. **Mode ordering.** The critical mode is $m = \lfloor N/2 \rfloor$ in every d (Proposition ??). The residual R_m is too small to reorder eigenvalues.
2. **Palindromic symmetry.** The symmetry $\sigma_m = \sigma_{N-m}$ holds exactly for all d (from the discrete cosine: $\cos(2\pi p m/N) = \cos(2\pi p(N-m)/N)$). On curved spaces, conformal inversion $\xi \leftrightarrow 1/\xi$ is geometric (a property of the Poincaré ball, not of the interaction), so the threshold equation retains palindromic form.
3. **Finite- c algebraic structure.** The $O(1/c)$ Virasoro block correction $\delta h_m = -f_m^2/c$ is a *polynomial in the Casimir* (equation ??), preserving the algebraic structure at every order.
4. **Havelock remainder shape.** The normalized anomaly profile $\delta_m/\|\delta\|$ has inner products > 0.98 between $\Delta = 0$ and $\Delta = 2$ for $N \leq 6$: the number-theoretic fingerprint transfers to $d = 3$ with $< 2\%$ distortion.

The $d = 2$ case is where the algebra is exact; the $d \geq 3$ case is where it is *approximate but structurally preserved*. The mechanism by which the logarithmic structure survives in 3+1D is the soft Havelock decomposition: the Casimir $m(N-m)/2$ is the dominant term at every d , with the interaction-specific residual R_m as a controlled perturbation.

Remark 10.1 (Transcendence of the transition dimensions). *The critical value $\Delta^*(N)$ where N_{crit} drops below N is expected to be transcendental for $N \geq 5$. The threshold equation involves $\sin^{-2\Delta}(\pi p/N)$ for different p with $\mathbb{Q}$ -linearly independent logarithms. For $N = 5$: the threshold equation involves $\sin^{-2\Delta}(\pi/5)$, an algebraic base raised to the power Δ . If Δ^* is algebraic and irrational, the Gelfond–Schneider theorem makes the individual terms transcendental, and Baker’s theorem on linear forms in logarithms would be needed to show no cancellation occurs. A rigorous proof requires Baker-type estimates. This contrasts with the palindromic thresholds $\xi^*(N)$ on $\mathbb{H}^2$, which lie in explicit algebraic number fields.*

For $N = 5$, the pentagon identity $\sin(2\pi/5)/\sin(\pi/5) = \varphi$ (the golden ratio) gives the eigenvalue as $\lambda_2(\Delta) = P_0(\Delta) + P_1(\Delta) \varphi^{-(2\Delta+2)}$, where P_0, P_1 are algebraic functions of Δ . The transition occurs at $\varphi^{2\Delta^*+2} = P_1/(-P_0) \approx 25.8$, giving $\Delta^* \approx 2.378$.

Theorem 10.3 (RG monotonicity of the Casimir ratio). *For any $N \geq 4$ and $m_{\text{crit}} = \lfloor N/2 \rfloor$, the generalized Casimir ratio $R(\Delta) = h(m_{\text{crit}}, N, \Delta)/h(1, N, \Delta)$ is strictly increasing on $[0, \infty)$, with exact limits*

$$R(0) = \frac{m_{\text{crit}}(N - m_{\text{crit}})}{N - 1}, \quad R(\infty) = \left[\frac{\sin(\pi m_{\text{crit}}/N)}{\sin(\pi/N)} \right]^2. \quad (10.3)$$

Proof. By the Chebyshev identity, the mode amplification factor $f(p) = (1 - \cos(2\pi m_{\text{crit}} p/N))/(1 - \cos(2\pi p/N))$ achieves its global maximum at $p = 1$ (and its palindromic twin $p = N-1$):

Even $N = 2k$: $\sin(\pi k p/N) = \sin(\pi p/2) \in \{0, \pm 1\}$, so $f(p) = 1/\sin^2(\pi p/N)$ for odd p and $f(p) = 0$ for even p . Since $\sin(\pi p/N) \geq \sin(\pi/N)$ for $p \geq 2$: $f(p) \leq f(1)$.

Odd $N = 2k+1$: The identity $\sin(k\pi p/N) = \sin(\pi p/2 - \pi p/(2N))$ (using $k/N = 1/2 - 1/(2N)$) gives explicit formulas: $f(p) = 1/(4\sin^2(\pi p/(2N)))$ for odd p , and $f(p) = 1/(4\cos^2(\pi p/(2N)))$ for even p . Since $f(1) = 1/(4\sin^2(\pi/(2N)))$: for odd $p \geq 3$, $\sin(\pi p/(2N)) \geq \sin(\pi/(2N))$ (monotonicity of $\sin$ on $(0, \pi/2)$), so $f(p) \leq f(1)$; for even $p \leq N-1$, $\cos(\pi p/(2N)) \geq \sin(\pi/(2N))$ (since $\pi p/(2N) + \pi/(2N) \leq \pi/2$ iff $p \leq N-1$), so $f(p) \leq f(1)$.

The normalized weight $\mu_p(\Delta) \propto \tilde{w}(p, \Delta) c_1(p)$ concentrates on $\{1, N-1\}$ as $\Delta \rightarrow \infty$, because $\tilde{w}(p)/\tilde{w}(1) = O([\sin(\pi/N)/\sin(\pi p/N)]^{2\Delta}) \rightarrow 0$ for $p \notin \{1, N-1\}$. Since $R(\Delta) = \mathbb{E}_\mu[f]$ and μ concentrates at the global maximum of f : $dR/d\Delta = \text{Cov}_\mu(d \log \tilde{w}/d\Delta, f) > 0$ (both f and $g = d \log \tilde{w}/d\Delta = -2 \log \sin(\pi p/N)$ are maximized at $p = 1$ and monotonically ordered on $\{1, \dots, \lfloor N/2 \rfloor\}$; the Chebyshev sum inequality gives $\text{Cov}_\mu(g, f) > 0$). $\square$

Corollary 10.4 (Stability monotonicity). *$N_{\text{crit}}(\Delta)$ is non-increasing in Δ : once a polygon type loses stability, it never regains it. The quantity $\log N_{\text{crit}}(\Delta)$ decreases from $\log 7$ ($\Delta = 0$, the Havelock regime) to $\log 3$ ($\Delta \rightarrow \infty$, only triangles survive). This is reminiscent of the Zamolodchikov c -theorem, though N_{crit} is a step function (not smooth) and Δ is not an RG parameter.*

11 Triangle universality and dynamical triangulation

The equilateral triangle ($N = 3$) is the *unique* regular polygon that is dynamically stable for all interaction ranges and quantum deformation levels.

Theorem 11.1 (Triangle universal stability). *On the flat plane, the equilateral triangle ($N = 3$) is the unique regular polygon stable for all interaction exponents $\Delta \geq 0$ (with $V(r) = r^{-2\Delta}$ for $\Delta > 0$ and $V(r) = -\log r$ at $\Delta = 0$). The triangular tessellation $\{3, 6\}$ is the unique dynamically stable regular 2D lattice for every $\Delta > 0$: the square $\{4, 4\}$ and hexagonal $\{6, 3\}$ tessellations are unstable for all $\Delta > 0$, while the triangular lattice is stable via Montgomery's theta-function minimization (?).*

Proof. Part A: Polygon stability of the triangle. Three independent arguments, each a proof:

1. *Equal chords.* For $N = 3$, all three pairwise distances are equal ($\sin(\pi/3) = \sin(2\pi/3) = \sqrt{3}/2$), so the energy $H = -\sum h(d_{jk})$ has no range-dependent competition between nearest and next-nearest neighbors. Any radial perturbation acts isotropically on all three chords, and the dependence on h factors out: stability is h -independent.
2. *Single-mode circulant.* The tangential Hessian at $N = 3$ is a $\mathbb{Z}_3$ -circulant with a single non-trivial mode $m = 1$ (and its palindromic conjugate $m = 2$). The Havelock eigenvalue $\lambda_1 = (N-1) - m(N-m)/2 = 2 - 1 = 1 > 0$ is strictly positive, and this is the only eigenvalue to check. Mode competition (required for a Havelock-threshold crossing) needs at least two independent non-critical modes, which occurs only at $N \geq 4$.
3. *Sign rule application.* Theorem 3.1 (Paper I) gives $H''(0) < 0$ for any h satisfying $h'(r) < 0$ and $(rh'(r))' \leq 0$. Both inverse-power $V(r) = r^{-2\Delta}$ (with $\Delta > 0$) and logarithmic $V(r) = -\log r$ satisfy these hypotheses. The sign rule therefore covers the entire family $\Delta \geq 0$ with a single proof.

Part B: Tessellation stability for $\Delta > 1$. The three regular 2D tessellations are triangular (coordination $z = 6$), square ($z = 4$), and hexagonal ($z = 3$). We show the first is stable and the other two are unstable.

Square lattice: permanently unstable. At the Brillouin-zone X -point $\mathbf{k} = (\pi/a, 0)$, only neighbors along the x -axis contribute ($1 - \cos(\pi) = 2$ for x -axis, $1 - \cos(0) = 0$ for y -axis). The transverse (yy) component of the dynamical matrix is

$$d_{yy}(X) = -\frac{2V'(a)}{a} \cdot (1 - \cos(\pi)) + (\text{further shells}) = -\frac{4V'(a)}{a} + \mathcal{O}(a^{-2\Delta-2}) < 0,$$

where $V'(r) = -2\Delta r^{-2\Delta-1}$ gives $-4V'(a)/a = 8\Delta a^{-2\Delta-2}$, but this is the transverse response to a longitudinal perturbation, which the product $\hat{R}_y \hat{R}_y = 0$ at the x -axis neighbors kills, leaving only the negative gradient contribution $-8\Delta a^{-2\Delta-2}$ at the y -axis pairs (here $\hat{R}_y = 1$, $V' < 0$). A single negative eigenvalue suffices for instability; $d_{yy}(X) < 0$ for all $\Delta > 0$.

Hexagonal lattice: permanently unstable. The hexagonal lattice has a two-atom basis, so the dynamical matrix at any $\mathbf{k}$ is 4×4 (two sites, two spatial components each). At the K -point, the optical-mode branch (sublattices oscillating anti-phase) has eigenvalue $d_{\text{opt}}(K) = 3V'(a)/a$. Since $V'(a) < 0$ for repulsive V , this is negative for all $\Delta > 0$: the two sublattices slide against each other with no longitudinal restoring force from the three coordination bonds. A single negative optical eigenvalue suffices for instability.

Triangular lattice: stable for all $\Delta > 0$ via Montgomery's theta-function minimization. For $\Delta > 1$, the lattice sum $U[\mathcal{L}] = \frac{1}{2} \sum_{\mathbf{R} \neq 0} |\mathbf{R}|^{-2\Delta}$ converges absolutely and equals the Epstein zeta function $\frac{1}{2} \zeta_{\mathcal{L}}(2\Delta)$ of the Bravais lattice $\mathcal{L}$. For $0 < \Delta \leq 1$, the sum diverges at long range, and the energy per particle is defined by analytic continuation of $\zeta_{\mathcal{L}}(s)$ (equivalently, by Ewald summation with the self-energy subtracted). In both regimes, the same theorem applies:

Montgomery's theorem. The triangular lattice A_2 is the unique minimizer, among 2D Bravais lattices of fixed covolume, of the theta function $\theta_{\mathcal{L}}(t) = \sum_{\mathbf{R} \in \mathcal{L}} e^{-\pi t |\mathbf{R}|^2}$ for every $t > 0$ (?). By the Mellin transform

$$\pi^{-s/2} \Gamma(s/2) \zeta_{\mathcal{L}}(s) = \int_0^\infty t^{s/2-1} (\theta_{\mathcal{L}}(t) - 1/A) dt + (\text{pole subtraction at } s = 2),$$

Montgomery's pointwise minimization of $\theta_{\mathcal{L}}(t)$ implies minimization of $\zeta_{\mathcal{L}}(s)$ for every real $s > 0$ in its (analytically continued) domain, with the triangular lattice as the unique minimizer. Substituting $s = 2\Delta$: the triangular lattice minimizes the (regularized) lattice energy for every $\Delta > 0$, including $\Delta < 1$ where direct summation fails.

Since the triangular lattice is the unique global minimizer of the Epstein zeta at every $\Delta > 0$, its Hessian (the dynamical matrix) is positive semidefinite at every Bloch wavevector $\mathbf{k}$, with zero eigenvalues only along the 2-dimensional space of global translations (acoustic modes at $\mathbf{k} = 0$). The triangular lattice is therefore dynamically stable for every $\Delta > 0$ without any numerical input. The boundary cases are: Cohn–Kumar universal optimality (?) gives the same conclusion for $\Delta > 1$ via completely-monotonic potentials; Bonsall–Maradudin (?) cover the Coulomb point $\Delta = 1$ via explicit Ewald summation on the 2D Wigner crystal; Montgomery 1988 covers the entire range $0 < \Delta < \infty$.

The triangular lattice is therefore the unique regular 2D tessellation that is dynamically stable for the entire family $V(r) = r^{-2\Delta}$, $\Delta > 0$. $\square$

Remark 11.1 (Dynamical triangulation). *In causal dynamical triangulations (CDT) and Regge calculus, spacetime is discretized into simplices (triangles in 2D). CDT imposes this triangulation by hand as a regularization. Theorem ?? proves that the triangulation is forced by stability: the triangular lattice is the unique dynamically stable regular 2D tessellation for the entire family $V(r) = r^{-2\Delta}$, $\Delta > 0$ (Montgomery ?), including the logarithmic limit at $\Delta \rightarrow 0^+$. The square $\{4, 4\}$ and hexagonal $\{6, 3\}$ tessellations are dynamically unstable at every $\Delta > 0$ (explicit X-point and K-point negative eigenvalues in the proof above). The gravitational path integral $Z = \sum_{\text{tilings}} e^{-S}$ therefore localizes on triangulated configurations: the square and hexagonal alternatives are saddle points with negative phonon modes and are exponentially suppressed in the semiclassical expansion. CDT's triangulation prescription is not a regularization choice; it is the unique stable lattice, forced by the universal optimality of A_2 . The extension to irregular triangulations (as used in full CDT) remains an open geometric problem, but the regular case is settled.*

12 The polygon partition function

The one-loop partition function of the vortex system organizes naturally by algebraic number field.

Proposition 12.1 (Polygon partition function). *The saddle-point partition function for the vortex-polygon system (a discrete sum over polygon number N , each contributing a single conical geometry, not a path integral over metrics) on $\mathbf{H}^2$ is*

$$Z_{\text{grav}} = \sum_{N=3}^{\infty} e^{-b(N)} Z_{\text{frozen}}(N), \quad (12.1)$$

where $b(N)$ (§??) is the gravitational action and $Z_{\text{frozen}} = 2^{3(N-2)/4}/(N-3)!!$ is the one-loop determinant (even N ; the odd- N case is computed from Casimir gaps). The series converges super-exponentially: $e^{-b(N)} \sim e^{-N^2/12}$. Numerically, $Z_{\text{grav}} = 1.3304$ and $N = 3, \dots, 8$ capture 99.99% of the total.

Proof. Convergence: $e^{-b(N)} = O(e^{-N^2/12})$ decays faster than any polynomial, while $Z_{\text{frozen}} = O(N^{O(N)})$ grows sub-exponentially; the product vanishes. The numerical value $Z = 1.3304$ is computed by direct summation to $N = 20$ (the tail $N > 20$ contributes $< 10^{-20}$). $\square$

The partition function decomposes by *trace field*, the algebraic number field generated by $\cos(2\pi/N)$:

Trace field	N values	Fraction of Z
$\mathbb{Q}$ (rational)	3, 4, 6	90.98%
$\mathbb{Q}(\sqrt{5})$ (golden)	5, 10	8.26%
$\mathbb{Q}(\sqrt{2})$ (silver)	8	0.42%
$\mathbb{Q}(\cos 2\pi/7)$ (cubic)	7	0.34%
All higher fields	≥ 9	$< 0.01\%$

The rational field dominates; the golden ratio field gives the leading correction; the entire algebraic tower is exponentially suppressed by $e^{-N^2/12}$.

The Baum-Connes assembly map $\mu: K_0^{\mathbb{Z}_N}(\mathbf{H}^2) \rightarrow K_0(C_r^*(\mathbb{Z}_N))$ sends the equivariant index $\text{ind}_m = f(m, N) + b(N)$ to the K-theory class with trace $c = 12b(N)$. (The three Kasparov module conditions for the bounded triple $\alpha_N = (E, \pi, F_\xi)$, self-adjointness, $F^2 - 1$ compact, and $[\pi(a), F]$ compact, are verified explicitly in Paper I, §??.) This is mode-independent (the same c for all m) and approximately integer: $c = N^2 + O(N)$ for all tested $N = 3, \dots, 24$. The discrepancy $c - N^2 \approx N + O(1)$ is the boundary self-energy ($N/12$) plus regularization terms.

13 The Onsager–Wheeler–DeWitt correspondence

The Feynman–Kac correspondence of §?? identifies the BO effective action’s Euclidean path integral with the WDW equation, a mathematical identity showing the polygon system’s statistical mechanics *is* a quantum system. We now extract the consequences.

Proposition 13.1 (Self-decoherence of the breathing mode). *The $N-2$ angular Havelock modes decohere the breathing mode ρ with decoherence rate*

$$\gamma = \frac{[\Phi'(\rho)]^2}{4} \sum_{m \neq m^*} \frac{1}{\lambda_m^2}. \quad (13.1)$$

The reduced density matrix after tracing out the bath is

$$\varrho_{\text{red}}(\rho, \rho') = \Psi_0(\rho) \Psi_0^*(\rho') e^{-\gamma(\rho - \rho')^2}.$$

For $\gamma \gg 1$ the off-diagonal elements are exponentially suppressed and ϱ_{red} is effectively diagonal: $\varrho_{\text{red}} \approx |\Psi_0(\rho)|^2 \delta(\rho - \rho')$.

Proof. The CL influence functional for the Gaussian bath with ρ -dependent frequencies $\omega_m(\rho) = \sqrt{\lambda_m(\rho)}$ gives a decoherence kernel $\exp(-\gamma(\rho - \rho')^2)$ with $\gamma = \sum_m (g_m/\omega_m)^2$, where $g_m = \partial\omega_m/\partial\rho = \Phi'/(2\sqrt{\lambda_m})$ is the system-bath coupling. Substituting: $\gamma = (\Phi')^2/4 \times \sum 1/\lambda_m^2$. For $N = 7$ (all modes stable on $\mathbf{H}^2$): $\gamma \approx 7$ –12 across the stable range. For $N = 11$ at the stability threshold $\rho_{11}^* \approx 0.88$ (the cosmologically relevant operating point, where all $N-2 = 9$ bath modes have $\lambda_m > 0$): $\gamma \gg 1$. Near the threshold, the critical mode m^* is excluded from the bath (it is the “system” variable). For even N , $m^* = N/2$ is self-palindromic ($N - m^* = m^*$), so only one mode is excluded. For odd N , the palindromic pair $\{m^*, N - m^*\}$ share the eigenvalue λ_{m^*} ; excluding only m^* leaves the partner $N - m^*$ in the bath with $\lambda_{N - m^*} \rightarrow 0$, so $1/\lambda^2 \rightarrow \infty$ and γ grows without

bound as the threshold is approached. The polygon is therefore *strongly decohered* at the physical operating point: the soft bath mode near the stability threshold drives $\gamma \gg 1$ structurally.

Irreversibility from near-integrability. The CMS angular bath is integrable *in isolation*. The coupling to the breathing mode ρ through the ρ -dependent frequencies $\lambda_m(\rho)$ breaks exact integrability at $O(1/c)$ (the W_N quartic coupling, $\sim 2\%$ at $N = 7$). This converts exact Poincaré recurrences (which would periodically restore coherence) into diffusive mixing. The Nekhoroshev theorem bounds the mixing timescale from below: $\tau_{\text{mix}} \geq \exp(c_0 c^a)$ for some $c_0 > 0$, $a > 0$ (with $\varepsilon = O(1/c)$ as the perturbation parameter). Even this exponentially long mixing time satisfies $\tau_{\text{mix}} \ll \tau_{\text{tunnel}} \sim e^{S_{\text{BO}}}$, provided $c_0 c^a < S_{\text{BO}}$ (for $c \approx 50\text{--}130$ and $S_{\text{BO}} \approx 38\text{--}103$, this is satisfied for generic Nekhoroshev exponents $a \leq 1/2$). A second bound comes from the Poincaré recurrence time of the full $(N-2)$ -mode system, $\tau_{\text{rec}} \sim \exp(O(N-2)) \sim e^5\text{--}e^9 \approx 150\text{--}8000$, again $\ll \tau_{\text{tunnel}}$. The decoherence is therefore *irreversible* on all cosmologically relevant timescales.

This is consistent with the exact BO result (Paper VI, eq. (??)): the mode-independence of Φ makes the BO potential exact (no non-adiabatic transitions), while the $O(1/c)$ W_N quartic, which breaks the exact CMS integrability, provides the mode mixing needed for irreversible decoherence. The two results operate at *different orders in* $1/c$ and are not contradictory: at $O(c^0)$, the CMS integrability gives exact BO separation (Havelock eigenvectors are ρ -independent, the angular modes decouple completely); at $O(1/c)$, the W_N quartic coupling breaks exact integrability and provides the system-bath coupling needed for decoherence. The decoherence is a perturbative $O(1/c)$ correction to an exactly decoupled $O(c^0)$ system, the standard situation in atomic/molecular BO theory, where non-adiabatic corrections at $O(m_e/m_p)$ allow energy transfer between electronic and nuclear sectors without contradicting the BO separation. $\square$

Implications. The CL decoherence selects the ρ -basis as the pointer basis ($\gamma \gg 1$ at the physical operating point), with $|\Psi_0(\rho)|^2 = e^{-\beta V}/Z$ as the Boltzmann weight (?). The polygon decoheres *itself*: no external observer or environment is required. The $N-2$ angular modes form a Caldeira–Leggett bath for the breathing mode ρ . The four CL conditions hold: (i) harmonic bath (Havelock diagonalization, exact); (ii) uniform parametric coupling ($\partial\lambda_m/\partial\rho = \Phi'(\rho)$ for all m); (iii) timescale separation ($\tau_{\text{tunnel}} \sim e^{S_{\text{BO}}} \gg \tau_m \sim O(1)$); (iv) FDT (automatic for Gaussian systems). The resulting Onsager–Machlup action is $S_{\text{OM}} = \int [\frac{c}{2}\dot{\rho}^2 + V(\rho)] d\tau$, where $\dot{\rho}^2$ is the noise kernel (not a kinetic energy).

The Markovian approximation requires $\tau_{\text{bath}} \ll \tau_{\text{tunnel}}$. The bath correlation time is $\tau_{\text{bath}} = 1/\omega_{\text{min}} = 1/\sqrt{\lambda_{\text{min}}^{(\text{bath})}}$ where $\lambda_{\text{min}}^{(\text{bath})}$ is the smallest *positive* eigenvalue among the bath modes (excluding the marginal mode(s) m^* and, for odd N , its palindromic partner $N-m^*$). At the palindromic threshold $C_1 = f(m^*, N)$: $\lambda_{\text{min}}^{(\text{bath})} = f(m^*, N) - f(m^*-1, N) = (N-2m^*+1)/2$, which equals 1 (odd N) or $1/2$ (even N). Thus $\tau_{\text{bath}} \leq \sqrt{2}$. The tunneling time $\tau_{\text{tunnel}} \sim e^{S_{\text{BO}}}$ with $S_{\text{BO}} \gg 1$ (Paper VI), so the separation $\tau_{\text{bath}}/\tau_{\text{tunnel}} = O(e^{-S_{\text{BO}}})$ is exponentially sharp, and non-Markovian corrections are $O(e^{-S_{\text{BO}}})$; the coefficient $c = 12b(N)$ is classical (Proposition ??). The Feynman–Kac transfer matrix gives $H\Psi_0 = 0$ (the WDW equation, Proposition ??), with Born rule $|\Psi_0|^2 = e^{-\beta V}/Z$ as a mathematical consequence:

Classical (Onsager)	Quantum (WDW)
Boltzmann weight $e^{-\beta V}$	$ \Psi_0 ^2$
Partition function Z	$\langle \Psi_0 \Psi_0 \rangle$
Thermal correlation $\langle \rho(\tau_1) \rho(\tau_2) \rangle$	$\langle \Psi_0 \hat{\rho} e^{-H\Delta\tau} \hat{\rho} \Psi_0 \rangle$

Derivation 13.2 (Quantum structure from polygon dynamics). *Starting from the classical Kirchhoff equations for N point vortices on $\mathbf{H}^2$, with no quantum postulate, the following quantum-mechanical structures are derived:*

- (Q1) Hilbert space. $\mathcal{H} = L^2(\mathbb{R}, d\rho)$ (Feynman–Kac, Proposition ??).
- (Q2) Hamiltonian. $H = -\frac{1}{2c} d^2/d\rho^2 + V(\rho)$ with $c = 12b(N)$ (CL mechanism + orbifold matching).
- (Q3) Physical state. The WDW constraint $H\Psi_0 = 0$ selects the ground state (the additive constant in V is fixed by $V(\rho^*) = 0$ at the BO equilibrium, giving $E_0 = 0$; excited states $E \neq 0$ do not satisfy the constraint).
- (Q4) Ground-state position distribution. $|\Psi_0(\rho)|^2 = e^{-\beta V}/Z$ (mathematical identity: the Boltzmann weight of the classical Onsager–Machlup path integral equals the quantum ground-state probability in the ρ -basis).
- (Q5) Pointer basis. The angular bath decoheres ρ with $\gamma \gg 1$ near the stability threshold (Proposition ??). The ρ -basis is the pointer basis.
- (Q6) Uncertainty principle. $\Delta\rho \Delta p = 1/2$ (minimum-uncertainty ground state near the turning point; $\Delta\rho = 1/\sqrt{2c\omega}$, $\Delta p = \sqrt{c\omega/2}$, $\omega^2 = |V''(\rho_1)| = \text{csch}^2(\rho_1)$).
- (Q7) Entanglement. The entanglement entropy between ρ and the angular bath is $S_{\text{ent}} \gg 1$ bit (from the CL squeezing parameters $r_m = (\Phi'/(2\lambda_m))^2$; see Code Availability).
- (Q8) Irreversibility. The breathing-angular coupling breaks exact CMS integrability at $O(1/c)$, converting exact recurrences into diffusive mixing. The Nekhoroshev bound gives $\tau_{\text{mix}} \geq \exp(c_0 c^a)$; even this exponentially long mixing time satisfies $\tau_{\text{mix}} \ll \tau_{\text{tunnel}} \sim e^{S_{\text{BO}}}$ for generic exponents. Decoherence is irreversible on all cosmologically relevant timescales.

Within the WDW framework ($H\Psi = 0$, unique state, no external time), items (Q1)–(Q5) are derived from the Feynman–Kac correspondence; items (Q6)–(Q8) are standard consequences of the WDW Hamiltonian structure (not specific to the polygon system).

Proof. Items (Q1)–(Q4) follow from the five-step chain above (Steps 1–5: Havelock $\rightarrow$ CL $\rightarrow$ Onsager–Machlup $\rightarrow$ Feynman–Kac $\rightarrow$ WDW). Item (Q5) is Proposition ?? with the mode-mixing irreversibility. Item (Q6): the WDW ground state near the turning point ρ_1 is $\Psi_0 \propto \exp(-c\omega(\rho - \rho_1)^2/2)$ with $\omega^2 = |V''(\rho_1)| = \text{csch}^2(\rho_1)$; the widths $\Delta\rho = 1/\sqrt{2c\omega}$ and $\Delta p = \sqrt{c\omega/2}$ give $\Delta\rho \Delta p = 1/2$ (minimum uncertainty). Item (Q7): the CL influence functional at zero temperature gives squeezing parameter $r_m = (\Phi'/(2\lambda_m))^2$ for each bath mode; the entanglement entropy is $S_{\text{ent}} = \sum_m [(r_m+1) \ln(r_m+1) - r_m \ln r_m]$. Item (Q8): the $O(1/c)$ W_N quartic coupling breaks exact CMS integrability; the Nekhoroshev bound gives $\tau_{\text{mix}} \geq \exp(c_0 c^a)$, and $\exp(c_0 c^a) \ll e^{S_{\text{BO}}}$ for generic exponents $a \leq 1/2$ at $c \approx 50$ –130, $S_{\text{BO}} \geq 38$. $\square$

Summary of the bridge.

Output	Mechanism	Input
Schrödinger (WDW)	Feynman–Kac	Onsager stat. mech.
CS Hilbert space	Vortex partition fn. at $k = c/6$	$b(N)$ from eigenvalues
3D Einstein	Havelock + Hadamard	Paper III, Thm. 5
4D vacuum Einstein	CS + KK	Paper IV, Theorem 14.3(a)
4D Einstein + matter	CS + KK + RT (derived from CFT ₂)	Paper IV, Theorem 14.3(b)
$SU(3) \times SU(2) \times U(1)$	McKay + Eightfold Way + KK	$N = 7$
$\sin^2 \theta_W = 3/11$	CS conformal weights	Paper IV, Derivation 9.1
$\delta_{\text{CKM}} = \arctan(\sqrt{7}) = 69.3^\circ$	$\mathbb{Z}_7$ Gauss sum	Paper IV, Derivation 13.1

Every row traces back to the classical vortex system. The quantum-mechanical description of the breathing mode enters through the Feynman–Kac theorem (a mathematical identity relating Brownian motion to the Schrödinger equation).

Remark 13.1 (Entropy bridge: Onsager coarse-graining to Bekenstein–Hawking). *The central charge $c = 12b(N)$ that appears as the Fisher–Rao information metric on the angular-mode Boltzmann distributions (eq. (??), Step 2 above) is the same c that enters the Cardy formula via Brown–Henneaux ($c = 3\ell/(2G_3)$, §??). The Bekenstein–Hawking entropy at the palindromic threshold is*

$$S_{\text{BH}} = \frac{\pi c}{3} \cosh \rho^* = \frac{\pi c}{3} T_{\text{GH}} L, \quad (13.2)$$

where $T_{\text{GH}} = \coth(\rho^*)/(2\pi)$ is the Gibbons–Hawking temperature and $L = 2\pi \sinh \rho^*$ is the horizon circumference. Three entropy levels illuminate the structure:

Level	Formula	$N=7$	Scaling
CL squeezing	$\sum_m [(r_m+1) \ln(r_m+1) - r_m \ln r_m]$	1.6	$O(1)$
One-loop (frozen det.)	$\frac{1}{2} \sum_{m \neq m^*} \ln \lambda_m(\rho^*) $	1.1	$O(N)$
Cardy (BH)	$(\pi c/3) \cosh \rho^*$	158	$O(e^{\rho^*})$

The Gaussian (Caldeira–Leggett) entanglement captures the same c but accounts for $O(1/c)$ of the total: the gap $S_{\text{BH}}/S_{1\text{-loop}} \approx 144$ (at $N = 7$, growing to 5.5×10^4 at $N = 15$) reflects the exponential density of Virasoro descendants inaccessible to the Gaussian approximation. The Onsager coarse-graining determines c ; the Cardy formula completes c to the full entropy.

Code availability. All numerical verifications are in the companion repository (<https://github.com/gspea/planetary-polygons-unified>). Parameter accounting and predictions appear in Paper VII.

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../figures/fig_wdw_potential.pdf

Figure 1: The Born-Oppenheimer potential $V(\rho)$ for $N = 7$ (left) and $N = 11$ (right). The shaded region is the classically forbidden tunneling zone. The turning point ρ^* (red dotted line) determines the WKB tunneling action S_{BO} , which fixes the electroweak hierarchy ($N = 7$) and the cosmological constant ($N = 11$).